Beyond the Label: Do Ecodesign Regulations Reduce Real-World Energy Use?

Kai Gao*

Job Market Paper

October 2026

Click here for the latest version.

Abstract

This paper studies how ecodesign regulation shapes product design and consumer welfare when compliance is tied to a single operating mode rather than overall real-world use. I examine the German washing machine market following the 2021 EU reform, which rescaled energy-efficiency labels and imposed a duration constraint on the regulated Eco program. Combining product-program-level data with a new laundry-habit survey, I document substantial scope for regulatory leakage, as the regulated Eco program accounts for only about 10 percent of reported wash cycles. Following the reform, Eco energy consumption declines, while energy consumption increases in hotter Cotton programs. To explain these patterns, I develop and estimate an equilibrium model of endogenous product entry, pricing, and program design that incorporates household wash-program choice. The estimates show that firms respond to regulatory incentives by improving measured Eco efficiency while offering more energy-intensive Cotton programs to attract households that prefer longer wash cycles. Counterfactuals imply that the Eco-duration cap raises reference-inclusive electricity consumption by 11.4 percent relative to retaining energy labels without the cap. Broadening the efficiency index to include Cotton 60°C reduces gross electricity consumption by 6.0 percent and increases consumer surplus relative to an uncapped Eco-only index.

Keywords: energy labels, product portfolio choice, information design
JEL codes: D83, L15, L68, Q48, Q58

---

*I am grateful to Panle Jia Barwick, Jean-François Houde, and Karam Kang for their guidance and support. I thank Ashley Swanson for helpful comments, and Lorenzo Magnolfi, Hyuk-Soo Kwon, Alan Sorensen, Jonathan Kroah, and participants in the UW–Madison IO Lunch and IO Seminar for valuable feedback. I thank Gundula Czyzewski of BSH and Reiner Steininger of the University of Bonn for answering technical and industry questions, and Andrea Liu for assistance with collecting product-manual data. This project was supported by the Robock Scholarship at the University of Wisconsin–Madison. All errors are my own. E-mail: kgao33@wisc.edu.

1 Introduction

Many product regulations rely on standardized tests, even though their benefits depend on how products perform in actual use. Automobile emissions provide a prominent example: Volkswagen's use of defeat-device software allowed vehicles to satisfy laboratory tests while producing substantially higher emissions on the road (Barrett et al., 2015). A related test–use distinction arises in medicine, where drug efficacy in controlled clinical trials can differ from effectiveness in routine practice (Eichler et al., 2011; Rothwell, 2005). These examples raise a broader question: when regulation rewards outcomes measured under standardized conditions, how do responses along both measured and unmeasured dimensions shape its realized benefits?[1]

This paper studies mechanisms through which measured and realized performance can diverge, using the German washing-machine market as a setting. European energy-efficiency regulation evaluates washing machines using a standardized Eco program, while households choose among multiple programs in use, including Cotton and Easy-Care. The regulated efficiency metric is prominently displayed at purchase, whereas energy use in other programs is less visible. Regulation may therefore strengthen incentives to reduce energy use in Eco while encouraging firms to enhance washing quality in unregulated programs through more energy-intensive designs. These design responses may also alter subsequent program choice by changing the relative attributes of different wash programs. Realized energy use can therefore depend not only on improvements in the regulated mode, but also on how firms design unregulated programs and how consumers use the product after purchase.

I exploit a major reform of European washing-machine regulation implemented in 2021 (European Commission, 2019a). The reform changed both how energy efficiency was measured and the design constraints firms faced. First, the energy label was rescaled from the old classes A^{+++} through D to classes A through G, and the formula used to measure energy efficiency was revised. Because most products had accumulated in the highest classes under the old system, the rescaling restored differentiation in the label and increased the market reward to lowering measured Eco energy use. Second, the reform imposed a maximum duration on the Eco cycle to make the program more convenient to use. This constraint may, however, reduce washing quality and restrict an important energy-saving margin: lowering wash temperature and compensating for the resulting loss in washing quality with a longer cycle.[2] Together, these changes increase the return to improving measured Eco efficiency while restricting how firms can achieve those improvements, providing useful variation for studying firms' design responses.

[1] A related incentive problem arises in education, where test-based accountability can shift effort toward tested outcomes and away from dimensions of learning that are not measured (Jacob, 2005).

[2] European washing machines heat wash water electrically within the machine. Water heating is by far the largest component of energy use during a wash cycle, making lower wash temperatures an important source of energy savings.

To trace these responses across product design and household use, I assemble a novel dataset that links product-market outcomes to program-level design and household behavior. The main data combine detailed model-level sales with hand-collected attributes from washing-machine manuals, including electricity use, water use, duration, and temperature for both regulated and unregulated wash programs. I complement these data with an original survey that I designed and commissioned in Germany, which measures households' program choices and their interpretation of the energy label. Together, these data allow me to study firms' design responses across programs, consumers' purchase and usage decisions, and the implications for realized energy consumption.

The data reveal three main findings. First, consumers respond strongly to the energy class but often misunderstand its scope. The 2021 label rescaling increases sales of models rated C or higher by 39.4 percent relative to those rated D or lower, while survey respondents often interpret the energy class as measuring the efficiency of the whole machine rather than the Eco program alone. Eco nevertheless accounts for only about 10 percent of reported wash cycles. Second, improvements in regulated Eco efficiency do not extend to unregulated programs. Within the same machine, Eco and Cotton energy use per cycle are negatively correlated, particularly for Cotton 60°C. Event-study estimates show that by the end of 2023, unweighted average energy use per cycle fell by 8.6 percent for Eco but rose by 11.3 percent for Cotton 60°C. Third, this divergence is consistent with firms shifting design toward unregulated programs. The duration cap makes Eco less attractive to consumers who value longer cycles, while the stricter energy-efficiency requirement limits compensation through more energy-intensive Eco inputs. These consumers instead turn toward Cotton, which can raise the marginal return to providing longer, more intensive Cotton cycles. This sorting creates incentives for energy-intensive design changes outside the program evaluated by the label. Consistent with this mechanism, a 10-percentage-point increase in exposure to the duration cap is associated with 5.9 percent higher energy use and 2.4 percent longer duration in Cotton 60°C.[3]

To explain these findings, I develop an equilibrium model of the washing machine market. Demand consists of two stages: product purchase and program use. At the purchase stage, consumers value the new energy class once the label becomes observable but do not directly value the energy efficiency of unregulated programs. They also anticipate the utility from the bundle of wash programs they will use. At the use stage, households choose among programs based on expected washing quality and convenience, which depend on program-specific inputs such as temperature, duration, and water use. The expected utility from program use enters product demand through a household-specific inclusive value, linking firms' program-design decisions to consumers' willingness to pay.

[3]Water use rises broadly across wash programs after the reform. The stricter rinsing-effectiveness requirement, which limits residual detergent after washing, may contribute to this increase.

On the supply side, firms make discrete product-entry and continuous program-design decisions. Firms choose which products to introduce and their energy-efficiency classes, then select program-specific inputs to achieve washing-quality targets, subject to class-specific energy-efficiency requirements and duration constraints for Eco. Firms trade off the demand benefits of higher energy ratings and improved program quality against design costs and regulatory constraints. By limiting firms' ability to maintain Eco washing quality, tighter regulatory constraints increase the marginal benefit of quality-enhancing inputs in unregulated programs.

I estimate demand in two steps. First, I estimate program-use preferences by simulated maximum likelihood, combining survey respondents' reported program-use frequencies with program inputs from product manuals matched to the washing machines they own. Variation in program inputs across machines and differences in household usage patterns identify preferences and their heterogeneity. Second, I use the estimated preferences to construct household-specific expected utility from each machine's bundle of wash programs and incorporate it into product demand. I identify product-demand preferences using variation in product characteristics and program designs generated by the entry of new models, which firms choose before observing demand shocks.

On the supply side, I first recover program-design cost parameters conditional on observed product portfolios. I use a minimum-distance estimator that matches observed program inputs to firms' optimal choices under engineering and regulatory constraints. Taking these costs as given, I estimate entry costs in a static discrete-choice game of incomplete information, in the spirit of Seim (2006). Estimation presents two computational challenges. First, the number of possible portfolios grows exponentially with the number of product lines. Rather than enumerate firm-level portfolio probabilities, I maximize a composite conditional likelihood based on product-line choices, conditioning on observed choices in the same firm's other lines and integrating over beliefs about rivals' portfolios. I use Gibbs sampling (Casella and George, 1992) to simulate rival portfolios from product-line conditional choice probabilities, avoiding explicit construction of their high-dimensional joint distributions. I iterate between estimating entry parameters and updating portfolio beliefs until the conditional choice probabilities converge (Aguirregabiria and Mira, 2007). Second, evaluating each candidate portfolio requires solving for optimal continuous program inputs. I train a neural-network surrogate to predict optimal inputs and profits net of design costs, avoiding repeated constrained optimization during entry estimation.

The estimated model reveals substantial heterogeneity in preferences for cycle duration. Relative to A^{+++} durations, a 30-minute reduction lowers Eco use by 5.5 p.p. among consumers who prefer longer cycles but raises it by 0.3 p.p. among those who prefer shorter cycles. On average, Eco's share falls by 2.2 p.p. and the combined Cotton share rises by 2.0 p.p. The product-demand estimates further show that consumers value both higher efficiency ratings and higher expected wash utility. These preferences give firms incentives to balance measured efficiency against the attractiveness

of the full menu of wash programs. Consistent with the sorting mechanism, shortening Eco also raises the marginal return to extending Cotton programs. Under the same duration comparison, the average marginal operating profit from extending Cotton programs rises by about 19 percent. These findings reveal a channel through which restrictions on the regulated program strengthen incentives to improve unregulated alternatives.

Policy Implications I use the estimated model to evaluate three alternative policies: removing existing regulatory restrictions, broadening the efficiency index to include additional wash programs, and introducing a carbon tax. In each case, firms adjust product introductions and program designs, while households adjust their purchase and program choices. I examine the implications for electricity consumption and consumer welfare over 2020–2023.

First, removing the Eco-duration cap while retaining energy labels increases consumer surplus by 22.5 percent relative to current policy and reduces total electricity consumption by 10.2 percent. Relaxing the cap allows firms to substitute longer washing time for higher temperatures, preserving washing quality while reducing electricity use. Longer Eco cycles also better serve households that value duration, weakening firms' incentives to attract these households through longer, more energy-intensive Cotton programs. Firms consequently offer longer, cooler Eco cycles and shorter Cotton 40°C and Cotton 60°C cycles, with little change in product introductions. Consumers benefit from both improved program offerings and lower lifetime electricity expenditure, with the latter accounting for 85.5 percent of the surplus gain. Nearly 99 percent of the gain accrues to households that prefer longer cycles. By comparison, removing efficiency requirements and labels while retaining the duration cap generates a smaller increase in consumer surplus and reduces electricity consumption by 2.5 percent. These results suggest that greater flexibility in program duration can improve consumer welfare and deliver larger energy savings while preserving the efficiency incentives provided by energy labels.

Second, I consider broadening the efficiency index to include Cotton programs. Assigning 90 percent of the weight to Eco and 10 percent to Cotton 60°C reduces total electricity consumption by 6.0 percent and increases consumer surplus relative to an Eco-only index, with the duration cap removed in both cases. Under an Eco-only index, firms can improve their energy ratings by reducing Eco energy consumption even when the associated design changes increase energy use in Cotton programs. Including Cotton 60°C in the index makes these increases costly in terms of the energy rating, encouraging firms to improve efficiency across programs rather than shift energy consumption toward unregulated uses. The results suggest that broadening the scope of energy-efficiency measurement can reduce such distortions while improving consumer welfare.

Third, I examine a carbon tax based on the electricity consumption of Eco and Cotton programs. Similar to a weighted efficiency index, the tax encourages firms to improve efficiency across

programs rather than shift energy consumption toward less regulated uses. Unlike energy labels, which reward measured efficiency through consumer purchase decisions, the tax directly raises firms' costs of energy-intensive designs. Taxing Cotton programs alone generates larger energy savings and consumer-welfare gains than taxing Eco alone, highlighting the importance of targeting programs that households use more frequently. Rebating tax revenue to consumers further increases the welfare gain. The largest gains occur at the highest tax rates considered, suggesting that the benefits of energy-based taxation may extend beyond the range examined.

Related Literature This paper contributes to three strands of literature. First, it relates to research on multitask incentives and performance measurement. Holmstrom and Milgrom (1991) and Baker (1992) show that incentives based on incomplete performance measures can distort the allocation of effort across tasks. Empirical studies examine how performance-based incentives and quality disclosure affect the allocation of effort and investment across quality dimensions (Jacob, 2005; Lu, 2012; Gupta, 2021; Vatter, 2022). I extend this literature to environmental product regulation and identify a demand-mediated channel for multitasking distortions. When regulation constrains one operating mode, consumers may shift toward unregulated alternatives or sort across products, changing firms' incentives to improve unregulated programs. These responses can induce more energy-intensive designs even as measured efficiency improves.

Second, the paper contributes to research on the equilibrium effects of product regulation and endogenous attributes. Existing studies examine labeling and certification policies (Barahona et al., 2023; Houde, 2018, 2026; Davis and Metcalf, 2016) and environmental standards and attribute-based regulations (Reynaert, 2021; Reynaert and Sallee, 2021; Ito and Sallee, 2018; Barwick et al., 2024). My setting allows me to observe both regulated and unregulated operating modes within the same product, together with household program-use information. These data allow me to examine how regulation targeting one operating mode affects the design of other modes and how consumers allocate use across them. By incorporating both margins, I evaluate whether improvements in measured efficiency translate into lower actual energy consumption and higher consumer welfare.

Third, the paper contributes to empirical entry games and endogenous product variety. Seim (2006) estimates an incomplete-information entry game using equilibrium choice probabilities, while Jia (2008) exploits strategic structure to solve a game with interdependent entry decisions. These approaches become computationally demanding when many firms choose among large portfolios of products. An alternative literature uses moment inequalities to estimate fixed costs without solving the full portfolio game, typically under complete information (Holmes, 2011; Houde et al., 2023; Eizenberg, 2014; Wollmann, 2018; Montag, 2023; Fan and Yang, 2025). While these approaches accommodate large action spaces, they can yield set identification and complicate counterfactual analysis, which may require additional equilibrium computations and assumptions about equilib-

rium selection. I develop an alternative approach that retains likelihood-based estimation in an incomplete-information game with large product portfolios. I use product-line conditional choice probabilities to reduce the dimensionality of firms' decisions and Gibbs sampling to construct the implied joint distribution of rival portfolios. A neural-network surrogate approximates optimal program inputs and profits, avoiding repeated constrained optimization. This approach permits point estimation and extends naturally to counterfactual analysis with endogenous portfolio choices and continuous product design.

The rest of the paper proceeds as follows. Section 2 describes the institutional setting and the 2021 label reform and presents the data and documents the main empirical patterns. Section 5 develops the equilibrium model. Section 6 describes the estimation strategy, and Section 7 presents the estimation results. Section 8 presents the counterfactual policy analysis. Section 9 concludes.

2 Institutional Background and Data

2.1 The 2021 EU Energy Label Reform

The EU energy-labeling framework covers a wide range of household appliances, including refrigerators, washing machines, dishwashers, and air conditioners. Its purpose is to provide consumers with standardized information on product performance, particularly energy efficiency, and to strengthen firms' incentives to improve product design. Retailers are required to display energy labels both in physical stores and in online listings. This paper focuses on household washing machines.

The EU has implemented three major energy-labeling regimes for household washing machines, introduced in 1995, 2011, and 2021. This paper studies the transition from the 2011 regime to the 2021 regime. The reform was announced in March 2019 and took effect on March 1, 2021. Transitional provisions effective in December 2019 allowed manufacturers to update control panels in anticipation of the new requirements, enabling firms to begin adapting their products in early 2020. Consumers, however, continued to see the old energy labels until March 2021, when the new labels were introduced. From March 2021, newly placed products had to comply with the revised ecodesign requirements, while existing inventory could remain on sale under transitional rules.[4]

The 2021 reform introduced two major changes. First, it rescaled energy-efficiency ratings and revised both the standardized testing procedure and the formula used to determine them. Second, it introduced additional restrictions on the design of the regulated program, particularly limits on cycle duration. Together, these changes altered both the information consumers observe when purchasing washing machines and the constraints firms face when designing wash programs. Figure 2 compares

[4]Retailers had 14 working days to replace existing labels. Certain discontinued models could continue to be sold with their old labels until November 30, 2021.

the labels under the two regimes.

Energy-Label Rescaling Washing machines offer multiple wash programs, including Cotton, Easy Care, and Quick Wash. The EU energy label, however, measures performance under standardized test cycles rather than across all available programs.[5] Throughout the paper, I refer to these regulated test cycles as the Eco program, whose definition differs across regimes. Under the 2011 regime, the Eco program refers to the Standard Cotton 60°C cycle at full and half load and the Standard Cotton 40°C cycle at half load.[6] Under the 2021 regime, the Eco program refers to Eco 40–60, designed to clean normally soiled cotton laundry washable at either 40°C or 60°C in a single cycle. Other wash programs, including regular Cotton, are available to consumers but do not enter the calculation of the energy-efficiency rating.

Energy-efficiency classes are determined by an Energy Efficiency Index (EEI), which compares energy consumption measured under the Eco program with a capacity-dependent reference level. By the late 2010s, most washing machines had clustered in the highest efficiency classes, reducing the label's ability to differentiate among products. To restore differentiation and allow room for future efficiency improvements, the 2021 reform revised the EEI formula and replaced the previous efficiency scale, ranging from A^{+++} to D, with a new scale ranging from A to G. Appendix A.1 provides the detailed formulas and classification thresholds under both regimes.

Two distinctions matter. First, neither the differences across wash programs nor the change in the Eco program across regimes imply fundamentally different engineering technologies. Wash programs generally share the same hardware but use different combinations of temperature, duration, water use, and mechanical action to serve different washing needs.[7] Temperature designations such as 40°C and 60°C are nominal program labels rather than mandated wash temperatures, and firms retain discretion over program inputs subject to regulatory constraints.[8] Second, energy classes are not directly comparable across regimes because both the test-load conditions and the EEI formula changed. A change in the displayed class therefore does not necessarily reflect a change in actual energy consumption. As the European Commission notes, a machine previously rated A^{+++} could receive a C rating under the new label without any change to the machine (European Commission, 2021).

[5] Using designated test cycles facilitates comparability across products and limits certification costs, but may not reflect consumers' actual program choices.

[6] The Standard Cotton programs are distinct from the regular Cotton programs, and both are available on the machine's control panel.

[7] For example, Cotton 60°C is intended for more intensive cleaning, Easy Care prioritizes gentler fabric treatment, and Quick Wash is designed for time convenience.

[8] In practice, Eco programs typically operate at wash temperatures between 30°C and 45°C, below their nominal temperature designations.

Attribute Restrictions Beyond energy labeling, EU regulations impose attribute restrictions on the Eco program that do not apply to other wash programs. These restrictions constrain how firms balance cleaning performance against energy consumption. According to Sinner's Circle (Sinner, 1960), wash performance depends on four interacting factors: temperature, mechanical action, duration, and detergent chemistry.[9] Higher temperatures, longer cycles, and stronger mechanical action can improve stain and odor removal but may also increase fabric wear. Water serves as an intermediate input: more water facilitates rinsing but dilutes detergent and reduces friction between garments, while less water increases mechanical action but may hinder stain removal. Program inputs also affect electricity consumption, although their effects differ. Water heating accounts for more than 70 percent of electricity consumption in a typical wash cycle, making temperature reduction a primary means of reducing energy consumption. Lower temperatures, however, may compromise cleaning performance unless offset by adjustments in other inputs. Regulatory restrictions on program attributes can alter this tradeoff.

The main change in attribute restrictions under the 2021 reform was the introduction of duration limits on the Eco program. Under the earlier regime, manufacturers could improve measured energy efficiency by lowering wash temperatures and compensating with longer cycles while maintaining wash performance (Alborzi et al., 2017). However, excessively long cycles may discourage consumers from using the Eco program, limiting energy savings in actual use. To improve time convenience and encourage Eco use, the 2021 reform imposed capacity-dependent duration caps, with separate limits at full, half, and quarter loads. The effect on consumer demand, however, is ambiguous. Consumers who value time convenience may prefer shorter Eco cycles, while those who associate longer cycles with better cleaning performance may find them less attractive and switch to other programs. The resulting effect on actual energy consumption depends on both firms' program-design responses and consumers' program choices.

The EU also imposes water-consumption and minimum wash-performance requirements under both regimes. The water constraint primarily aims to conserve water. The 2021 reform revised its measurement and introduced a separate rinsing-effectiveness requirement to limit residual detergent in clothing, which may cause skin irritation and allergic reactions (European Commission, 2019b). As an intermediate input, water serves multiple purposes, and its use may largely reflect the need to satisfy rinsing requirements rather than considerations of wash quality and energy efficiency alone. Beyond water constraints, both regimes require a minimum level of stain removal, measured by a washing-efficiency index using standardized stain strips relative to a reference machine. The minimum threshold remained unchanged across regimes, although testing conditions changed. Advances in washing technology have made it possible to meet this requirement even at low tem-

[9] Because detergent chemistry is standardized in regulatory tests, I treat it as exogenous to manufacturers' program-design decisions.

peratures (Josephy et al., 2015), suggesting that compliance need not be a major technological obstacle. The index is not displayed on the energy label but is reported in the product information sheet. The reported washing-efficiency index clusters near the regulatory threshold, consistent with limited incentives to report performance above the minimum requirement. I therefore treat wash performance as a pass-or-fail certification constraint rather than a continuous attribute valued by consumers. Appendix A.2 reports the detailed regulatory constraints.

2.2 Data

I combine three main data sources: a product-level sales panel from GfK, program-level consumption information collected from product manuals, and a household laundry-habits survey conducted in Germany by Norstat on my behalf.

Product-Level Sales and Characteristics My primary data are a monthly panel of washing-machine sales in Germany from 2018 to 2023, provided by GfK. The data cover more than 90 percent of sales through electrical retailers across six geographic regions. For each region-month-model observation, the data report unit sales, revenues, and product characteristics, including energy-efficiency class, capacity, maximum spin speed, spin-drying efficiency, and noise level. To construct monthly market shares, I divide model-level unit sales by the estimated number of potential purchasers in each region-month. I define market size as the number of households in each region multiplied by the national washing-machine ownership rate and a monthly replacement probability implied by an average product lifespan of 12 years. To account for income heterogeneity in product demand, I supplement the sales data with the Panel on Household Finances (PHF), conducted by the Deutsche Bundesbank. I use the PHF to construct the household income distribution for demand estimation.

I focus on seven firms operating eight brands: BSH (Bosch and Siemens), Electrolux (AEG), Samsung, Miele, LG, Whirlpool (Bauknecht), and Arçelik (Beko). These firms account for approximately 93 percent of sales. I restrict the sample to washing machines with commonly purchased capacities of 7, 8, 9, and 10 kilograms. I define a product line as a combination of firm, load capacity, and maximum spin speed (e.g., Miele, 8 kg, 1,400 rpm). I define a product as a combination of firm, load capacity, maximum spin speed, energy-efficiency class, and quarter of initial market appearance (e.g., Miele, 8 kg, 1,400 rpm, class A, 2022Q1). The average transaction price is calculated as total revenue divided by total unit sales, aggregated across individual models within each product and region-month. The final sample contains 42 product lines and 349 products.

Program-Level Attribute Data My second data source consists of model-program-level information collected from 2,087 washing-machine user manuals. These manuals report energy consump-

tion, load size, water consumption, duration, and temperature for individual wash programs. Unlike the energy label, which bases efficiency ratings on Eco performance, the manuals also report attributes of non-Eco programs, allowing me to compare regulated and unregulated programs within the same machine. The completeness of reporting differs across regulatory regimes. Under the 2021 regime, manufacturers are required to report Eco attributes at full, half, and quarter loads, including energy consumption, water use, duration, and temperature. Manuals also report commonly used non-Eco programs, including Cotton 20°C, Cotton 60°C, and Easy Care, while Cotton 40°C is reported by some brands. Under the earlier regime, reporting was less standardized, with temperature and other attributes frequently missing.

To improve coverage, I supplement the manuals with the European Product Registry for Energy Labelling (EPREL), using overlapping observations to validate reported attributes and fill missing values.[10] I impute the remaining missing attributes using the procedure described in Appendix B, obtaining complete program-level attributes for all models in the demand sample. I then aggregate these attributes to the product level by taking the unweighted average of each program's attributes across models belonging to the same product, as defined above.

Household Laundry-Habits Survey My third data source is a household laundry-habits survey that I commissioned in Germany in December 2025 and fielded through Norstat. The survey contains 508 completed responses from households that purchased washing machines under either the old or new labeling regime. I collect household demographics, appliance purchase history, and the model identifier and purchase date of each respondent's washing machine. To measure consumer information, I ask whether respondents considered the energy-efficiency class when purchasing their machine and whether they interpret the rating as reflecting the efficiency of specific wash programs or the entire machine. I also collect self-reported washing behavior, including the frequency of use of each available wash program and the frequency of washing lightly and heavily soiled laundry.[11]

Using the reported model identifiers, I match each respondent to the corresponding product manual and recover the attributes of the wash programs available on their machine. The matched data combine household characteristics, beliefs about energy labels, program-use frequencies, and program-specific energy consumption, temperature, water use, and duration.

Table 1 reports summary statistics for product characteristics, demand, and wash-program attributes. Panel A summarizes product characteristics and sales by energy-efficiency class. Higher-rated products tend to be more expensive and have larger capacities. Under the new label, average

[10] EPREL is the EU product database containing technical information submitted by manufacturers for energy-labelled appliances. Under the 2021 regime, its technical documentation includes Eco energy consumption, water use, duration, and temperature at full, half, and quarter loads.

[11] The English version of the questionnaire is provided in Appendix C.

prices increase from €409 for class D to €601 for class A, while average capacity rises from 7.73 to 8.51 kg. Panel B reports program attributes and self-reported usage. Eco consumes less energy than Cotton 40°C and Cotton 60°C, averaging 0.87 kWh per cycle compared with 1.04 and 1.49 kWh, respectively. Eco also operates at lower temperatures and uses less water but runs substantially longer, averaging 225 minutes compared with approximately 184 minutes for the two Cotton programs.

3 Regulation-Induced Energy Shifting

This section develops a stylized model, in the spirit of Holmstrom and Milgrom (1991) and Baker (1992), to illustrate how regulation targeting Eco can shift energy consumption toward unregulated programs, potentially increase total electricity use, and generate socially inefficient choices.

Consider a firm offering a washing machine with two programs, Eco and Cotton, with durations d_E and d_C. Their energy consumption is given by $e_E(d_E)$ and $e_C(d_C)$, respectively. Both functions are increasing in duration, while Eco consumes less energy at a given duration. Consumers observe Eco energy consumption at purchase but mistakenly perceive it as representative of both programs. Because perceived energy costs are identical across programs, they enter product-purchase demand but cancel from program-choice comparisons. Program utility is therefore $v_r = \beta d_r + \varepsilon_r$, where β captures the average preference for duration and ε_r follows a standard type-I extreme-value distribution. Expected program utility is

$$\mathcal{I}(d_E, d_C) = \log\left(e^{\beta d_E} + e^{\beta d_C}\right).$$

Taking product price P as given and normalizing the firm's markup to one, the firm chooses program durations to maximize demand net of separable design costs:

$$\max_{d_E, d_C} \quad Q\left(P + \kappa e_E(d_E) - \theta \mathcal{I}(d_E, d_C)\right) - C_E(d_E) - C_C(d_C) \qquad \text{s.t.} \qquad d_E \leq \bar{d}_E,$$

where $Q' < 0$, $\kappa > 0$ captures consumers' valuation of perceived energy consumption, $\theta > 0$ measures the contribution of expected program utility to product demand, and C_E and C_C are separable design costs.[12]

Program designs are substitutes in expected utility, since $\frac{\partial^2 \mathcal{I}}{\partial d_E \partial d_C} = -\beta^2 s_E s_C \leq 0$, where s_r denotes the program-choice probability. Shortening one program increases the marginal value of extending the other, regardless of the sign of β. When this substitution carries over to product demand, tightening the Eco-duration cap increases the marginal return to extending Cotton.

[12]The reduced-form demand specification follows from logit product choice when perceived energy consumption and expected program utility enter purchase utility additively, with coefficients expressed in price-equivalent units.

Proposition 1 (Regulation-Induced Energy Shifting)**.** *Suppose the Eco-duration cap binds and Cotton duration has an interior local optimum satisfying the second-order condition. If* $\frac{\partial^2 Q}{\partial d_C \partial d_E} < 0$*, tightening the Eco-duration cap increases optimal Cotton duration and energy consumption:*

$$\frac{\partial d_C^*}{\partial \bar{d}_E} < 0, \qquad \frac{\partial e_C(d_C^*)}{\partial \bar{d}_E} < 0.$$

Proof. See Appendix D.1. □

The proposition establishes that restricting Eco duration can induce firms to increase energy consumption in the unregulated Cotton program. Whether this shifting raises electricity consumption also depends on consumer program choices. Conditional on machine purchase, average electricity consumption per cycle is $\mathcal{E} = s_E e_E(d_E^*) + s_C e_C(d_C^*)$, where $s_E + s_C = 1$. Differentiating with respect to the Eco-duration cap yields

$$\frac{\partial \mathcal{E}}{\partial \bar{d}_E} = \underbrace{s_E \frac{\partial e_E}{\partial d_E}}_{\text{Direct Eco effect}} + \underbrace{s_C \frac{\partial e_C}{\partial d_C} \frac{\partial d_C^*}{\partial \bar{d}_E}}_{\text{Firm design response}} + \underbrace{\left[e_C(d_C^*) - e_E(d_E^*)\right] \frac{\partial s_C}{\partial \bar{d}_E}}_{\text{Consumer substitution}}.$$

where the Eco-duration cap binds. Tightening the cap reduces Eco energy consumption but increases Cotton energy consumption. When Cotton consumes more energy than Eco, tightening the cap shifts program use toward Cotton if the average preference for duration is positive, reinforcing the increase in energy consumption. Average electricity consumption can therefore increase if the Cotton energy and program-substitution effects outweigh the Eco savings.

The interaction between energy-efficiency and duration constraints can further affect firms' design responses. When firms can adjust other Eco inputs, such as temperature, a tighter duration cap creates room under a binding Eco energy target to increase these inputs, potentially offsetting the shift toward Cotton. A stricter energy target limits this adjustment and can strengthen incentives to improve Cotton instead. Appendix D.2 formalizes this interaction.

These design responses also have welfare implications when consumers have incomplete energy information. Consumers perceive Eco energy consumption as representative of both programs, overlooking differences in their actual electricity costs. A social planner instead accounts for program-specific energy consumption when choosing program designs and allocating washing cycles. Firms may therefore choose excessively long and energy-intensive Cotton programs relative to the social optimum. Details are shown in Appendix D.3.

4 Descriptive Evidence

This section presents three empirical patterns that motivate the structural model in Section 5 and relate to the mechanisms developed in Section 3. First, following the reform, Eco energy

consumption per cycle initially rises before declining, while Cotton energy consumption increases and remains elevated, documenting divergent responses across programs. Second, products more exposed to the Eco-duration cap exhibit longer and more energy-intensive Cotton cycles relative to old-label benchmarks, consistent with the proposed mechanism of regulation-induced energy shifting. Third, consumers respond to energy ratings at purchase but often interpret Eco-based ratings as measuring whole-machine efficiency, even though non-Eco programs account for most reported washing cycles.

4.1 Changes in Program-Level Energy Consumption

Because program characteristics are fixed over a product's lifetime, I examine changes in mean energy consumption across products offered within each product line, weighting products equally. These changes reflect product turnover rather than shifts in sales shares. I restrict the sample to product lines observed between 2018Q1 and 2023Q4. For comparability across labeling regimes, I measure energy consumption per kilogram and apply common load weights to Eco programs.[13] I estimate

$$\log \bar{e}_{\ell rt} = \sum_{r' \in \mathcal{R}} \sum_{k \neq -1} \beta_{kr'} \mathbf{1}\{t - \tau_0 = k\} \mathbf{1}\{r = r'\} + \mu_{\ell r} + \mu_t + \varepsilon_{\ell rt}, \tag{1}$$

where $\bar{e}_{\ell rt}$ is mean energy consumption per kilogram in product line ℓ, program r, and quarter t. The specification includes product-line-by-program and quarter fixed effects. The set $\mathcal{R}$ contains Eco, Cotton 20°C, Cotton 40°C, Cotton 60°C, and Easy Care. Wool serves as the reference program because it requires specialized care for delicate fabrics, making its design less likely to respond to policy-induced changes in program inputs. The coefficients capture changes relative to Wool.

Figure 3 shows divergent trends relative to Wool and the pre-policy reference quarter. Eco energy consumption initially rises before declining after 2021, whereas Cotton 60°C energy consumption rises and remains elevated. By 2023Q4, the estimated changes are −2.2 percent for Eco and 14.9 percent for Cotton 60°C. Cotton 40°C and Easy Care also increase by 8.4 and 7.9 percent, respectively.

To examine the relationship between energy consumption per cycle across programs within a machine, Figure 4 plots Eco against Cotton energy consumption by machine capacity. It suggests that greater efficiency in the regulated Eco program does not necessarily translate into lower energy consumption in unregulated programs. Across all three capacities, machines with lower Eco energy consumption tend to have higher Cotton 60°C energy consumption, while the relationship for Cotton 40°C is weaker.

[13] I apply the new-label weights, normalized across full and half load, to Eco 40–60 under the new regime and to Standard Cotton 60°C at full load and Standard Cotton 40°C at half load under the old regime.

4.2 Eco Duration-Cap Exposure and Non-Eco Program Design

Proposition 1 in Section 3 predicts that a binding Eco duration cap shifts performance-related inputs toward unregulated Cotton programs. To investigate this mechanism, I examine whether products more exposed to the cap have longer and more energy-intensive Cotton programs relative to pre-policy benchmarks.

The sample consists of dual-label products with observed old- and new-label energy classes. Because their non-Eco outcomes are not observed under both regimes, I compare them with matched old-label-only A^{+++} products. For each program r and outcome $q \in \{e, d, w\}$, denoting energy consumption, duration, and water use, respectively, I estimate

$$\log q_{jr}^{\text{new}} - \log \widehat{q}_{jr}^{\text{old}} = \beta_r^q \, \text{Exposure}_j + \mu_{k(j)}^{q,r} + \eta_{b(j)}^{q,r} + \varepsilon_{jr}^q, \tag{2}$$

where $\widehat{q}_{jr}^{\text{old}}$ is the mean outcome among matched old-label products, and $\mu_{k(j)}^{q,r}$ and $\eta_{b(j)}^{q,r}$ denote capacity and brand fixed effects. Exposure measures how much pre-policy regulated-program duration exceeds the new cap:

$$\text{Exposure}_j = \max\left\{0, \frac{\overline{d}_{f(j)k(j),\text{reg}}^{\text{pre}} - ub_{k(j)}}{ub_{k(j)}}\right\}, \tag{3}$$

where $\overline{d}_{f(j)k(j),\text{reg}}^{\text{pre}}$ is the mean pre-policy duration within a firm–capacity group and $ub_{k(j)}$ is the new duration cap. Identification comes from variation in exposure across firm–capacity groups, conditional on brand and capacity fixed effects.

Table 2 reports the estimates. A 10-percentage-point increase in exposure is associated with 4.5 percent higher energy consumption and 1.5 percent longer duration in Cotton 60°C relative to old-label benchmarks. Cotton 40°C energy consumption also increases by 1.9 percent, although its duration response is not statistically significant. Neither program exhibits a significant change in water use. These findings are consistent with Proposition 1, suggesting that a binding Eco duration cap shifts performance-related inputs toward Cotton programs, particularly Cotton 60°C.

4.3 Consumer Program Use, Label Beliefs, and Purchase Responses

The household survey reveals a substantial gap between how consumers use washing machines and what they believe energy ratings measure. Figure 6 shows that Eco accounts for only 9% of reported washing cycles among new-label households, compared with 25% for Cotton 40°C, 17% for Cotton 60°C, and 22% for Easy Care. Eco usage is also limited among old-label households, accounting for 13% of cycles. Despite this limited use, Figure 7 shows that 71% of new-label households mistakenly believe the energy class reflects energy consumption across all programs, even though it is determined by Eco performance alone. Moreover, 67% rely on the energy class rather

than reported kWh consumption to assess energy costs. These findings suggest that consumers may value energy ratings as indicators of whole-machine efficiency, even though the ratings do not reflect energy consumption in their most frequently used programs.

To investigate purchase responses, I exploit the 2021 label rescaling from A^{+++}–D to A–G. Before the reform, most products were rated A^{+++}, providing little differentiation across models. The rescaling redistributed these products across the new A–D classes, creating variation in the information conveyed by energy ratings. I examine whether products receiving more favorable ratings experienced larger sales gains when the new labels became visible at retail.

Importantly, the transition in product design preceded the retail label switch. Beginning in 2020, manufacturers started updating control panels and Eco programs to accommodate the new regulatory requirements, following permission to display the Eco designation from late December 2019.[14] The new energy labels, however, became visible at retail only on March 1, 2021. Thus, for products whose designs remained unchanged around the rollout, the label switch altered the information presented to consumers without changing their underlying energy consumption. This timing allows me to examine the demand response to the newly displayed energy ratings separately from the earlier adjustments in program design.

Using a pre-implementation snapshot, I predict each product's energy class under the new system and define $E_j = 1$ for products predicted to receive A, B, or C, and $E_j = 0$ for those predicted to receive D. I estimate an event study centered on the March 2021 retail rollout, using February 2021 as the reference month:

$$\log(q_{jmt}) = \sum_{k \neq -1} \beta_k 1\{t = k\} E_j + \beta_p \log(p_{jmt}) + \mu_{jm} + \mu_t + \epsilon_{jmt}, \tag{4}$$

where q_{jmt} and p_{jmt} denote product sales and prices, μ_{jm} are product–market fixed effects, and μ_t are month fixed effects. I censor each product's panel at its first observed post-rollout rating change to limit contamination from subsequent redesign. The coefficients β_k capture differential sales changes between products predicted to receive A–C and those predicted to receive D, holding product identity fixed.

Figure 8 reports the estimates. Pre-reform coefficients are close to zero, indicating little evidence of differential pre-trends. Following the retail rollout, products predicted to receive A–C experience an immediate and persistent increase in sales relative to D products. The average post-reform coefficient of 0.44 implies approximately 55% higher relative sales. Because product-design adjustments began before the retail rollout, the sharp increase in relative sales around March 2021 is consistent with a demand response to the newly salient energy ratings rather than contemporaneous changes in energy performance. Together with the survey evidence, these results suggest

[14] See Commission Regulation (EU) 2019/2023, Article 11, and Czyzewski (2022).

that consumers place substantial value on energy ratings, which they often mistakenly interpret as measuring whole-machine efficiency.

5 Model

I develop a static equilibrium model of washing-machine demand and supply to examine how energy labels and ecodesign requirements affect product design, consumer choices, and energy consumption. Within each period, the model follows four stages:

S1. Product Entry. Private portfolio-level entry shocks are realized. Firms simultaneously choose which product configurations to introduce, taking the distribution of rivals' entry decisions as given.

S2. Program Design. Market entry outcomes and design costs are realized and observed by all firms. Firms choose the wash-program inputs of entering products relative to a predetermined design frontier, subject to the energy-efficiency target and attribute constraints. Program inputs remain fixed over the product's lifetime.

S3. Price Setting and Product Choice. Product-level demand and marginal-cost shocks are realized. Firms set prices in Bertrand–Nash equilibrium, and households choose among the available washing machines and the outside option. Product demand internalizes the expected utility from subsequent program use.

S4. Program Use. Program-use shocks are realized before each wash occasion. Households then choose among the available wash programs of the purchased washing machine.

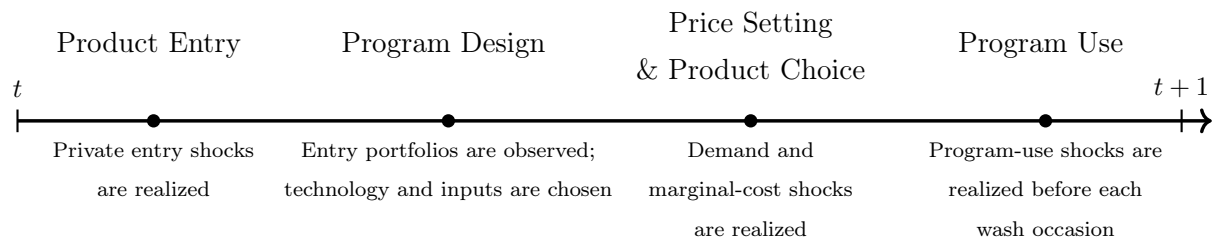


Figure 1: Timing within period t.

5.1 Demand

A market is indexed by (m, t), where m denotes region and t quarter. I model household demand as a two-stage decision problem, combining a differentiated-products demand system in the spirit of

Berry et al. (1995) with a subsequent usage decision, as in the appliance ownership and utilization framework of Dubin and McFadden (1984).

There is a continuum of households indexed by i. Household heterogeneity relevant for demand is summarized by income y_i and the household's share of high-stain laundry τ_i. Income shifts price sensitivity, while the high-stain share shifts preferences over wash-program temperature and duration. I begin with the program-use decision because it determines the continuation value associated with owning each product.

Program Use Let $r \in \mathcal{R}$ index a common set of inside wash programs: Cotton 20°C, Cotton 40°C, Cotton 60°C, Easy Care, and Eco. Let $r = 0$ denote an outside option, interpreted as the use of any other program, and let s index wash occasions. Conditional on owning product j, household i's utility from program $r \in \mathcal{R}$ on occasion s is

$$v_{ijrs} = \delta_{c(r)} + \rho_i^T T_{jr} + \rho_i^d d_{jr} + \rho_w w_{jr} + \rho_L L_{jr} + \varepsilon_{ijrs}^v, \tag{5}$$

where T_{jr}, d_{jr}, w_{jr}, and L_{jr} denote temperature, duration, water use, and load size;[15] ε_{ijrs}^v is i.i.d. type-I extreme value and is redrawn across wash occasions; and the outside option is normalized to $v_{ij0s} = \varepsilon_{ij0s}^v$. Preferences for temperature and duration vary with the household's share of high-stain laundry according to $\rho_i^q = \rho_0^q + \rho_1^q \tau_i + \sigma_q \nu_i^q$, for $q \in \{T, d\}$ and $\nu_i^q \sim N(0,1)$.

I assume that households do not observe the product-specific energy cost of an individual wash cycle when choosing programs. Instead, they have persistent perceptions of the relative operating costs of different program categories. The program-category fixed effect $\delta_{c(r)}$ therefore captures both baseline preferences and washing habits associated with category $c(r)$, as well as perceived differences in relative operating costs across program categories.[16]

Let $\bar{v}_{ijr} \equiv v_{ijrs} - \varepsilon_{ijrs}^v$ denote the deterministic component of utility from program r. Under the type-I extreme value assumption, the conditional probability that household i chooses program $r \in \mathcal{R}$ on product j is

$$P_{ijr} = \frac{\exp\left(\bar{v}_{ijr}\right)}{1 + \sum_{r' \in \mathcal{R}} \exp\left(\bar{v}_{ijr'}\right)}. \tag{6}$$

[15] Households may not observe individual program inputs directly, but they perceive the resulting washing quality. I represent perceived quality as a function of the underlying program inputs without requiring households to evaluate each input separately. Technology may also affect washing quality, but I assume that it is common across programs within a product and incorporate its effect in the product-choice stage.

[16] Program categories are Cotton, Easy Care, and Eco. In particular, Cotton 20°C, Cotton 40°C, and Cotton 60°C share the same category fixed effect.

The expected utility from product j's program bundle is

$$I_{ij} \equiv \mathbb{E}_{\varepsilon^v}\left[\max_{r\in\{0\}\cup\mathcal{R}} v_{ijrs}\right] = \log\left(1+\sum_{r\in\mathcal{R}}\exp\left(\bar{v}_{ijr}\right)\right), \tag{7}$$

up to the Euler–Mascheroni constant, which is common across products and normalized away.

Product Choice In each market (m,t), household i chooses among the set of available washing machines $\mathcal{J}_{mt}$ and an outside option. For product $j \in \mathcal{J}_{mt}$, indirect utility is

$$u_{ijmt} = -\alpha_i p_{jmt} + \sum_{c\in\mathcal{C}} \beta_c \mathbf{1}\{\text{ec21}(j)=c\}\mathbf{1}_{\text{new},t} + \beta^I I_{ij} + x_j'\beta^x + \delta_{b(j)} + \delta_m + \delta_t + \xi_{jmt} + \epsilon_{ijmt}, \tag{8}$$

where ϵ_{ijmt} is i.i.d. type-I extreme value and the outside-option mean utility is normalized to zero. Price sensitivity varies with household income according to $\alpha_i = \exp(\bar{\alpha} + \alpha^y y_i)$.

The indicator $\mathbf{1}_{\text{new},t}$ equals one under the new A–G labeling regime, while $\mathbf{1}\{\text{ec21}(j) = c\}$ indicates whether product j belongs to energy class $c \in \mathcal{C} = \{A, B, C, D\}$, with class D normalized to zero. I assume that consumers do not know the regulatory formula underlying the energy class and do not translate the label into product-specific energy costs. Instead, the displayed class enters utility directly as a coarse signal of energy efficiency, consistent with evidence that consumers may respond to energy certifications without fully processing the underlying energy-cost information (Houde, 2018). The coefficients β_c therefore capture consumers' valuation of each displayed energy class under the new label. I omit the pre-reform energy-class effects because almost all products were rated A^{+++} before the reform.

The inclusive value I_{ij} links program use to product choice by summarizing household i's expected value from product j's program bundle. Although I do not explicitly model reputation formation, this provides a parsimonious way to capture a reputation-like channel: consumers may learn about downstream product performance through reviews and other pre-purchase information (Reimers and Waldfogel, 2021), giving firms an incentive to design programs that generate higher expected use value. The coefficient β^I determines how strongly this expected program-use value enters the product-purchase decision.

The vector x_j contains other exogenous, time-invariant product characteristics, including capacity, spin speed, and technology. The terms $\delta_{b(j)}$, δ_m, and δ_t denote brand, region, and quarter fixed effects, while ξ_{jmt} captures product-market-time attributes observed by households but not by the econometrician.

Let $\bar{u}_{ijmt} = u_{ijmt} - \epsilon_{ijmt}$ denote the deterministic component of product utility. Conditional on

household characteristics, the probability that household i purchases product j is

$$s_{ijmt} = \frac{\exp\left(\bar{u}_{ijmt}\right)}{1 + \sum_{k \in \mathcal{J}_{mt}} \exp\left(\bar{u}_{ikmt}\right)}. \tag{9}$$

The market share of product j integrates this choice probability over the distribution of household heterogeneity:

$$s_{jmt} = \int s_{ijmt}\, dF_{mt}(y_i, \tau_i, \nu_i^T, \nu_i^d). \tag{10}$$

5.2 Supply

Actions. Let $\mathcal{L}_f$ denote firm f's set of product lines, where a line ℓ is defined by brand $\times$ capacity $\times$ spin speed. On each line, the firm can introduce at most one new product in quarter t. The line-level action is $a_{f\ell t} \in \mathcal{A}_{f\ell t} \subseteq \{0, A, B, C, D\}$, where $a_{f\ell t} = 0$ denotes no entry and $a_{f\ell t} \in \{A, B, C, D\}$ denotes entry of a product with the corresponding new-label energy class, subject to the line's feasibility restrictions. The firm's portfolio action is $a_{ft} = (a_{f\ell t})_{\ell \in \mathcal{L}_f}$, and $a_t = (a_{ft}, a_{-f,t})$ denotes the joint entry action of all firms. Let $\mathcal{J}_t(a_t)$ denote the resulting product portfolio.

For each entering product j, let q_{fjrt} denote the vector of inputs chosen for wash program r, including temperature, duration, water use, and other design inputs. I collect these program-design choices in q_{ft} and denote the industry-wide profile by $q_t = (q_{ft}, q_{-f,t})$. Program inputs of incumbent products remain fixed. Given the resulting product portfolio and program designs, firms then compete in prices across markets.

Price Setting. Given the realized product portfolio $\mathcal{J}_t(a_t)$ and program designs q_t, firms compete in prices across markets. Firm f chooses $\{p_{jmt}\}_{j \in \mathcal{J}_{ft}(a_t)}$ to maximize

$$\pi_{ft}\big(\mathcal{J}_t(a_t), q_t\big) = \sum_m \sum_{j \in \mathcal{J}_{ft}(a_t)} \big(p_{jmt} - mc_{jt}\big) s_{jmt}\big(p; \mathcal{J}_t(a_t), q_t\big) M_{mt}, \tag{11}$$

where s_{jmt} is product j's market share and M_{mt} is market size. Prices are determined in Bertrand–Nash equilibrium. Marginal cost is

$$\log mc_{jt} = x_j^{mc\prime}\gamma + \gamma_{b(j)} + \gamma_t + \omega_{jt}, \tag{12}$$

where x_j^{mc} contains capacity, spin speed, and technology, $\gamma_{b(j)}$ and γ_t are brand and quarter fixed effects, and ω_{jt} is an idiosyncratic marginal-cost shock observed by firms. Program inputs other than technology are control-algorithm parameters and therefore do not directly enter marginal production cost.

Program Design. After entry outcomes are realized and observed, firms simultaneously choose the wash-program inputs of entering products. For product j entering on line ℓ, let $q^{\text{ref}}_{f\ell r}$ denote its predetermined reference design. Duration references remain anchored to the matched A^{+++} benchmark; technology and temperature use the latest strictly earlier first-class introduction on the line, with an A^{+++} fallback. A repeated introduction of the same class copies that class's first design rather than resetting the duration reference. Define $\Delta q_{fjrt} = q_{fjrt} - q^{\text{ref}}_{f\ell r}$. The design cost of product j is

$$DC_{fjt} = \sum_{r \in \mathcal{R}} \left[\kappa_r' \Delta q_{fjrt} + \Delta q_{fjrt}' C_r \Delta q_{fjrt} \right], \tag{13}$$

where κ_r and $C_r \succeq 0$ govern the linear and quadratic costs of departing from the reference design.

Conditional on the realized portfolio a_t, firm f solves

$$\begin{aligned} W_{ft}(a_t) = \max_{q_{ft}} \mathbb{E}_{\xi,\omega} \left[\pi_{ft}\big(\mathcal{J}_t(a_t), q_t; \xi, \omega\big) \right] - \sum_{j \in \mathcal{J}^{\text{new}}_{ft}(a_{ft})} DC_{fjt} \\ \text{s.t.} \quad \text{EEI}_j(q_{j,\text{Eco}}) \in \mathcal{E}(e_j), \\ d_{jo}(q_{j,\text{Eco}}) \le \overline{d}_o(k_j), \qquad o \in \{F, H, Q\}. \end{aligned} \tag{14}$$

Here k_j denotes product j's rated capacity, $\mathcal{E}(e_j)$ is the EEI range associated with the energy class chosen at entry, and $\overline{d}_o(\cdot)$ denotes the applicable load-specific duration limits. The exact regulatory formulas are provided in the Appendix. Non-Eco programs are unconstrained by these requirements but affect program use, product demand, and operating profits.

Product Entry. Firms simultaneously choose their product introductions, taking the distribution of rivals' entry decisions, $\mathcal{P}_{-f,t}$, as given. Each firm receives private portfolio-specific payoff shocks, which capture unobserved factors affecting its introduction decisions. Firm f solves

$$V_{ft} = \max_{a_{ft} \in \mathcal{F}_f(J_{f,t-1})} \left\{ \mathbb{E}_{\mathcal{P}_{-f,t}} \left[W_{ft}(a_{ft}, a_{-f,t}) \right] - C^{\text{entry}}_{ft}(a_{ft}) + \sigma \eta_{ft}(a_{ft}) \right\}, \tag{15}$$

where $\eta_{ft}(a_{ft})$ follows a type-I extreme-value distribution and σ determines the scale of payoff shocks.

I specify entry costs to capture both the fixed cost of introducing a product and the additional cost of introducing a different energy class:

$$\begin{aligned} C^{\text{entry}}_{ft}(a_{ft}) = \sum_{\ell : a_{f\ell t} \neq 0} \big[\kappa_L + \gamma_{q(t)} + \beta_N \log(1 + N_{f\ell,t-1}) + \beta_G \log(1 + G_{f\ell t}) \\ + \kappa^{h(t)}_{a_{f\ell t}} \mathbf{1}\{a_{f\ell t} \notin \mathcal{C}^{\text{last}}_{f\ell,t}\} \big]. \end{aligned} \tag{16}$$

The common launch cost κ_L captures fixed expenses associated with introducing a new product, while $\gamma_{q(t)}$ allows these costs to vary seasonally, with $\gamma_1 = 0$. I allow launch costs to depend on the number of existing products on a line, $N_{f\ell,t-1}$, to capture differences in the cost of expanding an established product line, and on the number of quarters since its previous introduction, $G_{f\ell t}$, to account for variation in product replacement cycles. The switching premium $\kappa^{h(t)}_{a_{f\ell t}}$ captures additional costs of introducing an energy class not offered in the line's most recent introduction quarter, such as class-specific development and preparation costs beyond the continuous design adjustments. These premiums differ between the transition and post-policy periods, with classes C and D sharing a coefficient. The specification allows firms to incur switching costs when returning to an older class.

5.3 Discussion

The model makes two main simplifying assumptions. First, I treat consumer purchase and firm entry decisions as static. I assume that consumers replace washing machines only upon failure, rather than optimally timing replacement as in Goettler and Gordon (2011). Unlike microprocessors, where consumers may delay upgrades in anticipation of technological improvements and price declines, washing-machine replacement is more likely driven by necessity. Firms likewise maximize current-period profits, even though products remain on the market for approximately 2.5 years on average. Following Wollmann (2018), static entry decisions can approximate dynamic investment decisions when current profitability predicts future returns, with estimated entry costs reflecting an implicit hurdle rate. Although these assumptions abstract from dynamic replacement and entry incentives, such considerations are less central to the paper's main mechanism of regulation-induced changes in program design and household substitution across wash programs.

Second, I simplify how firms adjust program designs across product introductions. The data show minimal changes in program inputs across cohorts within the same product line and energy class. I therefore allow firms to redesign all programs when introducing a new energy class on a line, while subsequent introductions of the same class inherit the existing design. The model includes a fixed cost of product entry but abstracts from program-level fixed costs of modifying existing designs. In practice, such costs may discourage firms from adjusting unregulated programs and lead them to concentrate redesign efforts on Eco. This simplification captures the observed stability of designs within line-class combinations but does not explicitly model firms' decisions about which programs to redesign.

6 Estimation

I estimate program preferences, product demand, and the engineering relationship first; recover marginal costs from pricing conditions; then estimate design and entry costs. Two neural-network approximations make repeated portfolio and input evaluations feasible. The first predicts operating profits and own-firm input derivatives. The second predicts optimized inputs and operating and net profits after solving the design game. Appendix F.1 details their training. This use of profit approximation is related to Bodéré (2026); the static entry estimator below differs from his dynamic model and fixed-rival-beliefs estimation.

6.1 Program Use, Demand, and Marginal Cost

For survey household i, let n_{ir} count reported uses of program r. Simulated maximum likelihood integrates household-specific tastes jointly across its reported washes:

$$\max_{\rho} \sum_{i} \log \left[\frac{1}{R} \sum_{b=1}^{R} \prod_{r} P_{ijr}(\nu_i^b; \rho)^{n_{ir}} \right], \qquad R = 16{,}000. \tag{17}$$

The count multinomial constant can be omitted. Averaging inside the household likelihood preserves persistent preference heterogeneity. I then compute household-specific inclusive values for product demand.

Product demand is estimated by aggregate-share GMM using the inversion of Berry et al. (1995) and the implementation of Conlon and Gortmaker (2020). With demand residual $\xi(\theta)$ and instruments Z, the criterion is

$$g(\theta) = \frac{1}{n} Z'\xi(\theta), \qquad \widehat{\theta} = \arg\min_{\theta} g(\theta)' W g(\theta). \tag{18}$$

The current specification uses a one-step weighting matrix and aggregate moments, without additional micro moments. Excluded instruments include same-firm and rival characteristic sums, their interactions with market-quarter mean income, and mean program inclusive value. Characteristic sums use capacity, spin speed, and displayed electricity consumption. Technology is an included demand characteristic, not an additional BLP characteristic-sum instrument. Income, stain intensity, and temperature and duration tastes are integrated with 625 weighted agents per market. The validity of the instruments and the exogeneity of included technology are maintained identifying assumptions.

Given demand estimates and product ownership, the multiproduct Bertrand first-order conditions recover markups and marginal costs. I regress log recovered marginal cost on the variables in equation (12). Its residual and the demand residual determine the joint shock distribution used to

integrate design-stage profits. All profit labels and entry costs are converted to euros, consistently with the price normalization in demand.

Instruments. The instruments follow the timing of the model. Product characteristics and program designs are chosen before the product-market demand shock ξ_{jmt} and marginal-cost shock ω_{jt} are realized, whereas prices are set after both shocks are observed. Price is therefore endogenous, while predetermined product and program characteristics are orthogonal to the contemporaneous demand shock.

I instrument for price using standard BLP instruments constructed from the capacity and spin speed of products offered by the same firm and by rival firms. Because price sensitivity varies with household income, I additionally interact these instruments with mean household income in market (m, t). These interactions provide excluded variation for the price–income interaction and therefore identify α^y separately from average price sensitivity.

The coefficient β^I is also nonlinear because the inclusive value I_{ij} varies across households. I use the market-average inclusive value

$$Z^I_{jmt} = \int I_{ij}\, dF_{mt}(i) \tag{19}$$

as an additional instrument. Under the model timing, this object is determined by program designs chosen before ξ_{jmt} is realized and therefore provides exogenous variation in expected program-use value. This use of excluded variation to identify nonlinear preference heterogeneity follows the general identification logic of Berry and Haile (2014).

Collecting the excluded instruments and exogenous product characteristics in Z_{jmt}, the aggregate moments satisfy

$$\mathbb{E}\left[Z_{jmt}\xi_{jmt}(\theta)\right] = 0. \tag{20}$$

6.2 Supply Estimation

I estimate design and entry costs conditional on the demand, engineering, and marginal-cost estimates. Design costs determine how firms adjust program characteristics, while entry costs govern their decisions to introduce new products. Both estimations use approximations to equilibrium profits to reduce computational complexity. Appendix F provides details on the estimation procedures and inference.

Design costs. I estimate ten design-cost parameters: a linear coefficient and a positive curvature for each of five input groups—Cotton 40°C duration, Cotton 60°C duration, Eco duration, Eco

temperature, and technology. Following equation (13), each input has a linear-quadratic adjustment cost relative to a reference design. Full- and half-load Eco inputs share cost coefficients within each group.

I estimate these parameters by minimum distance, choosing the costs that best reproduce observed program designs. Let $g_N(\theta)$ denote the differences between observed and model-predicted moments. The estimator solves

$$\widehat{\theta} = \arg \min_{\theta \in \mathbb{R}^5 \times \mathbb{R}^5_{++}} g_N(\theta)' W g_N(\theta). \tag{21}$$

The moments include eight restrictions on baseline A^{+++} designs, 20 class-specific mean-input moments, and 416 changes in program inputs associated with product introductions. Each input group receives equal total weight, with the remaining weights distributed across baseline restrictions, class means, and individual changes as described in Appendix F. Energy consumption is not directly targeted and provides an additional measure of model fit.

Identification comes from how firms adjust program characteristics in response to changing regulatory incentives. Baseline A^{+++} designs discipline the linear cost coefficients by linking firms' initial choices to their marginal operating profits. Changes in program designs following the reform then identify the curvatures, which determine how costly it is to move away from these initial designs. For example, a large adjustment in Eco temperature in response to a relatively small change in profitability implies a low adjustment cost. Matching both average characteristics across energy classes and changes within product lines helps distinguish systematic design differences from the costs of modifying existing products.

For each candidate set of parameters, I solve firms' optimal program inputs conditional on observed product portfolios and reference designs using sequential least squares programming (SLSQP) (Kraft, 1988). Duration adjustments are measured relative to the matched A^{+++} design, while temperature and technology adjustments use the most recent preceding first introduction of an energy class, with A^{+++} as a fallback. Repeated introductions of the same class retain its initial design. I verify the resulting designs using local best-response checks and select the cost parameters that minimize the distance between observed and predicted inputs. Appendix F.3 provides details on the numerical procedure and convergence limitations.

Entry costs. I estimate a common launch-cost intercept, two state coefficients, a common payoff-shock scale, and six switching premiums. Launch costs depend on the existing number of products on a line and the time since its previous introduction. Switching premiums capture the additional cost of introducing an energy class that was absent from the line's most recent introduction quarter. I allow these premiums to differ between the transition and post-policy periods, estimating separate

costs for classes A and B and a common cost for classes C and D in each period.

The estimation uses 469 eligible product-line–quarters across seven firms. Eligibility requires positive sales in the preceding quarter and an available design reference, regardless of whether a new product is introduced. I exclude line–quarters with multiple introductions that cannot be represented by the model's choice set, while retaining their products when calculating other firms' profits. Expected profits already account for design costs, so only the additional discrete entry costs are subtracted.

Entry costs are identified by comparing the expected profitability of alternative product introductions with firms' observed decisions. Variation in profitability across energy classes, product lines, and quarters helps distinguish the costs of introducing new products from those of switching energy classes. Given the type-I extreme-value payoff shocks, the probability that firm f chooses portfolio a_{ft} is

$$\mathcal{P}_{ft}(a_{ft};\theta,\mathcal{P}_{-f,t}) = \frac{\exp\left\{\overline{W}_{ft}(a_{ft};\mathcal{P}_{-f,t})/\sigma\right\}}{\sum_{b_{ft}\in\mathcal{F}_f(J_{f,t-1})}\exp\left\{\overline{W}_{ft}(b_{ft};\mathcal{P}_{-f,t})/\sigma\right\}}, \tag{22}$$

where $\overline{W}_{ft}(a_{ft};\mathcal{P}_{-f,t}) = \mathbb{E}_{\mathcal{P}_{-f,t}}[W_{ft}(a_{ft},a_{-f,t})] - C^{\text{entry}}_{ft}(a_{ft})$ is expected profit net of entry costs, θ collects the entry-cost parameters and σ, and $\mathcal{P}_{-f,t}$ denotes the distribution of rival portfolios.

Direct estimation faces two computational challenges. First, each product line has up to five possible actions, so the denominator of equation (22) can involve millions of portfolio combinations. Second, evaluating the expected profit of each portfolio requires solving the constrained program-design game, which becomes computationally prohibitive when integrating over rival portfolios.

To address the large action space, I exploit the portfolio logit structure and condition on the firm's decisions on its other product lines. By Bayes rule, the resulting conditional choice probability is

$$\mathcal{P}_{f\ell t}(a \mid a_{f,-\ell,t};\theta,\mathcal{P}_{-f,t}) = \frac{\exp\left\{\overline{W}_{ft}((a,a_{f,-\ell,t});\mathcal{P}_{-f,t})/\sigma\right\}}{\sum_{b\in\mathcal{A}_{f\ell t}(a_{f,-\ell,t})}\exp\left\{\overline{W}_{ft}((b,a_{f,-\ell,t});\mathcal{P}_{-f,t})/\sigma\right\}}. \tag{23}$$

Here $a_{f,-\ell,t}$ denotes the firm's actions on its other lines. Unlike the full portfolio probability, the conditional probability requires evaluating at most five actions in the denominator. I estimate the entry-cost parameters by maximizing the conditional composite log likelihood (Varin et al., 2011),

$$\ell(\theta;\mathcal{P}) = \sum_{(f,\ell,t)\in\mathcal{S}^{\text{entry}}} \log \mathcal{P}_{f\ell t}(a^{\text{obs}}_{f\ell t} \mid a^{\text{obs}}_{f,-\ell,t};\theta,\mathcal{P}_{-f,t}). \tag{24}$$

Conditioning reduces the computational burden without assuming independent decisions across a firm's product lines.

The conditional probabilities still depend on the distribution of rival portfolios, $\mathcal{P}_{-f,t}$, which must be determined in equilibrium. I approximate these distributions using Gibbs sampling, sequentially updating product-line actions according to their conditional probabilities to generate draws of complete firm portfolios (Casella and George, 1992). Following the nested pseudo-likelihood approach of Aguirregabiria and Mira (2007), I alternate between estimating entry-cost parameters given rival beliefs and updating those beliefs using simulated portfolios. I repeat these steps until the parameters and conditional choice probabilities stabilize.

To avoid repeatedly solving the program-design game for each simulated portfolio, I train a neural-network surrogate on solutions of the design problem to approximate optimized profits across portfolio configurations. This allows me to evaluate expected payoffs during entry estimation without repeatedly solving for optimal program inputs. Appendix F describes the surrogate and simulation procedures. I report conditional heteroskedasticity-robust standard errors, with firm-clustered sensitivity estimates and further inference details in Appendix F.5.

7 Estimation Results

7.1 Engineering, Program Use, and Product Demand

Table A1 reports the estimated relationship between electricity consumption and program inputs. In the pooled engineering specification, temperature has the largest elasticity (1.599), followed by duration (0.360), water (0.333), and load (0.168). Holding technology and other inputs fixed, a 1 percent increase in duration raises electricity use by 0.36 percent. Longer cycles therefore save energy only when accompanied by changes in temperature, water, or technology.[17]

Table 3 shows substantial heterogeneity in preferences for program duration. For a household with no heavy-stain laundry, extending a cycle by ten minutes lowers mean utility by 0.097, holding other attributes fixed. The duration penalty is smaller for households with more heavy-stain laundry, who also place less value on temperature. The estimated standard deviation of duration preferences is 2.248 per hour. Thus, while a duration cap may benefit households that prefer shorter cycles, it can restrict a feature valued by others.

The expected value of the wash-program menu also affects appliance purchases. Its coefficient in product demand is 0.147 (Table 4). Relative to class D, the utility coefficients are 0.956 for class A, 0.186 for B, and 0.172 for C. Holding other characteristics fixed, the class-A coefficient raises purchase odds relative to the outside option by a factor of $\exp(0.956) = 2.60$. This is a conditional valuation of the label, not the equilibrium sales response to upgrading a machine,

[17]The supply and counterfactual exercises use earlier engineering estimates and intermediate profit checkpoints rather than the latest pooled regression. Appendix F.1 documents these differences.

which also changes its design and price. Because price sensitivity falls with income, a given utility improvement translates into a higher monetary valuation for richer households.

More efficient technology improves product appeal but increases production costs. The coefficient on the log energy-requirement index is -1.147 in demand and -0.116 in log marginal cost (Table 5). Reducing energy requirements by 10 percent at fixed program inputs increases utility by approximately 0.121 and marginal production cost by 1.23 percent, holding other characteristics fixed. Firms therefore balance the benefits of efficiency against higher production and design costs.

7.2 Design Costs and Model Fit

Table 6 reports the estimated costs of departing from a reference design. For input k, a deviation of h costs $(\kappa_r)_k h + (C_r)_{kk} h^2$ euros per introduction, with curvature reported as $2(C_r)_{kk}$. A ten-minute change costs € 2,045 for Cotton 40°C duration and € 4,057 for Cotton 60°C duration, compared with € 50 for one Eco-duration coordinate. Extending both full- and half-load Eco cycles by ten minutes costs € 100. These amounts are costs per introduction, not per machine sold, and exclude discrete entry costs and changes in operating profits. They measure deviations from the reference design rather than incremental costs from an already modified design.

Eco duration is thus a relatively inexpensive margin of adjustment. Although the quadratic cost rises with the size of the change, firms can lengthen Eco cycles at much lower design cost than Cotton cycles. This difference helps explain why removing the duration cap produces a large Eco response. Equilibrium responses also depend on regulatory constraints, household preferences, and other design choices.

Technological improvements are considerably more expensive. A change in the log energy-requirement index of $\log(0.9)$, equivalent to a 10 percent reduction in energy requirements at fixed inputs, costs approximately € 3.05 million per introduction. This calculation uses the index in its original units rather than the table's 10^{-3} display scale. It holds other design coordinates fixed and excludes entry costs and changes in marginal production cost. The magnitude helps explain why firms may prefer to adjust program inputs instead of making large technological improvements.

The model fits Eco duration closely: observed and predicted levels are 190.1 and 190.0 minutes, and the corresponding reductions from the reference are 41.4 and 41.6 minutes. It is less successful at explaining increases in Cotton duration and improvements in technology. For the selected technology-improvement sample, the predicted change in the energy-requirement index is approximately zero, compared with -0.078 in the data (Tables A3 and A4; Figure F2).

7.3 Entry Costs and Model Fit

Table 7 reports a common launch-cost intercept of € 2.12 million, before adjustments for season, product stock, time since the previous introduction, and energy-class switching. The estimates imply lower launch costs for established lines and higher costs following longer gaps between introductions. Increasing the preceding product count from one to three reduces launch costs by € 0.53 million, while increasing the gap from one to three quarters raises them by € 0.93 million, holding other characteristics fixed. These patterns are consistent with economies from maintaining an active product line and costs of restarting introductions after a period of inactivity.

Switching premiums apply when a firm introduces a class absent from the line's most recent introduction quarter. During the transition, the estimated premiums are € 2.58 million for class A, € 3.43 million for B, and € 1.61 million for C/D. After the transition, they are € 1.89 million, € 3.49 million, and € 4.73 million, respectively. These premiums favor persistence in recently introduced classes, particularly C/D after the transition. They are additional to the launch cost and do not apply when the chosen class is already in the reference set.

The size of these estimates reflects the definition of an introduction and the static nature of the model. A modeled product can cover several commercial variants across markets, so its launch may entail tooling, certification, marketing, and distribution expenses beyond program design. The data do not identify these expenditures separately. Moreover, the estimated costs may absorb omitted continuation values, the option to delay introduction, and errors in predicted portfolio profits. They should therefore be interpreted as effective costs that rationalize observed entry decisions, not direct measures of engineering expenditures.

The estimated payoff-shock scale is € 1.62 million. The model predicts introductions in 33.8 percent of eligible line–quarters, compared with 36.0 percent in the data. It underpredicts class-A entry, slightly overpredicts B and C, and closely matches the class-D share (Figure F3; Table A5). These fit measures condition on the observed decisions on a firm's other lines; the counterfactuals instead allow its portfolio history to evolve.

8 Counterfactuals

I examine three ways of changing regulatory incentives: removing existing ecodesign or labeling restrictions, extending the efficiency index to Cotton programs, and imposing an energy-based producer tax. These policies operate through different mechanisms. The duration cap limits Eco design, the energy label rewards measured efficiency, and the tax raises the marginal cost of energy-intensive products. I compare their implications for product entry, program design, electricity use, and consumer welfare.

8.1 Common Counterfactual Approach

I simulate 20 market paths for each policy over 2020Q1–2023Q4. The removal experiments share the current-policy baseline. The weighted-EEI and tax experiments have separate baselines—an uncapped Eco-only index and zero tax, respectively—and each evaluates 50 policy specifications, including its baseline. The tax experiments use the carbon-model approximation and its corresponding Cotton engineering relationship. All exercises retain the separate Eco-duration cost coefficient. Appendices F.1 and G document the parameter snapshots and computational checks.

In each quarter, I update the incumbent portfolio, determine eligible introductions from the previous state, simulate firms' entry choices, solve entrants' program designs, and determine prices and household choices. Under labeling, entrants choose among classes A–D; without labels, they use a common physical template with no class premium. Duration costs remain anchored to the matched A^{+++} design, and subsequent introductions of a class on the same line retain its first design. Initial products follow their observed availability, while simulated entrants have no estimated exit hazard, subject to compatibility restrictions. The frozen joint-removal benchmark differs: it solves the initial market once and repeats that outcome rather than allowing the market to evolve.

Electricity and welfare accounting. Let s_{ijt} be household i's purchase probability for product j, and let E_{ijt} be its quarterly electricity use, averaging over program choices; market indices are suppressed. For the removal experiments, I represent the purchase outside option by an equally weighted basket of observed 2020Q1 A^{+++} machines, evaluated at each household's preferences. Let E_{i0t} be its reference electricity use and W_{it} the population and integration weight. I define

$$G_t = \sum_i W_{it} \sum_j s_{ijt} E_{ijt}, \qquad D_t = \sum_i W_{it} \Big(\sum_j s_{ijt} \Big) E_{i0t}, \tag{25}$$

$$R_t = G_t - D_t, \qquad T_t = R_t + B_t, \qquad B_t = \sum_i W_{it} E_{i0t}. \tag{26}$$

Here G_t measures gross electricity use by purchased machines, D_t is the use displaced from the outside reference, and R_t is the net change relative to that reference. Adding reference use for all households, B_t, gives total electricity use T_t. Because the population and outside reference are held fixed across the evolving removal experiments, $\Delta T_t = \Delta R_t$. The outside alternative in wash-program choice is distinct and is assigned zero electricity use.

Following Train (2015), I compute consumer surplus from the purchase logsum, remove utility attributable solely to energy labels, and subtract discounted lifetime electricity costs:

$$CS_{it} = \frac{100}{\alpha_i} \left[\log \left(1 + \sum_j e^{\bar{u}_{ijt}} \right) - \sum_j s_{ijt} \ell_{ijt} \right] - 0.30 A \sum_j s_{ijt} (E_{ijt} - E_{i0t}), \tag{27}$$

Here $\bar{u}_{ijt}$ is systematic purchase utility and ℓ_{ijt} its label component. The lifetime electricity factor is $A = \sum_{k=0}^{47}(1.05)^{-k/4} = 36.554$, based on a 12-year product life and a 5% annual discount rate. Electricity costs EUR 0.30/kWh, and prices enter utility in hundreds of euros. The purchase logsum already includes the expected value of program choice.

The weighted-EEI and tax grids use a different electricity and welfare accounting convention. They report gross electricity G and surplus CS^G, which deducts gross rather than reference-adjusted electricity expenditure. In aggregate, $CS^G = CS - 0.30AD$; consumer welfare including tax rebates is $CW^G = CS^G +$ rebates. I therefore compare each grid only with its own baseline. Lower gross electricity does not necessarily imply lower reference-inclusive electricity, and the welfare changes are not directly comparable with those from the removal experiments. Harmonizing the measures would require re-evaluating household-level choices, not merely subtracting an aggregate constant.

The reported totals cover 16 quarterly purchase cohorts. Electricity measures the quarterly use associated with each cohort, rather than the installed stock's total use during 2020–2023; welfare includes discounted lifetime electricity expenditure. Policy differences use matched simulation paths. The completed runs retain 16 Gibbs draws after 20 burn-in sweeps and have not been subjected to the later independent-seed check. Consumer welfare excludes producer surplus and monetized environmental damages.

8.2 Counterfactual Policies and Product Design

Removing the duration cap or labeling requirements. Removing the duration cap gives firms more flexibility in Eco design while preserving incentives from the efficiency index. By 2023Q4, full-load Eco duration rises from 211.7 to 381.2 minutes, and half-load duration rises from 162.5 to 346.1 minutes; full-load temperature falls from 31.4 to 29.7°C (Table 9). Cotton 40°C and Cotton 60°C durations decline by 12.0 and 5.4 minutes, respectively. Total introductions change little, from 89.5 to 87.1, while the class-A share falls from 45.7% to 43.7%. The main response is therefore in program design rather than product variety or energy-class composition. Longer cycles do not themselves reduce electricity use in the engineering equation; the savings arise from accompanying changes in other inputs.

Removing the efficiency-index restrictions and label utility, while retaining the duration cap, has a different effect. Introductions fall to 66.1, a 26.1% reduction. Eco duration remains close to its capped level, but full-load temperature increases to 37.3°C. Cotton 40°C and Cotton 60°C durations fall to 174.6 and 170.4 minutes. These averages reflect both changes in entrant design and changes in portfolio composition. Because joint removal is evaluated using a frozen market, that experiment cannot be treated as the fourth cell of a common factorial comparison.

Broadening the EEI test. The weighted efficiency index adds Cotton programs to the Eco-based measure. Specifically, I replace the Eco-only numerator with $w_E e_{jE} + w_{40} e_{j40} + w_{60} e_{j60}$, retaining the capacity adjustment and existing class thresholds. The tested specifications place 90–100% of the weight on Eco and at most 10% on Cotton 40°C and Cotton 60°C together. All remove the duration cap. These weights measure regulatory performance, not household program-use shares, and the thresholds are not recalibrated to hold regulatory stringency constant. Including Cotton in the index makes energy use outside Eco relevant to the rating.

Putting 10% of the weight on Cotton 60°C, $(w_E, w_{40}, w_{60}) = (0.90, 0, 0.10)$, reduces introductions from 87.1 to 80.8 relative to the uncapped Eco-only index. Full-load Eco and Cotton 60°C durations fall by 24.1 and 6.6 minutes, respectively, while Cotton 40°C duration rises by 3.0 minutes (Table 10). Eco temperature also declines. The policy changes the joint design of several programs, rather than improving Cotton performance while holding Eco fixed. With equal Cotton weights, $(0.90, 0.05, 0.05)$, introductions fall to 84.9 and both Cotton durations decline.

An energy-based producer tax. The producer-tax experiment removes the efficiency index, label utility, and duration cap and adds the following charge to each machine's marginal cost:

$$\mathcal{T}_j = \frac{\tau_E e_{jE} + \tau_C (e_{j40} + e_{j60})}{3} \tag{28}$$

I evaluate 50 pairs of coefficients in $[0, 100]^2$, including zero and taxes on either program group alone. The coefficients translate assessed electricity use per cycle into euros per machine; they are not carbon prices per tonne. Household electricity prices remain unchanged. Firms adjust introductions and designs, and households adjust purchases and program use. Tax revenue is rebated equally to households within each market, including nonbuyers.

With $(\tau_E, \tau_C) = (100, 100)$, introductions decline from 62.6 to 59.4 relative to the zero-tax benchmark. Average full-load Eco duration changes little, from 345.1 to 344.0 minutes, while half-load duration falls from 278.3 to 272.7 minutes. Cotton durations also change little. Thus, the substantial decline in gross electricity discussed below is not driven by equally large changes in average cycle length. Product-level means do not distinguish improvements in energy efficiency from changes in program use or purchase composition.

8.3 Electricity Consumption and Welfare

Removing existing restrictions. Relative to current policy, removing the duration cap raises consumer surplus by € 603.2 million and reduces reference-relative electricity use by 47.0 GWh (Table 11). With the outside reference held fixed, total electricity use also falls by 47.0 GWh, from 459.7 to 412.7 GWh, a 10.2% reduction. Equivalently, the cap increases total electricity use by

11.4% relative to a policy with labels but no cap. Removing efficiency-index and label incentives while keeping the cap produces smaller gains: € 100.9 million in consumer surplus and 11.6 GWh in electricity savings. Figure 9 traces these differences across purchase cohorts.

Accounting for displaced electricity is important to this comparison. Removing efficiency-index and label incentives lowers gross purchase electricity by 57.5 GWh—more than removing the cap—but also reduces displaced reference use by 45.9 GWh. The resulting reduction in reference-relative electricity is only 11.6 GWh (Figure 10). Changes in purchasing therefore reverse the ranking based on gross electricity alone. For the same reason, gross-use results from the policy grids cannot be interpreted as changes in reference-inclusive electricity.

The welfare gains are concentrated among households that prefer longer cycles. Although these households account for 23.3% of weighted decision observations, they receive 98.9% of the surplus gain from removing the cap and 94.3% of the gain from removing labeling incentives (Table 12). Their gains average € 264.65 and € 42.21 per observation, respectively, compared with € 0.90 and € 0.77 for other households. For cap removal, lower reference-relative electricity costs account for € 515.7 million of the gain, and the corrected purchase logsum for € 87.5 million. The corresponding components for removing labeling incentives are € 127.2 million and a € 26.3 million decline (Appendix G).

The frozen joint-removal benchmark increases consumer surplus by € 1,192.5 million and reduces reference-relative electricity by 15.6 GWh. However, it also changes the reference population and aggregate reference use: total electricity falls by 19.3 GWh, and the horizon contains approximately 80,646 fewer weighted decision observations. These differences preclude interpreting its larger surplus gain as evidence that joint removal dominates the evolving-market alternatives.

Reweighting EEI. Among the tested efficiency-index specifications, weights $(0.90, 0, 0.10)$ produce the largest consumer-welfare gain and the greatest reduction in gross purchase electricity. Relative to the uncapped Eco-only index, gross electricity falls by 10.00 GWh (6.0%) and consumer welfare rises by € 91.6 million (Table 13). Giving equal weight to both Cotton programs, $(0.90, 0.05, 0.05)$, reduces electricity by 6.63 GWh and raises welfare by € 64.2 million; putting all the Cotton weight on Cotton 40°C reduces electricity by only 1.48 GWh and raises welfare by € 21.2 million. Within the tested specifications, assigning weight to Cotton 60°C is more effective.

The preferred specification places the maximum tested weight on Cotton and is the only tested point that is not dominated in both mean welfare and gross electricity (Figure 11). This boundary result does not establish a global optimum. Both duration-preference groups benefit: gains average € 17.23 per observation for households that prefer longer cycles and € 7.12 for others. Lower electricity expenditure contributes € 109.7 million, partly offset by an € 18.1 million de-

cline in the corrected purchase-logsum component. This breakdown is an accounting identity, not a decomposition of the underlying design mechanisms.

Producer taxation. The largest tested tax, $(\tau_E, \tau_C) = (100, 100)$, produces the highest consumer welfare and lowest gross electricity within the tax grid. Relative to zero tax, gross electricity falls by 37.34 GWh (29.0%), while consumer surplus before rebates rises by € 166.2 million. Returning € 173.3 million in tax revenue raises the consumer-welfare gain to € 339.5 million. The pre-rebate gain shows that this effect is not entirely a transfer. Lower electricity bills contribute € 409.4 million, while the corrected purchase logsum declines by € 243.3 million, reflecting changes in prices, product availability, and characteristics.

A tax on Cotton alone, $(0, 100)$, reduces gross electricity by 29.02 GWh and raises rebate-inclusive consumer welfare by € 274.2 million. Taxing Eco alone, $(100, 0)$, yields corresponding changes of 8.97 GWh and € 87.2 million. This difference underscores the importance of energy use outside the standardized Eco program. Under $(100, 100)$, gains average € 56.68 per observation among households that prefer longer cycles and € 28.59 for other households (Table 14). The best outcome occurs at the highest tax rates tested, not at an identified optimal tax. The gross-electricity and transfer-inclusive welfare measures also do not support a social-welfare ranking across policy families.

9 Conclusion

This paper studies how energy labels and ecodesign constraints affect product design when regulatory tests capture only one of several operating modes. I use the 2021 EU washing-machine reform in Germany, which rescaled energy labels and imposed an Eco-duration cap. The evidence shows that energy consumption declined in the regulated Eco program but rose in hotter Cotton programs. Because Eco accounts for only a small share of household wash cycles, improvements in the tested mode need not translate into comparable savings in actual use.

The estimated model links these responses to firms' design incentives and households' program choices. Counterfactuals show that relaxing the Eco-duration cap can lower electricity use and improve consumer welfare while retaining labeling incentives. Including Cotton energy use in the efficiency index also improves outcomes under the tested specifications, as does an energy-based producer tax. Taken together, the results suggest that effective regulation should account for both firms' flexibility in meeting standards and consumers' use of programs outside the test environment.

References

Aguirregabiria, V. and Mira, P. (2007). Sequential estimation of dynamic discrete games. *Econometrica*, 75(1):1–53.

Alborzi, F., Boyano, A., Cordella, M., Espinosa, N., Graulich, K., Hook, I., Rüdenauer, I., Stamminger, R., and Villanueva, A. (2017). Ecodesign and energy label for household washing machines and washer dryers: Preparatory study – final report. Technical Report EUR 28809 EN, Publications Office of the European Union, Luxembourg. JRC108604.

Baker, G. P. (1992). Incentive contracts and performance measurement. *Journal of political Economy*, 100(3):598–614.

Barahona, N., Otero, C., and Otero, S. (2023). Equilibrium effects of food labeling policies. *Econometrica*, 91(3):839–868.

Barrett, S. R. H., Speth, R. L., Eastham, S. D., Dedoussi, I. C., Ashok, A., Malina, R., and Keith, D. W. (2015). Impact of the volkswagen emissions control defeat device on us public health. *Environmental Research Letters*, 10(11):114005.

Barwick, P. J., Kwon, H.-S., and Li, S. (2024). Attribute-based subsidies and market power: an application to electric vehicles. Technical report, National Bureau of Economic Research.

Berry, S., Levinsohn, J., and Pakes, A. (1995). Automobile prices in market equilibrium. *Econometrica*, 63(4):841–890.

Berry, S. T. and Haile, P. A. (2014). Identification in differentiated products markets using market level data. *Econometrica*, 82(5):1749–1797.

Bodéré, P. (2026). Dynamic spatial competition in early education: An equilibrium analysis of the preschool market in pennsylvania. Working paper, July 22.

Casella, G. and George, E. I. (1992). Explaining the gibbs sampler. *The American Statistician*, 46(3):167–174.

Conlon, C. and Gortmaker, J. (2020). Best practices for differentiated products demand estimation with pyblp. *The RAND Journal of Economics*, 51(4):1108–1161.

Czyzewski, G. (2022). Energy savings 2021: New european energy label and ecodesign requirements for washing machines. In Bertoldi, P., editor, *Energy Efficiency in Domestic Appliances and Lighting: Proceedings of the 10th International Conference (EEDAL'19)*, pages 1–11. Springer, Cham.

Davis, L. W. and Metcalf, G. E. (2016). Does better information lead to better choices? evidence from energy-efficiency labels. *Journal of the Association of Environmental and Resource Economists*, 3(3):589–625.

Dubin, J. A. and McFadden, D. L. (1984). An econometric analysis of residential electric appliance holdings and consumption. *Econometrica*, 52(2):345–362.

Eichler, H.-G., Abadie, E., Breckenridge, A., Flamion, B., Gustafsson, L. L., Leufkens, H., Rowland, M., Schneider, C. K., and Bloechl-Daum, B. (2011). Bridging the efficacy–effectiveness gap: a regulator's perspective on addressing variability of drug response. *Nature Reviews Drug Discovery*, 10(7):495–506.

Eizenberg, A. (2014). Upstream innovation and product variety in the U.S. home PC market. *The Review of Economic Studies*, 81(3):1003–1045.

European Commission (2019a). Commission regulation (eu) 2019/2023 of 1 october 2019 laying down ecodesign requirements for household washing machines and household washer-dryers pursuant to directive 2009/125/ec of the european parliament and of the council, amending commission regulation (ec) no 1275/2008 and repealing commission regulation (eu) no 1015/2010 (text with eea relevance). Official Journal of the European Union, OJ L 315, 5.12.2019, p. 285. EUR-Lex.

European Commission (2019b). Impact assessment accompanying the ecodesign and energy labelling regulations for household washing machines and household washer-dryers. Commission Staff Working Document SWD(2019) 349 final, European Commission.

European Commission (2021). European energy labels: rescaling and transition periods. https://commission.europa.eu/system/files/2021-04/rescaled_eu_energy_labels_and_transition_period.pdf. Guidance document on rescaled EU energy labels.

Fan, Y. and Yang, C. (2025). Estimating discrete games with many firms and many decisions: An application to merger and product variety. *Journal of Political Economy*, 133(6):1886–1931.

Goettler, R. L. and Gordon, B. R. (2011). Does amd spur intel to innovate more? *Journal of Political Economy*, 119(6):1141–1200.

Gupta, A. (2021). Impacts of performance pay for hospitals: The readmissions reduction program. *American Economic Review*, 111(4):1241–1283.

Holmes, T. J. (2011). The diffusion of Wal-Mart and economies of density. *Econometrica*, 79(1):253–302.

Holmstrom, B. and Milgrom, P. (1991). Multitask principal–agent analyses: Incentive contracts, asset ownership, and job design. *The Journal of Law, Economics, and Organization*, 7(special_issue):24–52.

Houde, J.-F., Newberry, P., and Seim, K. (2023). Nexus tax laws and economies of density in E-Commerce: A study of Amazon's fulfillment center network. *Econometrica*, 91(1):147–190.

Houde, S. (2018). How consumers respond to product certification and the value of energy information. *The RAND Journal of Economics*, 49(2):453–477.

Houde, S. (2026). The incidence of coarse certification: Evidence from the energy star program. *The RAND Journal of Economics*, 57(3):545–585.

Ito, K. and Sallee, J. M. (2018). The economics of attribute-based regulation: Theory and evidence from fuel economy standards. *Review of Economics and Statistics*, 100(2):319–336.

Jacob, B. A. (2005). Accountability, incentives and behavior: The impact of high-stakes testing in the chicago public schools. *Journal of Public Economics*, 89(5–6):761–796.

Jia, P. (2008). What happens when wal-mart comes to town: An empirical analysis of the discount retailing industry. *Econometrica*, 76(6):1263–1316.

Josephy, B., Bush, E., Attali, S., and Zuloaga, F. R. (2015). Cold wash: The cool and modern way to launder. In *Proceedings of the 2015 ECEEE Summer Study on Energy Efficiency: First Fuel Now*, pages 1527–1535. European Council for an Energy Efficient Economy.

Kraft, D. (1988). A software package for sequential quadratic programming. Technical Report FB 88-28, DFVLR.

Lu, S. F. (2012). Multitasking, information disclosure, and product quality: Evidence from nursing homes. *Journal of Economics & Management Strategy*, 21(3):673–705.

Montag, F. (2023). Mergers, foreign competition, and jobs: Evidence from the us appliance industry. *George J. Stigler Center for the Study of the Economy & the State Working Paper*, (326).

Reimers, I. and Waldfogel, J. (2021). Digitization and pre-purchase information: The causal and welfare impacts of reviews and crowd ratings. *American Economic Review*, 111(6):1944–1971.

Reynaert, M. (2021). Abatement strategies and the cost of environmental regulation: Emission standards on the european car market. *The Review of Economic Studies*, 88(1):454–488.

Reynaert, M. and Sallee, J. M. (2021). Who benefits when firms game corrective policies? *American Economic Journal: Economic Policy*, 13(1):372–412.

Rothwell, P. M. (2005). External validity of randomised controlled trials: "to whom do the results of this trial apply?". *The Lancet*, 365(9453):82–93.

Seim, K. (2006). An empirical model of firm entry with endogenous product-type choices. *The RAND Journal of Economics*, 37(3):619–640.

Sinner, H. (1960). *Über das Waschen mit Haushaltswaschmaschinen*. Haus und Heim Verlag, Hamburg, Germany.

Train, K. (2015). Welfare calculations in discrete choice models when anticipated and experienced attributes differ: A guide with examples. *Journal of Choice Modelling*, 16:15–22.

Van Holsteijn en Kemna B.V., Vlaamse Instelling voor Technologisch Onderzoek NV, Viegand Maagøe A/S, and Wuppertal Institut für Klima, Umwelt, Energie GmbH (2014). "omnibus" review study on cold appliances, washing machines, dishwashers, washer-driers, lighting, set-top boxes and pumps: Final report. Technical report, Prepared for the European Commission, DG ENER-C3, Brussels/Delft. 12 March 2014; specific contract no. ENER/C3/2012-418 LOT2/03/SI2.654805.

Varin, C., Reid, N., and Firth, D. (2011). An overview of composite likelihood methods. *Statistica Sinica*, 21:5–42.

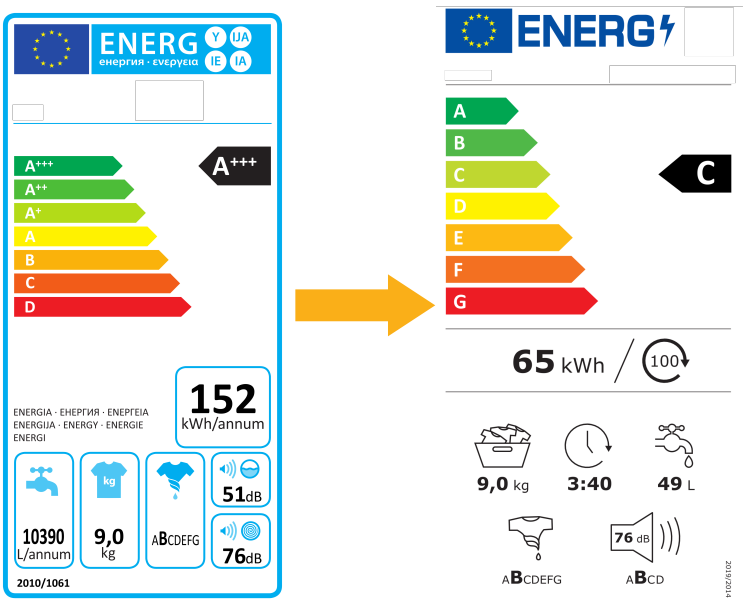
Vatter, B. (2022). Quality disclosure and regulation: Scoring design in medicare advantage. *Available at SSRN 4250361*.

Wollmann, T. G. (2018). Trucks without bailouts: Equilibrium product characteristics for commercial vehicles. *American Economic Review*, 108(6):1364–1406.

Zaheer, M., Kottur, S., Ravanbakhsh, S., Póczos, B., Salakhutdinov, R., and Smola, A. (2017). Deep sets. In *Advances in Neural Information Processing Systems*, volume 30.

Figures

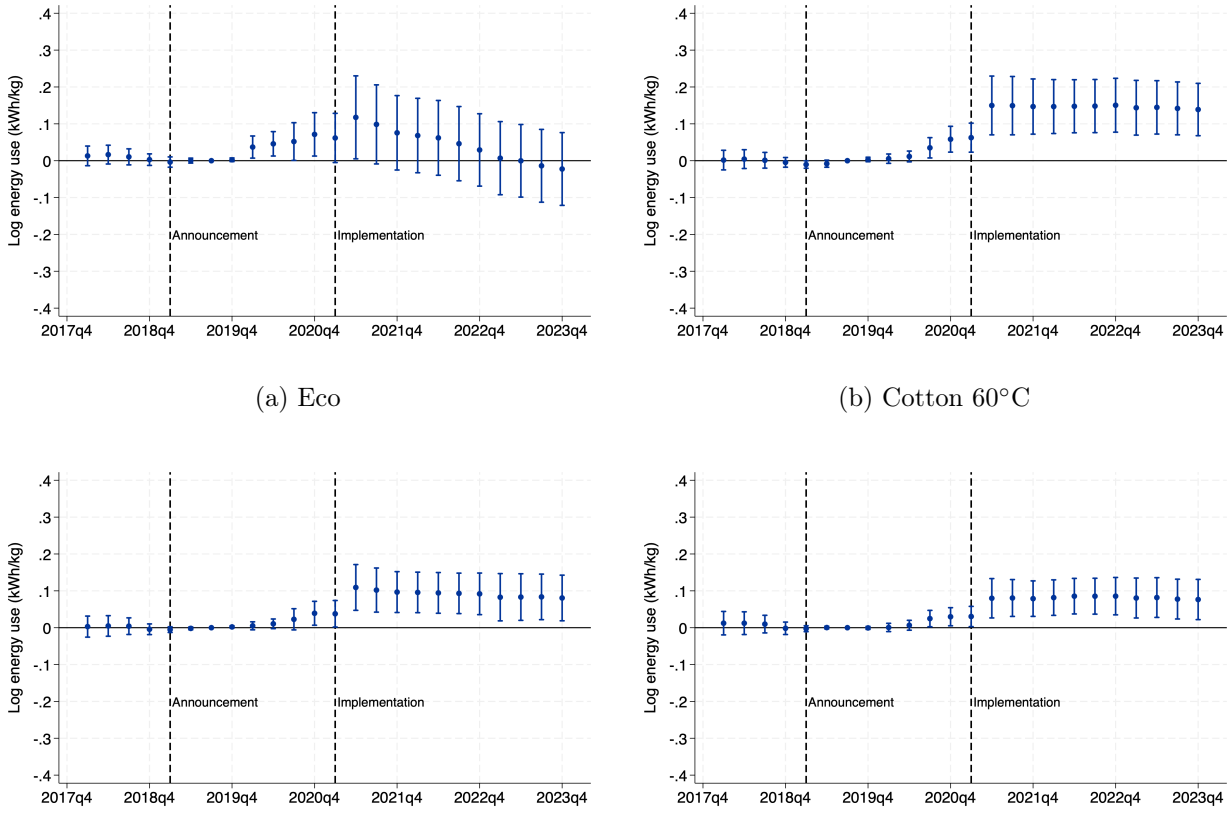
Figure 2: Energy Labels of Washing Machines in the EU



(a) 2011 Energy Label (b) 2021 Energy Label

Notes: The figure compares two versions of the EU energy label for washing machines. Both labels are shown for the same washing machine model and report the attributes of its energy-saving wash program. To preserve anonymity, brand and model identifiers are removed. The labels report energy-efficiency class and selected program attributes, including energy use, capacity, water consumption, program duration, and noise.

Figure 3: Changes in Program-Level Energy Consumption



(a) Eco

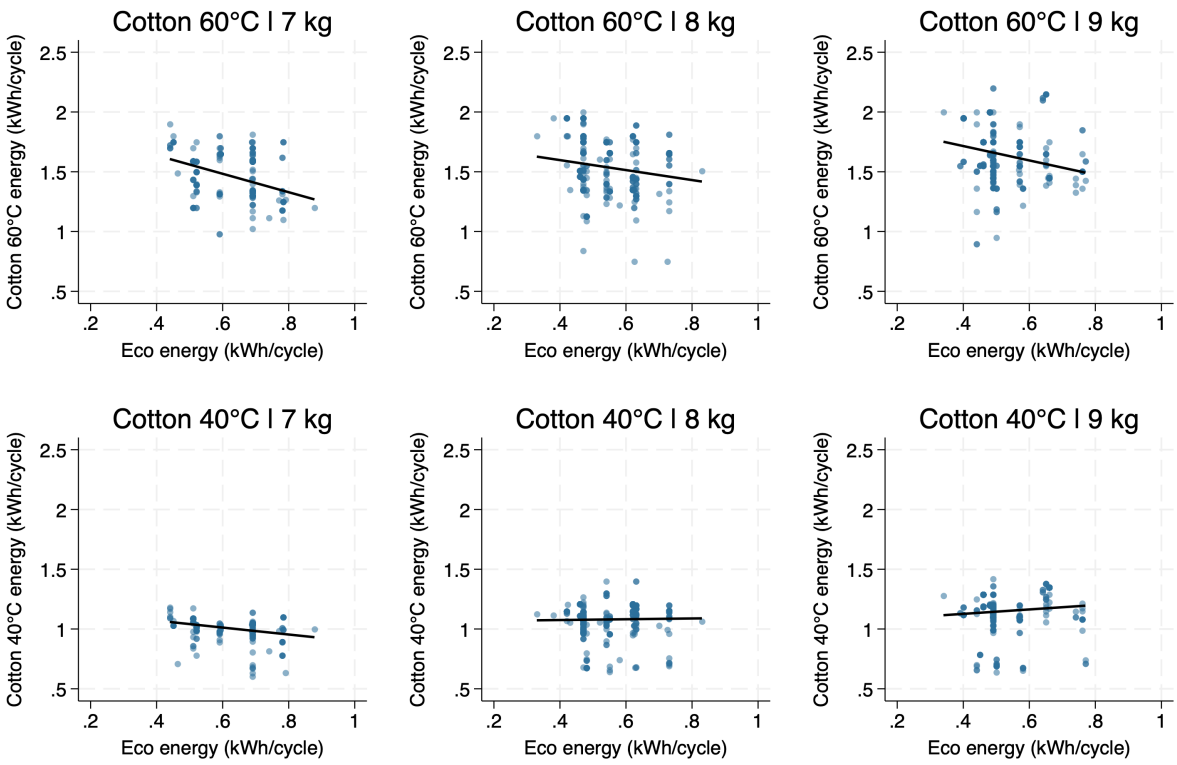
(b) Cotton 60°C

(c) Cotton 40°C

(d) Easy Care

Notes: Panels (a)–(d) report changes in log energy consumption per kilogram for Eco, Cotton 60°C, Cotton 40°C, and Easy Care, respectively, relative to Wool. Dots denote the estimated coefficients $\beta_{kr'}$ from equation (1), normalized to zero in 2019Q3. The sample contains 31 product lines (firm × capacity × spin speed), observed in every quarter from 2018Q1 to 2023Q4, with products weighted equally within each line. Bars denote 95 percent confidence intervals based on standard errors clustered by product line. Dashed vertical lines mark the reform's announcement in March 2019 and implementation in March 2021.

Figure 4: Eco and Cotton Energy Consumption by Capacity



Notes: Each point represents a washing-machine model carrying the new energy label. The horizontal axis reports certified Eco energy consumption, and the vertical axis reports Cotton energy consumption, both in kWh per cycle. The top and bottom rows show Cotton 60°C and Cotton 40°C, respectively; columns distinguish capacities of 7, 8, and 9 kg. Solid lines show separate unadjusted linear fits within each panel. The sample includes models with nonmissing energy consumption for both programs, including imputed Cotton values.

Figure 5: Product turnover by new energy class

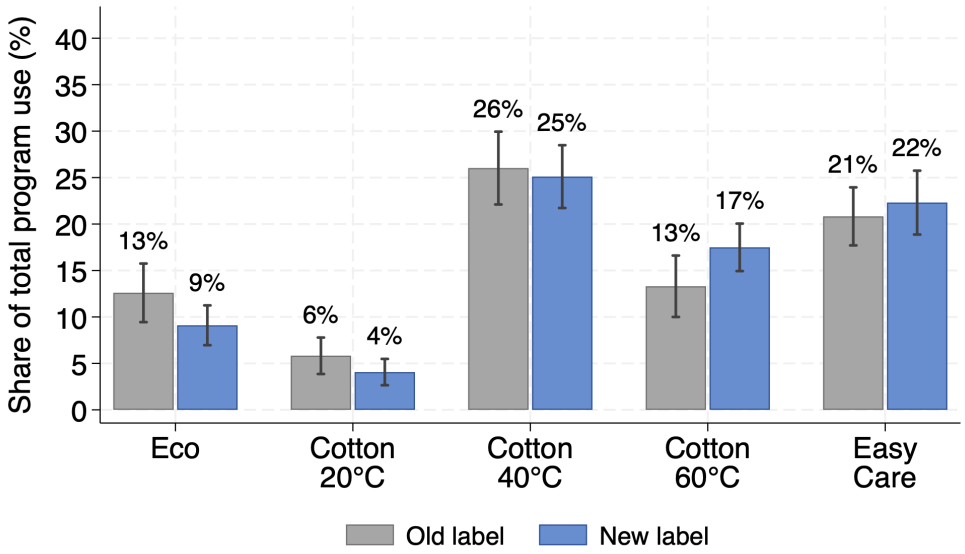


(a) Entry

(b) Exit

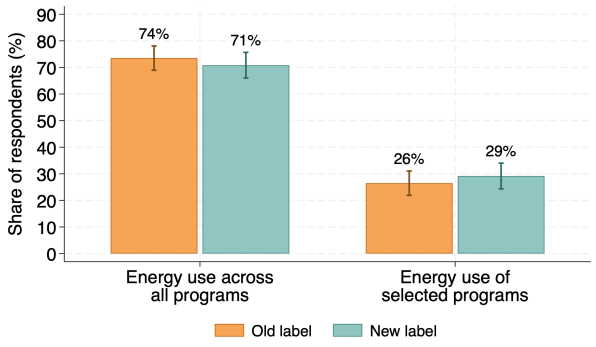
Notes: Quarterly national product entries and recorded exits, 2020q1–2023q4, for new-label classes *A*–*E*. A product is the current demand-data group: firm × capacity/spin-speed line × entry-quarter cohort × energy-label class (new and legacy class identifiers). Each product is counted once across markets and wash programs, rather than once per physical model. Entry and exit use the group's recorded dates; a group remains active if any constituent model has no recorded exit. The sample endpoint is not treated as an exit. Bars retain the same *A*–*E* colors in both panels; the dashed line marks 2021q1 label rescaling.

Figure 6: Share of program use

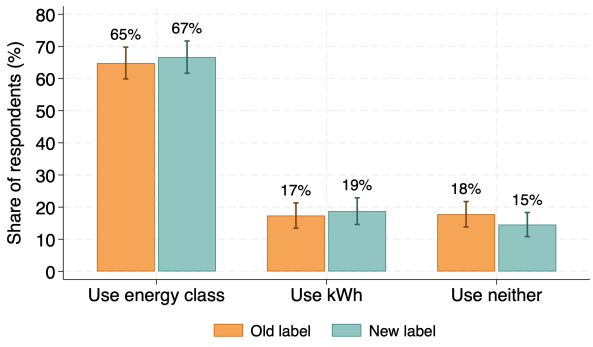


Notes: The figure reports the distribution of wash-program choices in the household survey, separately for owners of old- and new-label washing machines. Shares are calculated as the fraction of total reported wash cycles assigned to each program. Eco accounts for 13 percent of cycles among old-label households and 9 percent among new-label households. Cotton 40°C, Cotton 60°C, and Easy Care are the most frequently used programs. Remaining cycles belong to other programs not shown in the figure.

Figure 7: Consumer Beliefs About Energy Ratings and Energy Costs

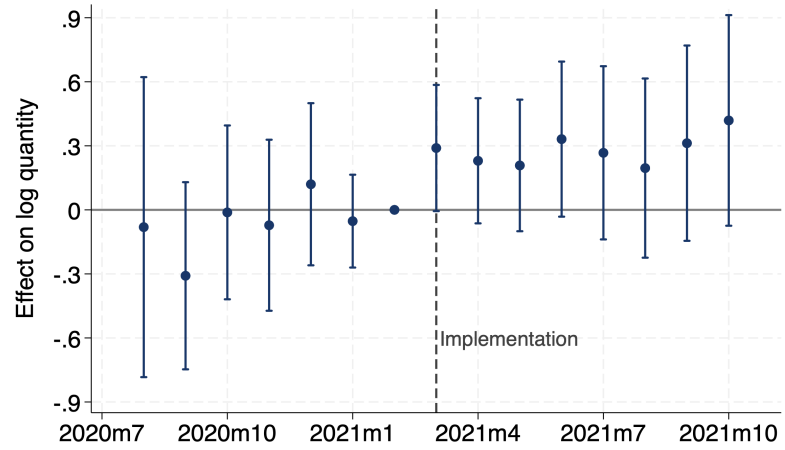


(a) Interpretation of Energy Ratings

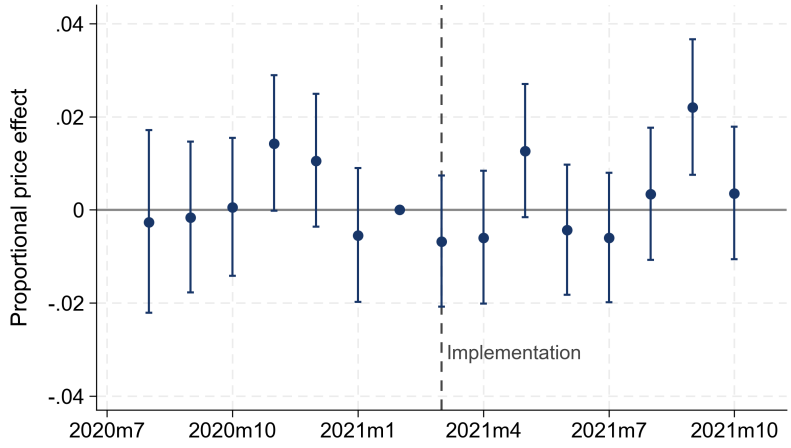


(b) Information Used to Assess Energy Costs

Notes: The figure reports household survey responses separately for owners of old- and new-label washing machines. Panel (a) shows whether respondents believe the energy rating reflects energy consumption across all programs or only selected programs. Panel (b) reports whether respondents use the energy class, reported kWh consumption, or neither to assess energy costs. Most respondents in both groups interpret the energy rating as measuring whole-machine efficiency, while approximately two-thirds rely on the energy class rather than reported kWh consumption.



(a) Demand effect



(b) Price effect

Figure 8: Demand and Price Effects

Notes: Event-study estimates around the March 2021 energy-label rescaling. Panel (a) reports coefficients for log product sales, controlling for log price. Panel (b) reports proportional price effects, $\exp(\widehat{\beta}_k) - 1$, from a log-price regression; confidence limits are transformed in the same way. Both specifications include product–market and calendar-month fixed effects and omit February 2021. Points are estimates and bars are 95 percent confidence intervals. The demand regression clusters standard errors by product; the price regression uses conventional standard errors. Each dashed line marks implementation. The two panels retain their respective estimation samples.

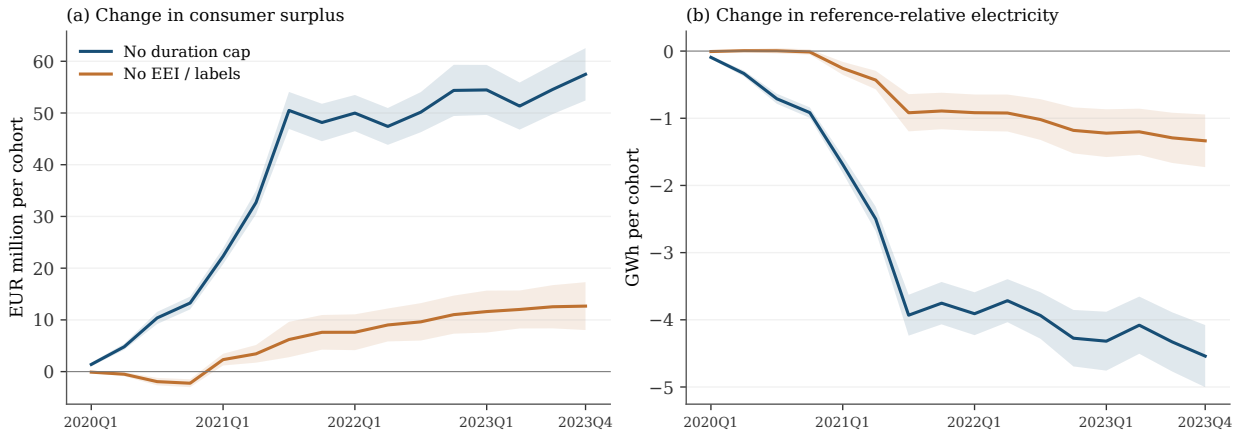


Figure 9: Quarterly effects relative to current policy

Means of 20 matched path differences. Shaded bands are plus or minus 1.96 Monte Carlo standard errors and exclude parameter and approximation uncertainty. Each point is a decision-cohort outcome: welfare includes discounted lifetime electricity costs, whereas electricity is quarterly use for that cohort. The frozen benchmark is omitted because it does not follow the same evolving market.

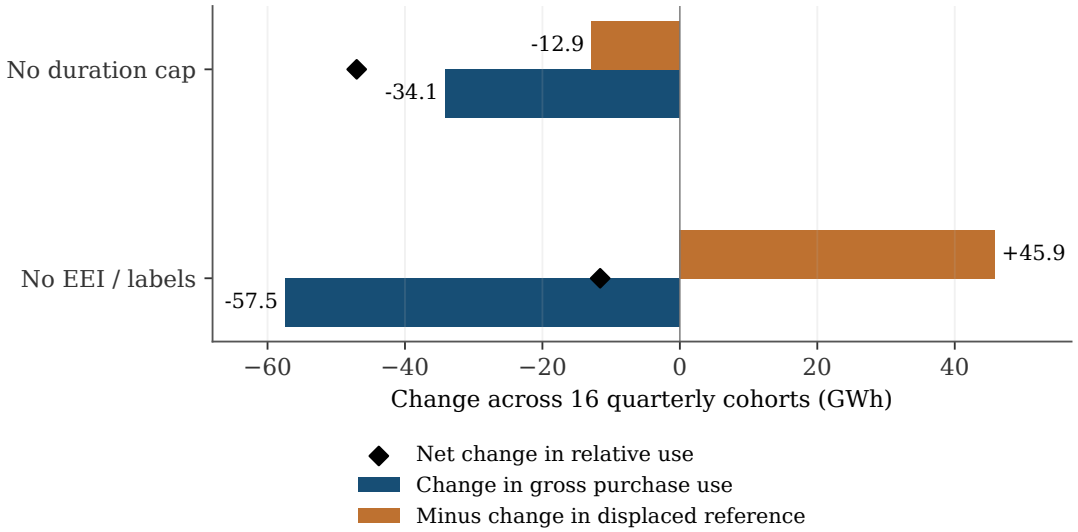


Figure 10: Accounting for the change in electricity

Bars show ΔG and $-\Delta D$; their sum is the net change ΔR, shown by the diamond. Negative values reduce reference-relative electricity. Components are means of horizon totals across matched paths. This is an accounting identity, not a causal allocation to design, program switching, or purchase decisions.

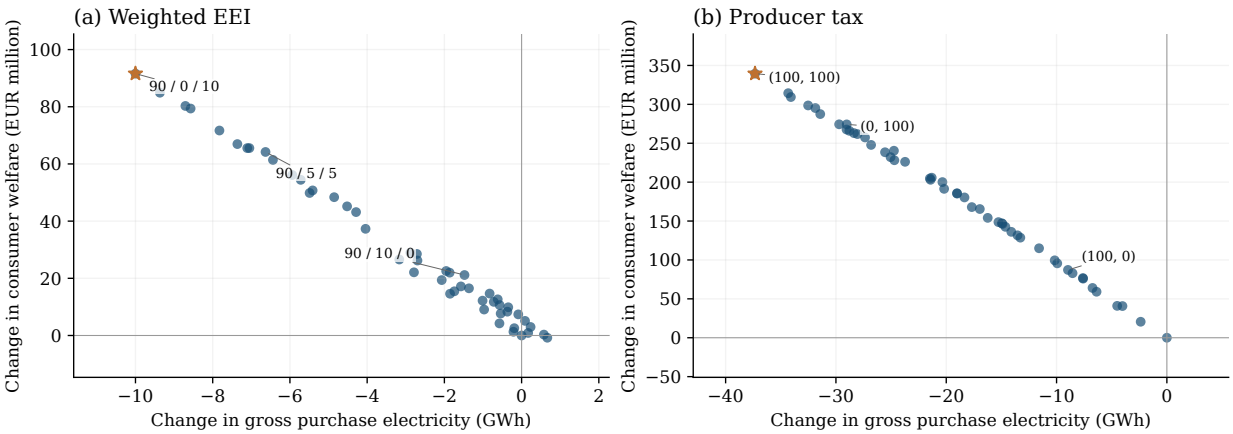


Figure 11: Consumer welfare and gross electricity across tested policies

Each panel shows 50 path-mean changes relative to its own baseline, using gross-use accounting. Stars identify the sole nondominated tested point in each family; upper-left outcomes have higher consumer welfare and lower gross purchase electricity. Labels give EEI percentages or tax-coefficient pairs. The panels do not share a policy baseline or a common Cotton engineering calibration.

Tables

Table 1: Summary Statistics

Panel A. Products					
Energy class	A^{+++}	*D*	*C*	*B*	*A*
Average price (€)	616.41	408.94	512.27	522.63	600.95
Average monthly unit sales	839.84	562.75	673.09	640.14	809.89
Average capacity (kg)	7.99	7.73	8.36	8.05	8.51
Number of products	139	44	40	41	85

Panel B. Program attributes and usage

Program	Energy (kWh)	Water (L)	Duration (min)	Load size (kg)	Temperature (°C)	Reported cycles
Eco	0.87	56.38	224.53	8.12	37.98	7,304
Cotton 20°C	0.34	81.47	171.31	8.12	21.62	3,247
Cotton 40°C	1.04	83.00	184.30	8.12	41.26	17,317
Cotton 60°C	1.49	81.94	183.70	8.12	56.12	10,580
Easy Care	0.66	57.82	132.57	3.77	39.67	14,732

Notes: Product statistics use national-market observations for the products retained in demand estimation, from 2018m1 to 2023m12. Panel A reports unweighted product-month means and distinct aggregated product counts by energy class. A^{+++} includes only old-label products; *D*–*A* use the new label. Prices are CPI-deflated euros. Panel B is pooled across energy classes and reports unweighted product means of program attributes, averaging within products over time first and excluding missing values separately for each attribute. Eco combines old-label Standard 60 and new-label Eco at full load. Attributes include existing imputations. Reported cycles are totals from both survey waves, with old and new Eco combined; survey counts are not restricted to full-load cycles or to products in the demand sample.

Table 2: Duration-Cap Exposure and Cotton Program Design

	Cotton 60°C			Cotton 40°C		
	Energy	Duration	Water	Energy	Duration	Water
Exposure	0.452***	0.154***	-0.051	0.186***	0.042	-0.043
	(0.068)	(0.056)	(0.048)	(0.055)	(0.048)	(0.054)
Capacity FE	Yes	Yes	Yes	Yes	Yes	Yes
Brand FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	244	244	244	244	244	244
R-squared	0.519	0.318	0.369	0.376	0.532	0.362

Notes: Outcomes are log differences in energy, duration, or water use between dual-label products and matched old-label-only donors (EC11 observed, EC21 missing). Donors are matched by brand, Standard 60 full-load duration, and load. Inputs include imputed values. Exposure is the excess of Standard full-load duration over the Eco cap, normalized by the cap and bounded below at zero. All regressions include capacity and brand fixed effects. Conventional standard errors are in parentheses; samples vary by outcome. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Program-Use Estimates

Parameter	Estimate	Standard error
Temperature (°C) (ρ_0^T)	0.046	(0.004)
Duration (hours) (ρ_0^d)	-0.582	(0.064)
Water (litres) (ρ_w)	0.016	(0.001)
Load (kg) (ρ_L)	0.373	(0.027)
Duration × heavy-stain share (ρ_1^d)	0.362	(0.171)
Temperature × heavy-stain share (ρ_1^T)	-0.041	(0.012)
Old Eco/Standard effect	-4.937	(0.177)
New Eco effect	-4.024	(0.144)
Cotton effect	-5.638	(0.223)
Easy Care effect	-3.163	(0.121)
Temperature taste standard deviation (σ_T)	0.064	(0.002)
Duration taste standard deviation (σ_d)	2.248	(0.068)

Notes: Household count likelihood with 16,000 simulated taste draws. Program-effect rows report $\delta_{c(r)}$. These effects are relative to the other-program option; Cotton 20°C/40°C/60°C share one effect. Stain share is uncentered. Temperature and duration random tastes are independent standard normals multiplied by the reported standard deviations. All brands are included.

Table 4: Demand Estimates: I_{ij} Model

Variable	Parameter	Estimate	Std. Error
Linear demand coefficients			
Technology (shared Eco/Standard–Cotton)	$(\beta^x)_{\text{tech}}$	-1.147	(0.227)
Capacity: 7 kg	$(\beta^x)_{7\text{kg}}$	0.309	(0.251)
Capacity: 8 kg	$(\beta^x)_{8\text{kg}}$	0.220	(0.185)
Capacity: 9 kg	$(\beta^x)_{9\text{kg}}$	0.412	(0.102)
Spin speed: 1400 rpm	$(\beta^x)_{1400}$	0.140	(0.125)
Spin speed: 1600 rpm	$(\beta^x)_{1600}$	0.392	(0.126)
Energy class A	β_A	0.956	(0.131)
Energy class B	β_B	0.186	(0.064)
Energy class C	β_C	0.172	(0.049)
Household-specific coefficients			
Log price-sensitivity intercept	$\bar{\alpha}$	1.619	(1.057)
Log-income effect on price sensitivity	α^y	-0.578	(0.278)
Program value	β^I	0.147	(0.045)
Price coefficient specification			
$-\alpha_i$ in utility		$-\exp(\bar{\alpha} + \alpha^y y_i)$	
Observations		7,496	
Markets		96	
Products		349	

Notes: Prices are measured in hundreds of euros; parentheses report robust standard errors. Price sensitivity is negative exponential in unstandardized log PHF income; residual price heterogeneity is fixed at zero. 5 market-quarter income quantile bins and program-taste nodes form 625 agents per market. Program value is the household-specific inclusive value I_{ij} computed from product-program inputs and survey-based program-use estimates. Energy class D is omitted; class effects apply from 2021Q1. Technology enters utility linearly; BLP characteristic sums use capacity, spin speed, and displayed-label energy use. Displayed runtime and Eco composite water do not enter directly in this saved specification; their program attributes affect utility through program value. Excluded instruments include traditional BLP characteristic sums, their interactions with market-quarter mean income, and mean program value. The estimates use aggregate moments only. Demand covers 2020Q1–2023Q4. The income covariate y_i is the unstandardized logarithm of PHF income measured in hundreds of euros. Class coefficients multiply $\mathbf{1}_{\text{new},t}$ in the model.

Table 5: Marginal Cost Estimates

Variable	Estimate	Std. Error
Shared technology	-0.116	(0.047)
Capacity: 7 kg	-0.552	(0.016)
Capacity: 8 kg	-0.347	(0.015)
Capacity: 9 kg	-0.146	(0.015)
Spin speed: 1400 rpm	-0.179	(0.028)
Spin speed: 1600 rpm	-0.009	(0.029)
Firm fixed effects	Yes	
Quarter fixed effects	Yes	
Observations	7,496	
R^2	0.754	

Notes: The dependent variable is log marginal cost recovered from the estimated demand system and firms' pricing first-order conditions. Technology is the shared firm–entry-quarter–energy-class effect estimated using Eco and legacy Standard observations and nonimputed Cotton observations. Lower values indicate lower energy use conditional on program inputs. The regression includes an intercept and firm and calendar-quarter fixed effects. Parentheses report HC1 heteroskedasticity-robust standard errors, conditional on the estimated demand and technology measures. Omitted categories are 10 kg and 1200 rpm.

Table 6: Design-Cost Estimates

Input adjustment	Linear term $(\kappa_r)_k$	Curvature $2(C_r)_{kk}$
Cotton 40°C duration (min)	27.14	35.47
	(7.04)	(7.50)
Cotton 60°C duration (min)	59.20	69.31
	(10.64)	(74.49)
Eco duration (min)	-1.55	1.31
	(3.25)	(1.75)
Eco temperature (°C)	45.92	97.61
	(6.44)	(304.51)
Technology (10^{-3} index units)	-263.09	544.76
	(40.42)	(3,581.98)
Parameters	10	
Moment restrictions	444	

Notes: For input component k, cost is $(\kappa_r)_k \Delta q_k + (C_r)_{kk}(\Delta q_k)^2$, in euros per introduction. The curvature column reports the second derivative, $2(C_r)_{kk}$, rather than the quadratic coefficient itself. Parentheses report conditional nonclustered HC1 standard errors. Eco duration and Cotton 40°C have separate curvatures. Full- and half-load Eco inputs share group coefficients. The technology row defines one input unit as 10^{-3} of the underlying index: its displayed linear and curvature coefficients are the original coefficients multiplied by 10^{-3} and 10^{-6}, respectively. Estimates are the best verified candidate in a finite-budget search, not a converged outer optimum. Appendix F.5 reports clustered sensitivity and qualifications.

Table 7: Entry-Cost Estimates

Parameter	Estimate	S.E.
Common parameters		
Launch intercept κ_L	2.118	1.259
Quarter 2 γ_2	0.067	0.474
Quarter 3 γ_3	-0.637	0.502
Quarter 4 γ_4	-0.520	0.483
$\log(1+N)$ coefficient β_N	-0.759	0.391
$\log(1+G)$ coefficient β_G	1.344	0.855
Shock scale σ	1.617	0.853
Transition switching premiums		
Class A	2.576	1.574
Class B	3.434	2.124
Classes C and D	1.609	0.959
Post-policy switching premiums		
Class A	1.892	0.768
Class B	3.494	2.048
Classes C and D	4.735	2.602

Notes: Amounts are millions of euros; 469 line–quarters and 7 firms. Quarter 1 is omitted. Switching premiums apply to classes absent from the latest preceding introduction on the line. S.E.s are nonclustered HC1 sandwich estimates, treating line–quarters as independent, projected onto the tangent space of the binding constraints, conditional on the active constraint set, fitted beliefs, and surrogate. They exclude uncertainty about which constraints bind and upstream estimation uncertainty.

Table 8: Counterfactual policy definitions

Policy	Regulatory treatment	Comparator and reported accounting
Current policy	Eco-only EEI, labels, duration cap	Common removal baseline; R, CS
No duration cap	Eco-only EEI and labels	Current policy; R, CS
No EEI / labels	Duration cap retained	Current policy; R, CS
Frozen joint removal	No EEI, labels, or cap; initial state repeated	Descriptive comparison; R, CS
Weighted EEI	50 Eco/Cotton weight vectors; labels; no cap	Uncapped Eco-only; G, CW^G
Producer tax	50 Eco/Cotton coefficient pairs; no EEI, labels, or cap	Carbon-model zero tax; G, CW^G

Notes: Each evolving policy has 20 paths; the frozen benchmark is solved once. Both grids include their baseline among the 50 policies. "No EEI / labels" removes constraints and explicit label utility; it does not restore the old label scale. The grid and removal accounting conventions differ as described in Section 8.1.

Table 9: Entry and product-design responses

	Current policy	No duration cap	No EEI / labels
Entry and availability			
Product introductions, 2020–2023	89.45	87.10	66.10
Active products, 2023Q4	89.45	87.10	66.10
Introductions per quarter	5.59	5.44	4.13
Final-quarter program design: minutes or degrees Celsius			
Eco duration: full load	211.7	381.2	212.2
Eco duration: half load	162.5	346.1	162.9
Eco duration: quarter load (fixed)	192.6	190.9	197.8
Cotton 40°C duration	183.4	171.4	174.6
Cotton 60°C duration	175.3	169.9	170.4
Eco temperature: full load	31.4	29.7	37.3
Final-quarter class composition: percent of products			
Class *A*	45.7	43.7	—
Class *B*	14.5	14.6	—
Class *C*	18.1	19.8	—
Class *D*	21.8	21.9	—

Notes: Means across 20 paths. Program attributes and class shares are equal-product averages within each path's active 2023Q4 portfolio, then averaged across paths; they are not sales weighted. Product introductions count simulated entry actions over 16 quarters. Final active-product counts coincide with cumulative introductions in these saved paths. Quarter-load Eco attributes are fixed at the product level, so changes in their mean reflect composition. Class shares are omitted for unlabeled entry: the *D* template is a physical implementation device, not a counterfactual rating.

Table 10: Product responses under selected grid policies

	Weighted EEI		Producer tax	
	100 / 0 / 0	90 / 0 / 10	(0, 0)	(100, 100)
Introductions, 2020–2023	87.10	80.75	62.55	59.40
Active products, 2023Q4	87.10	80.75	62.55	59.40
Eco duration: full load (min)	381.2	357.2	345.1	344.0
Eco duration: half load (min)	346.1	319.9	278.3	272.7
Cotton 40°C duration (min)	171.4	174.4	166.2	165.2
Cotton 60°C duration (min)	169.9	163.3	168.4	168.3
Eco temperature: full load (°C)	29.7	26.2	30.4	30.1
Eco temperature: half load (°C)	28.7	25.4	28.3	27.9

Notes: Means across 20 paths. Design attributes are equal-product means in each active 2023Q4 portfolio, then averaged across paths. Entry counts sum introductions over 16 quarters. EEI weights are percentages in the order Eco, Cotton 40°C, Cotton 60°C. Tax columns use the carbon model; their levels are not a common-baseline comparison with the EEI family.

Table 11: Changes in consumer surplus and electricity

	ΔCS (EUR m)	MCSE (EUR m)	ΔR (GWh)	MCSE (GWh)
Current policy	0.00	0.00	0.00	0.00
No duration cap	603.22	20.55	-47.02	1.86
No EEI / labels	100.87	18.93	-11.60	1.64
Frozen joint removal	1,192.49	24.53	-15.57	1.65

Notes: Changes relative to current policy, averaged across matched paths. CS uses the purchase outside reference; R sums reference-relative quarterly use across 16 cohorts. MCSE is the paired simulation standard error, not parameter uncertainty. Current-policy levels are EUR 588.35 million for CS and 15.89 GWh for R. The frozen benchmark changes fundamentals and population. Grid outcomes use distinct accounting and appear in Table 13.

Table 12: Distribution of consumer-surplus changes

	Baseline CS (EUR m)	ΔCS (EUR m)	MCSE (EUR m)	ΔCS/obs. (EUR)
No duration cap				
All households	588.35	603.22	20.55	62.44
Positive duration preference	176.52	596.53	20.56	264.65
Nonpositive duration preference	411.83	6.69	4.88	0.90
No EEI / labels				
All households	588.35	100.87	18.93	10.44
Positive duration preference	176.52	95.14	16.45	42.21
Nonpositive duration preference	411.83	5.72	7.39	0.77

Notes: The groups are defined by the sign of the household's program-duration coefficient, not by observed cycle length. They contain 2.254 million positive-preference and 7.406 million nonpositive-preference weighted decision observations over the horizon; these are not counts of unique households. Per-observation gains divide each group's aggregate change by its own population sum. Baseline values are repeated to make the two policy comparisons self-contained.

Table 13: Selected policy-grid outcomes relative to matched baselines

Policy	ΔG	ΔCS^G	Rebates	ΔCW^G
	GWh	EUR m	EUR m	EUR m
Weighted EEI: Eco / Cotton 40°C / Cotton 60°C				
90 / 5 / 5	-6.63	64.19	0.00	64.19
90 / 10 / 0	-1.48	21.15	0.00	21.15
90 / 0 / 10	-10.00	91.59	0.00	91.59
Producer tax: (τ_E, τ_C)				
(50, 50)	-19.02	80.48	104.81	185.29
(100, 0)	-8.97	44.74	42.44	87.18
(0, 100)	-29.02	122.74	151.51	274.25
(100, 100)	-37.34	166.17	173.30	339.48

Notes: Means of paired differences over 20 paths and 16 cohorts. EEI compares with uncapped Eco-only; tax compares with zero tax in the carbon model. Baseline gross electricity is 165.89 and 128.87 GWh, respectively. CS^G deducts gross purchase electricity expenditure; CW^G adds rebates. Revenue equals aggregate rebates and is added once. These are distinct accounting columns from Table 11. Full grids and Monte Carlo errors appear in Tables A11 and A12.

Table 14: Distribution of selected grid-policy welfare changes

Group	ΔCS^G	Rebates	ΔCW^G	ΔCW^G/obs.
	EUR m	EUR m	EUR m	EUR
EEI weights 90 / 0 / 10				
All households	91.59	0.00	91.59	9.48
Positive duration preference	38.84	0.00	38.84	17.23
Nonpositive duration preference	52.75	0.00	52.75	7.12
Producer tax (100, 100)				
All households	166.17	173.30	339.48	35.14
Positive duration preference	87.32	40.44	127.75	56.68
Nonpositive duration preference	78.86	132.86	211.72	28.59

Notes: Changes relative to each family's baseline under gross-use accounting. Groups and population weights are the same as in Table 12. Rebates are equal per household within each market. Aggregate subgroup changes sum to the all-household outcome; per-observation effects divide by each group's own horizon population.

Appendix

Appendix Contents

A Regulatory Details **57**

A.1 Energy-Efficiency Index Calculations 57

A.2 Program-Design Constraints 58

B Data Imputation **62**

C Laundry-habits survey **64**

D Proofs and Extensions for Regulation-Induced Energy Shifting **69**

D.1 Proof of Proposition 1 69

D.2 Extension: Joint Eco Constraints 70

D.3 Socially Efficient Program Design 72

E Descriptive Evidence **73**

E.1 Energy Equation 73

F Estimation and Computational Implementation **74**

F.1 Parameter Snapshots and Profit Approximation 74

F.2 Reference Designs, Samples, and Moments 77

F.3 Constrained Design Solver 81

F.4 Portfolio Conditionals, Gibbs Sampling, and NPL 81

F.5 Conditional Inference 82

G Counterfactual Computation and Validation **83**

G.1 Policy implementation and state transitions 83

G.2 Numerical protocol used in the completed runs 85

G.3 Aggregation, decomposition, and reproducibility 87

G.4 Completed Policy Grids: Implementation and Validation 88

A Regulatory Details

This appendix summarizes the energy-efficiency calculations and program-design constraints under the 2011 and 2021 EU regulations for household washing machines.

A.1 Energy-Efficiency Index Calculations

A.1.1 2011 Labeling Regime

Under the 2011 regime, the Energy Efficiency Index (EEI) measures weighted annual energy consumption relative to a capacity-based benchmark:

$$\mathrm{EEI}_{2011} = 100 \times \frac{\mathrm{AE}_c}{\mathrm{SAE}_c}, \qquad \mathrm{SAE}_c = 47c + 51.7, \tag{A.1}$$

where AE_c denotes annual energy consumption in kWh/year and c is the rated capacity in kg, defined as the lower of the full-load capacities of the standard Cotton 60°C and Cotton 40°C programs.

Annual energy consumption is calculated as

$$\mathrm{AE}_c = 220E_t + \frac{(P_o + P_l)(525600 - 200T_t)}{2 \times 60 \times 1000}, \tag{A.2}$$

where E_t is weighted energy consumption per cycle (kWh), P_o and P_l are weighted power consumption in off-mode and left-on mode (W), and T_t is weighted program duration (minutes). The calculation assumes 220 standard washing cycles per year and includes energy consumed in low-power modes.[18]

Weighted energy consumption is calculated from three standardized Cotton test cycles:

$$E_t = \frac{3E_{60,\mathrm{full}} + 2E_{60,\mathrm{half}} + 2E_{40,\mathrm{half}}}{7}, \tag{A.3}$$

where $E_{60,\mathrm{full}}$ and $E_{60,\mathrm{half}}$ denote energy consumption of the standard Cotton 60°C program at full and half load, respectively, and $E_{40,\mathrm{half}}$ denotes energy consumption of the standard Cotton 40°C program at half load. The same weights apply to P_o, P_l, and T_t.

The resulting EEI determines the energy-efficiency class according to the following thresholds.

[18] The expression applies to the standard calculation without a separate adjustment for automatic power management. For machines equipped with a power-management system, the regulation provides an alternative formula incorporating the duration of left-on mode.

Table A15: Energy efficiency classes

Energy efficiency class	**Energy Efficiency Index**
A+++ (most efficient)	EEI < 46
A++	$46 \leq$ EEI < 52
A+	$52 \leq$ EEI < 59
A	$59 \leq$ EEI < 68

A.1.2 2021 Labeling Regime

Under the 2021 regime, the EEI is calculated from weighted energy consumption of the Eco 40–60 program:

$$\mathrm{EEI}_{2021} = 100 \times \frac{E_W}{\mathrm{SCE}_W}, \qquad \mathrm{SCE}_W = -0.0025c^2 + 0.0846c + 0.3920, \tag{A.4}$$

where E_W is weighted energy consumption in kWh per cycle and c is the rated capacity of the Eco 40–60 program in kg.

Weighted energy consumption combines measurements at full, half, and quarter load:

$$E_W = AE_{W,\mathrm{full}} + BE_{W,\mathrm{half}} + CE_{W,\mathrm{quarter}}, \tag{A.5}$$

where the capacity-dependent weights for machines with rated capacity above 3 kg are

$$\begin{aligned} A &= -0.0391c + 0.6918, \\ B &= -0.0109c + 0.3582, \\ C &= 1 - A - B. \end{aligned} \tag{A.6}$$

For machines with rated capacity at or below 3 kg, the regulation assigns $A = 1$ and $B = C = 0$.

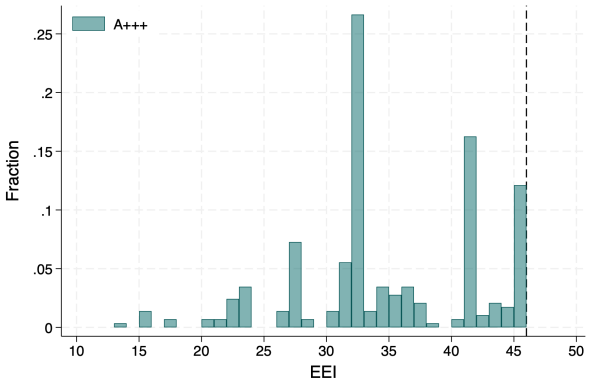
The resulting EEI determines the energy-efficiency class according to the following thresholds.

A.2 Program-Design Constraints

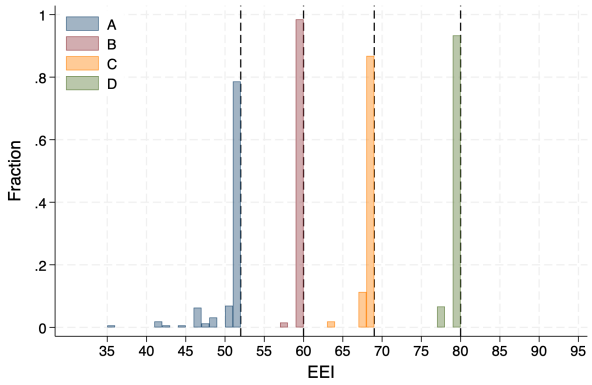
A.2.1 Cycle-Duration Limits under the 2021 Regime

The 2021 ecodesign regulation imposes capacity-dependent upper bounds on the duration of the Eco 40–60 program. The limits differ between full and partial loads:

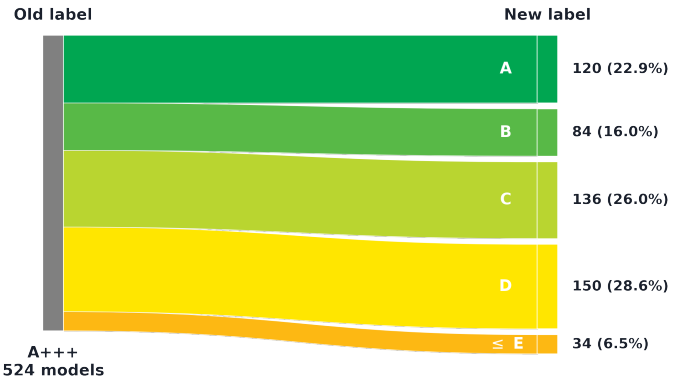
$$\begin{aligned} t_{\mathrm{cap}}^{\mathrm{full}}(c) &= \min\{137 + 10.2c,\ 240\}, \\ t_{\mathrm{cap}}^{\mathrm{half}}(c) = t_{\mathrm{cap}}^{\mathrm{quarter}}(c) &= \min\{120 + 6c,\ 180\}, \end{aligned} \tag{A.7}$$



(a) Old label (EEI11)



(b) New label (EEI21)



(c) Portfolio transformation

Figure A12: Energy Efficiency Indices and Portfolio Transformation

Notes: Panel (a) shows the distribution of EEI11 among A^{+++} products sold between March 2019 and December 2020. Panel (b) shows the distribution of EEI21 among products classified A–D under the new labeling regime. Panel (c) shows portfolio transformation across labeling regimes. Because the reform changed the underlying efficiency calculation and testing requirements, EEI11 and EEI21 are not directly comparable.

Table A16: Energy efficiency classes

Energy Efficiency Class	**Energy Efficiency Index**
A	$\text{EEI} \leq 52$
B	$52 < \text{EEI} \leq 60$
C	$60 < \text{EEI} \leq 69$
D	$69 < \text{EEI} \leq 80$
E	$80 < \text{EEI} \leq 91$
F	$91 < \text{EEI} \leq 102$
G	$\text{EEI} > 102$

where c is rated capacity in kg and duration is measured in minutes. Table A17 reports the corresponding limits for common washing-machine capacities.

Table A17: Eco 40–60 Duration Limits under the 2021 Regulation

Rated capacity	Full load	Half/quarter load
6 kg	198.2 min	156 min
7 kg	208.4 min	162 min
8 kg	218.6 min	168 min
9 kg	228.8 min	174 min
10 kg	239.0 min	180 min
$\geq$10.5 kg	240 min	180 min

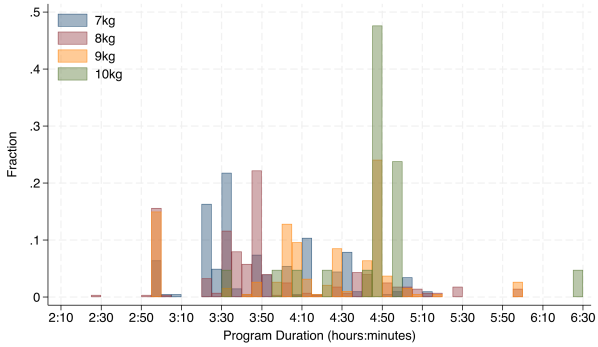
A.2.2 Water-Consumption Limits under the 2011 and 2021 Regimes

Under the 2011 ecodesign regulation, water-consumption limits applied to the standard Cotton 60°C program at full load. From December 1, 2011, the requirement was

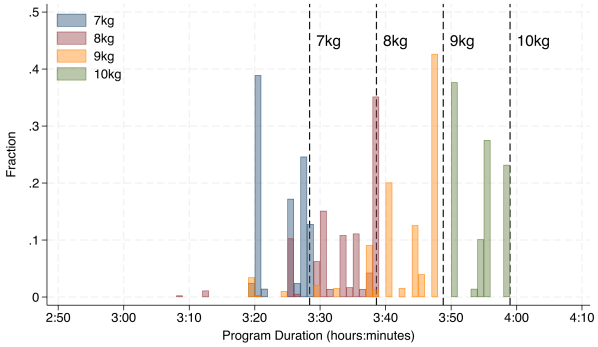
$$W_t \leq 5c + 35, \tag{A.8}$$

where W_t is the water consumption of the standard Cotton 60°C program at full load (liters per cycle), and c is the lower of the rated capacities of the standard Cotton 60°C and Cotton 40°C programs at full load.

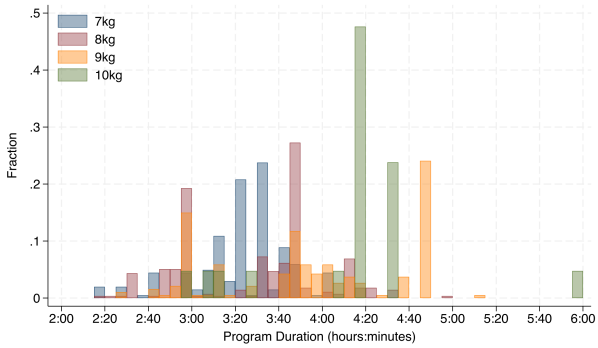
Figure A13: Eco program duration distributions under the old and new labels



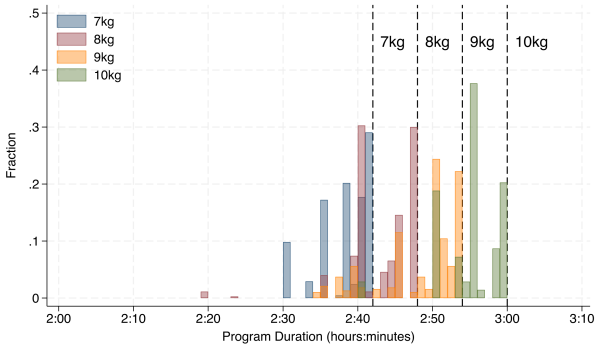
(a) Old label: Cotton 60°C, full load



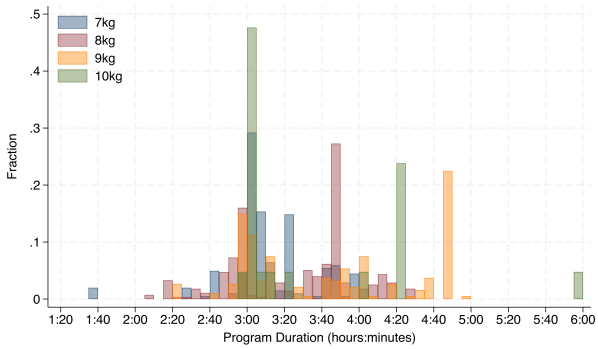
(b) New label: Eco, full load



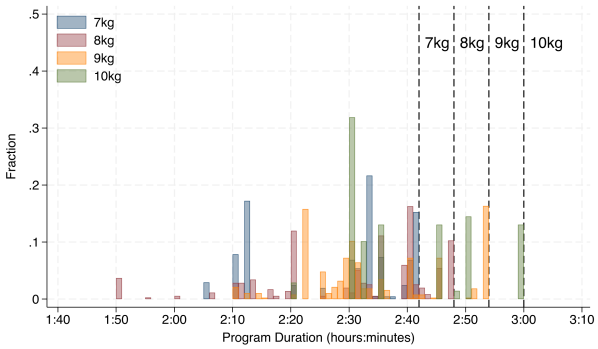
(c) Old label: Cotton 60°C, half load



(d) New label: Eco, half load



(e) Old label: Cotton 40°C, half load



(f) New label: Eco, quarter load

Notes: The figure shows program-duration distributions separately by rated capacity, with equal weight assigned to each model within each capacity group. The left column reports the three standard Cotton test conditions used under the old label; the right column reports Eco at full, half, and quarter load under the new label. Rows therefore do not always represent identical test conditions across regimes. Vertical dashed lines in the right column indicate capacity-specific regulatory duration limits. For plotting only, reported durations exceeding their applicable limits are adjusted to the largest whole minute strictly below the limit; the underlying data are unchanged.

From December 1, 2013, the requirement was tightened to

$$W_t \leq 5c_{1/2} + 35, \tag{A.9}$$

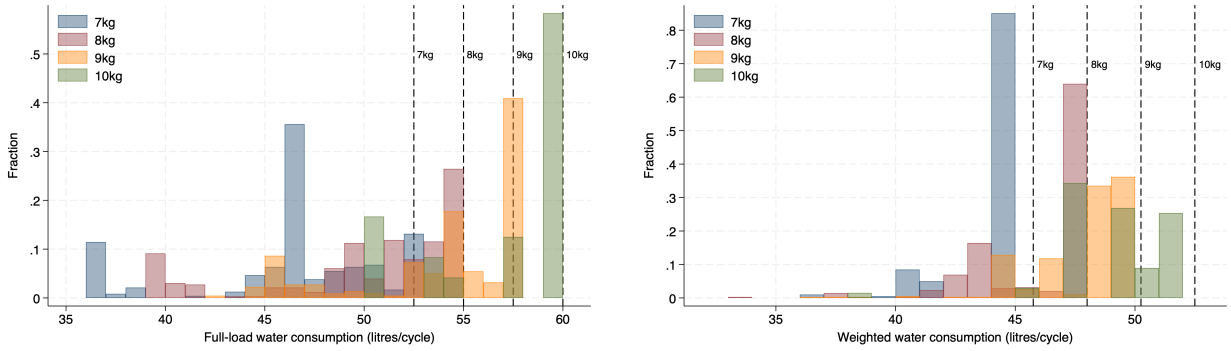
where $c_{1/2}$ is the lower of the rated capacities of the standard Cotton 60°C and Cotton 40°C programs at partial load.

The 2021 regulation instead constrains weighted water consumption of the Eco 40–60 program:

$$W_W \leq 2.25c + 30, \tag{A.10}$$

where W_W is weighted water consumption in liters per cycle, calculated across full, half, and quarter loads using the same capacity-dependent weights A, B, and C as in the energy-efficiency calculation.

Thus, the two regimes differ in both the regulated test programs and the measurement basis used to impose water-consumption limits.



(a) Old label: full-load water consumption

(b) New label: weighted water consumption

Figure A14: Distribution of Water Consumption

Notes: The figure shows water consumption in litres per cycle, separately by rated capacity, with equal weight assigned to each model within each capacity group. Panel (a) reports standard Cotton 60°C water consumption at full load under the old label. Panel (b) reports weighted Eco water consumption across full, half, and quarter loads under the new label. Vertical dashed lines indicate the capacity-specific regulatory limits. For plotting only, reported values exceeding their applicable limits are adjusted to the largest whole litre strictly below the limit. The two panels use different test conditions and aggregation rules and are therefore not directly comparable.

B Data Imputation

Standard-program temperature. Under the pre-2021 labeling regime, product manuals identify the regulated test cycles as the standard Cotton 60°C program at full and half load and the standard Cotton 40°C program at half load, but they generally do not report the temperature

actually reached during these cycles. I do not assign the nominal program temperature to these observations. This distinction is empirically important because the nominal program name need not correspond to the realized wash temperature. Industry and consumer testing has documented substantial departures from the nominal temperature. In particular, Czyzewski (2022), writing from the perspective of BSH and the CENELEC working group on laundry appliances, reports that measured sump temperatures in Cotton 60°C programs vary substantially across products and that most machines do not reach 60°C; temperatures experienced by the laundry are lower still. An earlier Stiftung Warentest market test provides a quantitative benchmark: among 14 tested washing machines, none reached 60°C in the tested 60°C cotton program, with reported temperatures generally around 45–50°C and the lowest observation approximately 44°C (Van Holsteijn en Kemna B.V. et al., 2014).

I therefore impute realized temperatures from the physical characteristics of the wash cycle rather than from its nominal program name. I estimate an XGBoost regression using Eco 40–60 cycles for which temperature is observed. The estimation sample pools full-, half-, and quarter-load Eco cycles and excludes observations whose temperature was itself previously model-imputed. The predictors are program load, energy consumption, water consumption, program duration, product capacity, and the ratio of program load to rated capacity. All predictors are required to be observed and strictly positive. Importantly, the specification does not include the nominal program temperature, program identity, energy-label class, manufacturer, or the certified aggregate energy measure. The identifying assumption for the transfer is that, conditional on these physical inputs, the relationship between temperature and the other dimensions of program design is common across the Eco and standard test programs.

I evaluate the prediction rule using a product-level holdout sample: 20 percent of brand–model combinations are excluded from estimation, so that cycles from the same product cannot appear in both the training and validation samples. Within the Eco programs, the holdout regression yields an R^2 of 0.988, an RMSE of 0.79°C, and a mean absolute error of 0.42°C. I then re-estimate the model on all eligible Eco observations and apply it to the three old-regime standard programs using their observed physical inputs. The procedure imputes 19,521 standard-program temperatures; observations with incomplete or nonpositive physical inputs remain missing.

The resulting temperatures are well below the nominal program settings. The median imputed temperature is 36.1°C for the standard Cotton 60°C full-load cycle, 34.7°C for the standard Cotton 60°C half-load cycle, and 32.1°C for the standard Cotton 40°C half-load cycle. These estimates are lower than the roughly 45–50°C range reported in the 2013 Stiftung Warentest test, but both the external tests and the imputation imply the same economically important feature of the old testing regime: the nominal 40°C or 60°C designation is a poor measure of the temperature actually delivered by an energy-efficient cycle. The lower imputed values should therefore not be

interpreted as reproducing the Stiftung Warentest distribution mechanically; rather, the external evidence supports the central feature of the imputation—substantial temperature reductions relative to the nominal setting.

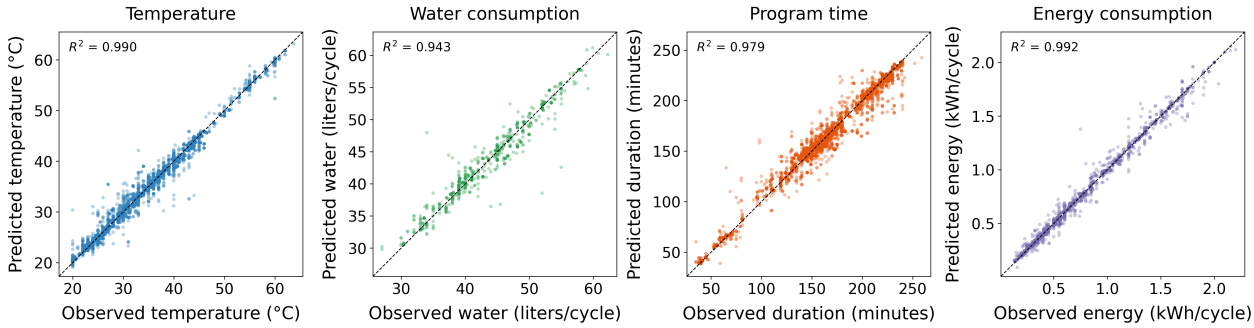


Figure B15: Program inputs out-of-sample fit

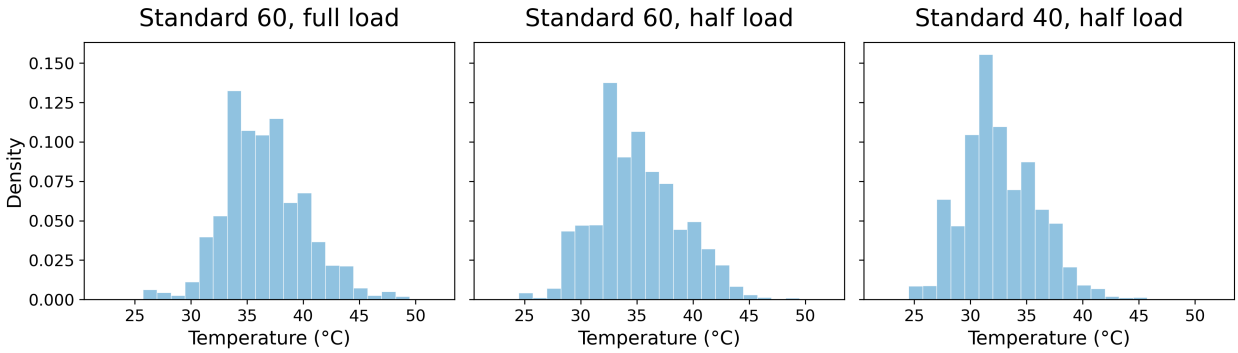


Figure B16: Distribution of standard program temperature

C Laundry-habits survey

Dear Sir or Madam,

This questionnaire has been designed by the University of Wisconsin–Madison (USA). It is intended for: (1) consumers who own a washing machine with the post-2021 A–G energy label, and (2) the primary person responsible for doing the laundry in the household. Completing the questionnaire will take approximately 5–10 minutes.

1. Who is mainly responsible for doing the laundry in your household? Please choose one option.

   - ☐ I am mainly responsible.
   - ☐ I share responsibility with someone else.

- ☐ Someone else in the household is mainly responsible.

2. In which federal state (Bundesland) do you live? Please choose one option.

   - ☐ Baden-Württemberg
   - ☐ Bavaria (Bayern)
   - ☐ Berlin
   - ☐ Brandenburg
   - ☐ Bremen
   - ☐ Hamburg
   - ☐ Hesse (Hessen)
   - ☐ Lower Saxony (Niedersachsen)
   - ☐ Mecklenburg-Vorpommern
   - ☐ North Rhine-Westphalia (Nordrhein-Westfalen)
   - ☐ Rhineland-Palatinate (Rheinland-Pfalz)
   - ☐ Saarland
   - ☐ Saxony (Sachsen)
   - ☐ Saxony-Anhalt (Sachsen-Anhalt)
   - ☐ Schleswig-Holstein
   - ☐ Thuringia (Thüringen)

3. How many people are living in your household, including yourself? Please choose one option.

   - ☐ 1 person
   - ☐ 2 people
   - ☐ 3 people
   - ☐ 4 people
   - ☐ More than 4 people

4. Approximately what is your household's monthly income?

---

5. What is the **brand and model number** of your washing machine?
   (*You can usually find the model number on the energy label or on the* Typenschild */ rating plate—often inside the door opening/door frame, on the back of the machine, or in the user manual/receipt. For Bosch/Siemens, it may be listed as the "E-Nr."; for AEG/Electrolux, it may be the "PNC".*)

   _____

6. What is the energy efficiency class of your washing machine (*A*–*G* label)? Please choose one option.

   ☐ A
   ☐ B
   ☐ C
   ☐ D
   ☐ E
   ☐ I'm not sure.

7. What is the rated load capacity of your washing machine (in kg)? Please choose one option.

   ☐ 6 kg or less
   ☐ 7 kg
   ☐ 8 kg
   ☐ 9 kg
   ☐ 10 kg or more
   ☐ I'm not sure.

8. When did you purchase your washing machine? Please choose one option.

2021:	☐Q1	☐Q2	☐Q3	☐Q4
2022:	☐Q1	☐Q2	☐Q3	☐Q4
2023:	☐Q1	☐Q2	☐Q3	☐Q4
2024:	☐Q1	☐Q2	☐Q3	☐Q4
2025:	☐Q1	☐Q2	☐Q3	☐Q4

9. When purchasing a washing machine, which information do you use to judge energy efficiency? Please choose the option that best describes you.

☐ I use the energy efficiency class (*A*–*G*) as a quick filter.

☐ I mainly compare the energy consumption values (kWh per 100 cycles) on the energy label and choose the lower value.

☐ I look at both the class (*A*–*G*) and the kWh value, but I do not calculate electricity costs.

☐ I convert the kWh value into an electricity cost estimate.

☐ I rarely or never use the energy efficiency class or the kWh value.

☐ I'm not sure.

10. *Only if you selected in the previous question that you convert the kWh value into an electricity cost estimate:*
    Approximately how much do you think your washing machine costs in electricity per year? Please choose one option.

    ☐ Less than €10

    ☐ €10–€20

    ☐ €21–€30

    ☐ €31–€40

    ☐ €41–€60

    ☐ €61–€80

    ☐ More than €80

    ☐ I'm not sure.

11. What would you expect a washing machine with the highest energy efficiency class to save on? Please choose one option.

    ☐ Energy in all programmes

    ☐ Energy and water in all programmes

    ☐ Energy in some selected programmes

    ☐ Energy and water in some selected programmes

12. How often do your laundry loads have the following stain levels? *"Stains" include visible marks (e.g., food, grease, mud) and/or strong odours (e.g., sweat). Please choose one option per row.*

	Not at all	**Once per 4 weeks**	**Once per 3 weeks**	**Once per 2 weeks**	**1–2x per week**	**3–4x per week**	**5–6x per week**	**> 6x per week**
Low	☐	☐	☐	☐	☐	☐	☐	☐
Medium	☐	☐	☐	☐	☐	☐	☐	☐
High	☐	☐	☐	☐	☐	☐	☐	☐

Low stains: No visible stains; lightly worn; no strong odour.

Medium stains: Some visible dirt/marks or noticeable odour (typical everyday laundry).

High stains: Multiple items with heavy/visible stains (e.g., mud/grease/food) and/or strong odour.

13. How often do you use the following washing programmes for washing your clothes? Please choose one option per row.

	Not at all	Once per 4 weeks	Once per 3 weeks	Once per 2 weeks	1–2x per week	3–4x per week	5–6x per week	>6x per week
Cotton 20 °C / Cold wash	☐	☐	☐	☐	☐	☐	☐	☐
Cotton 30 °C	☐	☐	☐	☐	☐	☐	☐	☐
Cotton 40 °C	☐	☐	☐	☐	☐	☐	☐	☐
Cotton 60 °C	☐	☐	☐	☐	☐	☐	☐	☐
Cotton 90 °C	☐	☐	☐	☐	☐	☐	☐	☐
Synthetic / Easy care 30 °C / 40 °C	☐	☐	☐	☐	☐	☐	☐	☐
Eco 40-60 (Eco) / Energy saving	☐	☐	☐	☐	☐	☐	☐	☐
Silk/Wool	☐	☐	☐	☐	☐	☐	☐	☐
Quick wash / Short	☐	☐	☐	☐	☐	☐	☐	☐
Mix (all fabrics)	☐	☐	☐	☐	☐	☐	☐	☐
Other	☐	☐	☐	☐	☐	☐	☐	☐

D Proofs and Extensions for Regulation-Induced Energy Shifting

D.1 Proof of Proposition 1

Under logit program choice, expected utility is

$$\mathcal{I}(d_E, d_C) = \log\left(e^{\beta d_E} + e^{\beta d_C}\right).$$

Let s_r denote the corresponding program-choice probability. Differentiating yields

$$\frac{\partial \mathcal{I}}{\partial d_E} = \beta s_E, \qquad \frac{\partial \mathcal{I}}{\partial d_C} = \beta s_C, \qquad \frac{\partial^2 \mathcal{I}}{\partial d_C \partial d_E} = -\beta^2 s_E s_C.$$

Product demand is $Q(z)$, where

$$z = P + \kappa e_E(d_E) - \theta \mathcal{I}(d_E, d_C),$$

with $Q' < 0$, $\kappa > 0$, and $\theta > 0$. Throughout, partial derivatives of Q with respect to program inputs refer to the composite demand function $Q(z)$. Applying the chain rule gives

$$\frac{\partial^2 Q}{\partial d_C \partial d_E} = \theta\beta s_C \left[\beta s_E \left(\theta Q''(z) + Q'(z)\right) - \kappa e_E'(d_E) Q''(z)\right].$$

The sign depends on the curvature of product demand and the direct contribution of Eco energy consumption. Proposition 1 imposes $\frac{\partial^2 Q}{\partial d_C \partial d_E} < 0$ as a sufficient condition for decreasing differences in demand.

With the Eco-duration cap binding, $d_E^* = \bar{d}_E$. The Cotton first-order condition is

$$\frac{\partial Q}{\partial d_C} = C_C'(d_C).$$

Implicit differentiation with respect to $\bar{d}_E$ yields

$$\frac{\partial d_C^*}{\partial \bar{d}_E} = \frac{\frac{\partial^2 Q}{\partial d_C \partial d_E}}{C_C'' - \frac{\partial^2 Q}{\partial d_C^2}} < 0,$$

where the inequality follows from decreasing differences in demand and the strict second-order condition $C_C'' - \frac{\partial^2 Q}{\partial d_C^2} > 0$.

Because Cotton energy consumption satisfies $e_C'(d_C) > 0$,

$$\frac{\partial e_C(d_C^*)}{\partial \bar{d}_E} = e_C'(d_C^*) \frac{\partial d_C^*}{\partial \bar{d}_E} < 0.$$

Thus, tightening the Eco-duration cap increases optimal Cotton duration and energy consumption, establishing Proposition 1. □

D.2 Extension: Joint Eco Constraints

I extend the stylized model to allow firms to choose a second Eco input x_E, such as temperature, while Cotton duration remains its only design input. Eco energy consumption is $e_E(d_E, x_E)$, with $\frac{\partial e_E}{\partial d_E} > 0$ and $\frac{\partial e_E}{\partial x_E} > 0$. Consumer utility from Eco becomes $v_E = \beta d_E + \beta_x x_E + \varepsilon_E$, while Cotton utility remains $v_C = \beta d_C + \varepsilon_C$. Expected program utility is

$$\mathcal{I}(d_E, x_E, d_C) = \log\left(e^{\beta d_E + \beta_x x_E} + e^{\beta d_C}\right).$$

Taking price as given and normalizing the markup to one, the firm solves

$$\begin{aligned} \max_{d_E, x_E, d_C} \quad & Q\left(P + \kappa e_E(d_E, x_E) - \theta \mathcal{I}(d_E, x_E, d_C)\right) - C_E(d_E, x_E) - C_C(d_C) \\ \text{s.t.} \quad & d_E \le \bar{d}_E, \qquad e_E(d_E, x_E) \le \bar{e}. \end{aligned}$$

Proposition 2 (Energy Shifting under Joint Eco Constraints)**.** *Suppose both Eco constraints bind and Cotton duration has an interior local optimum satisfying the second-order condition. If* $\frac{\partial^2 Q}{\partial d_C \partial d_E} < 0$ *and* $\frac{\partial^2 Q}{\partial d_C \partial x_E} < 0$*, tightening the Eco-energy cap increases optimal Cotton duration and energy consumption, holding the duration cap fixed. Jointly tightening both caps also increases Cotton duration and energy consumption if the second Eco input does not increase.*

Proof. Under logit program choice, the cross-partials of expected utility are

$$\frac{\partial^2 \mathcal{I}}{\partial d_C \partial d_E} = -\beta^2 s_E s_C, \qquad \frac{\partial^2 \mathcal{I}}{\partial d_C \partial x_E} = -\beta \beta_x s_E s_C.$$

As in Proposition 1, the corresponding demand cross-partials also depend on the curvature of Q and the direct contribution of Eco energy consumption. The proposition assumes both are negative.

Because both constraints bind,

$$d_E^* = \bar{d}_E, \qquad e_E(\bar{d}_E, x_E^*) = \bar{e}.$$

Differentiating the energy constraint gives

$$\mathrm{d}x_E^* = \frac{\mathrm{d}\bar{e} - \frac{\partial e_E}{\partial d_E}\mathrm{d}\bar{d}_E}{\frac{\partial e_E}{\partial x_E}}. \tag{D.1}$$

The Cotton first-order condition is

$$\frac{\partial Q}{\partial d_C} = C_C'(d_C).$$

Implicit differentiation yields

$$\left(C_C'' - \frac{\partial^2 Q}{\partial d_C^2}\right) \mathrm{d}d_C^* = \frac{\partial^2 Q}{\partial d_C \partial d_E} \mathrm{d}\bar{d}_E + \frac{\partial^2 Q}{\partial d_C \partial x_E} \mathrm{d}x_E^*.$$

Substituting equation (D.1) gives

$$\begin{aligned} \mathrm{d}d_C^* = \frac{1}{C_C'' - \frac{\partial^2 Q}{\partial d_C^2}} \Bigg[\left(\frac{\partial^2 Q}{\partial d_C \partial d_E} - \frac{\frac{\partial e_E}{\partial d_E}}{\frac{\partial e_E}{\partial x_E}} \frac{\partial^2 Q}{\partial d_C \partial x_E} \right) \mathrm{d}\bar{d}_E \\ + \frac{\frac{\partial^2 Q}{\partial d_C \partial x_E}}{\frac{\partial e_E}{\partial x_E}} \mathrm{d}\bar{e} \Bigg]. \end{aligned} \tag{D.2}$$

Holding the duration cap fixed,

$$\frac{\partial d_C^*}{\partial \bar{e}} = \frac{\frac{\partial^2 Q}{\partial d_C \partial x_E}}{\frac{\partial e_E}{\partial x_E}\left(C_C'' - \frac{\partial^2 Q}{\partial d_C^2}\right)} < 0.$$

The inequality follows from decreasing differences between Cotton duration and the second Eco input, increasing Eco energy consumption in x_E, and the strict second-order condition. Thus, tightening the energy cap reduces x_E and increases optimal Cotton duration.

Now consider jointly tightening both caps, with $\mathrm{d}\bar{d}_E < 0$ and $\mathrm{d}\bar{e} < 0$. Equation (D.1) implies that the second Eco input does not increase if

$$\mathrm{d}\bar{e} \leq \frac{\partial e_E}{\partial d_E} \mathrm{d}\bar{d}_E.$$

Under this condition, $\mathrm{d}d_E^* < 0$ and $\mathrm{d}x_E^* \leq 0$. Because both demand cross-partials are negative, the Cotton first-order condition implies $\mathrm{d}d_C^* > 0$.

Finally, since $e_C'(d_C) > 0$, any increase in Cotton duration also increases Cotton energy consumption. This establishes both claims. □

Holding the energy cap fixed instead gives

$$\left.\frac{\partial d_C^*}{\partial \bar{d}_E}\right|_{\bar{e}} = \frac{\frac{\partial^2 Q}{\partial d_C \partial d_E} - \frac{\frac{\partial e_E}{\partial d_E}}{\frac{\partial e_E}{\partial x_E}} \frac{\partial^2 Q}{\partial d_C \partial x_E}}{C_C'' - \frac{\partial^2 Q}{\partial d_C^2}}.$$

The sign is ambiguous because tightening the duration cap permits an offsetting increase in the second Eco input. Joint constraints strengthen energy shifting when they prevent such compensation within Eco.

D.3 Socially Efficient Program Design

I extend the stylized model to compare private and socially efficient program designs. Consumers perceive Eco energy consumption as representative of both programs. To isolate the resulting information distortion, suppose the social cost of electricity equals its valuation at purchase. Since θ converts program utility into price-equivalent units, define $\eta = \kappa/\theta$.

Under consumers' beliefs, expected program utility net of perceived electricity costs is

$$\mathcal{I}^P(d_E, d_C) = \mathcal{I}(d_E, d_C) - \eta e_E(d_E).$$

A social planner instead accounts for actual program-specific electricity consumption:

$$\mathcal{I}^S(d_E, d_C) = \log\left(e^{\beta d_E - \eta e_E(d_E)} + e^{\beta d_C - \eta e_C(d_C)}\right).$$

The socially efficient Cotton usage share is

$$s_C^S = \frac{e^{\beta d_C - \eta e_C(d_C)}}{e^{\beta d_E - \eta e_E(d_E)} + e^{\beta d_C - \eta e_C(d_C)}}.$$

Thus, private and socially efficient program choices differ by

$$\log\frac{s_C}{s_E} - \log\frac{s_C^S}{s_E^S} = \eta\left[e_C(d_C) - e_E(d_E)\right].$$

When Cotton consumes more electricity than Eco, private choices allocate too many cycles to Cotton relative to the social benchmark.

To characterize socially efficient design, let

$$z^P = P + \kappa e_E(d_E) - \theta\mathcal{I}(d_E, d_C), \qquad z^S = P - \theta\mathcal{I}^S(d_E, d_C).$$

Let consumer surplus under full information be

$$CS(z^S) = \int_{z^S}^{\infty} Q(u)\, du.$$

Maintaining the unit-markup normalization and abstracting from other distortions, the social planner chooses program durations to maximize total surplus:

$$\max_{d_E, d_C} \quad CS(z^S) + Q(z^S) - C_E(d_E) - C_C(d_C) \qquad \text{s.t.} \qquad d_E \le \bar{d}_E.$$

This objective includes consumer surplus, operating profits, and design costs, with actual electricity costs incorporated into $\mathcal{I}^S$.

The socially optimal Cotton duration satisfies

$$\theta \left[Q(z^S) - Q'(z^S) \right] s_C^S \left[\beta - \eta \frac{\partial e_C}{\partial d_C} \right] = C_C'(d_C^S). \tag{D.3}$$

By contrast, the firm's private optimum satisfies

$$-\theta Q'(z^P) \beta s_C = C_C'(d_C^P). \tag{D.4}$$

The social planner accounts for the additional electricity consumed when Cotton duration increases. The firm does not face this direct energy penalty because consumers perceive Cotton energy consumption as equal to Eco energy consumption. However, the two objectives also differ in their valuation of product demand and consumer surplus.

To compare the optimal designs, hold Eco duration fixed at the binding regulatory cap. Suppose the social objective is strictly concave in Cotton duration. Evaluated at the private optimum d_C^P, a sufficient condition for excessive Cotton duration is

$$\left[Q(z^S) - Q'(z^S) \right] s_C^S \left[\beta - \eta \frac{\partial e_C}{\partial d_C} \right] < -Q'(z^P) \beta s_C. \tag{D.5}$$

Under this condition, the social marginal return to extending Cotton is smaller than the private marginal return at d_C^P. Strict concavity then implies

$$d_C^S < d_C^P, \qquad e_C(d_C^S) < e_C(d_C^P).$$

For example, when $\beta > 0$, the condition is satisfied if $\eta \frac{\partial e_C}{\partial d_C} \geq \beta$ at the private optimum: the social marginal electricity cost of extending Cotton is at least as large as its marginal duration benefit. More generally, whether private designs are excessively energy intensive depends on the relative marginal benefits in equation (D.5).

E Descriptive Evidence

E.1 Energy Equation

I estimate the pooled energy equation

$$\log e_{jo} = z_{f(j),t_j^e,c_j} + \delta_{g(o)} + \boldsymbol{\theta}' \log \mathbf{x}_{jo} + \varepsilon_{jo}, \tag{E.1}$$

where $\mathbf{x}_{jo} = (L_{jo}, T_{jo}, d_{jo}, w_{jo})$ contains load, temperature, duration in hours, and water use. Input slopes are common across programs. Program effects $\delta_{g(o)}$ distinguish Eco/Standard from Cotton.

Technology $z_{f(j),t_j^e,c_j}$ is a firm–entry-quarter–energy-class effect shared across categories; lower values imply lower energy use at fixed inputs.

I use unweighted fixed-effects OLS. The sample includes eligible imputed Eco/Standard observations and fully nonimputed Cotton20/40/60 observations. Wool and Easy Care are excluded. Table A1 reports conventional OLS standard errors and adjusted R^2. Predicted energy is obtained by exponentiating the fitted equation with $\varepsilon_{jo} = 0$, without a retransformation correction.

Table A1: Pooled Energy Equation Estimates

	Estimate	Std. error
Log load	0.168	0.008
Log temperature	1.599	0.007
Log duration (hours)	0.360	0.013
Log water	0.333	0.015
Program FE	✓	
Firm–cohort–class FE	✓	
Observations	8,221	
Adjusted R^2	0.942	

Notes: Unweighted OLS with common input slopes, program fixed effects, and firm–entry-quarter–energy-class fixed effects. Program fixed effects distinguish two groups: Eco/Standard and Cotton, with Eco/Standard omitted. Cohort denotes entry quarter. Eco/Standard observations may include imputed inputs; Cotton20/40/60 observations must be nonimputed on all five variables. Easy Care and Wool are excluded. Standard errors are conventional homoskedastic OLS standard errors, accounting for absorbed fixed effects.

F Estimation and Computational Implementation

F.1 Parameter Snapshots and Profit Approximation

The latest engineering regression pools Eco/Standard and nonimputed Cotton observations with common slopes, as described in Appendix E.1. The completed supply and counterfactual calculations instead retain the frozen engineering runtime used to construct their profit labels. Its Eco/Standard elasticities for load, temperature, duration, and water are 0.1984, 1.6492, 0.1625, and 0.3439. Cotton energy is evaluated with the separately saved Cotton relation. Consequently, the completed policy results are conditional on this earlier engineering mapping, not a recomputation

Table A2: Quarterly Energy Use and Input Levels Relative to 2020q2

	Energy use		Input level				
Quarter	Level	Δ	Load size	Time	Temp.	Water	Technology
Panel A: Eco							
2020q2	0.610	—	4.88	2.94	34.1	46.3	0.025
2020q4	0.624	0.015	4.89	2.97	34.1	45.8	0.024
2021q2	0.615	0.005	4.91	2.98	33.9	45.9	0.017
2021q4	0.591	-0.018	4.94	2.98	33.2	46.0	0.007
2022q2	0.577	-0.033	4.97	2.99	32.8	46.2	0.001
2022q4	0.556	-0.053	4.95	2.98	32.3	46.1	-0.002
2023q2	0.544	-0.066	4.96	2.98	31.9	46.2	-0.008
2023q4	0.524	-0.086	4.99	2.98	31.4	46.4	-0.015
Panel B: Cotton 60° C							
2020q2	1.483	—	8.20	3.14	55.1	75.3	0.052
2020q4	1.532	0.049	8.13	3.23	56.0	80.6	0.036
2021q2	1.548	0.066	8.16	3.24	56.2	81.4	0.030
2021q4	1.563	0.080	8.23	3.31	56.0	81.3	0.018
2022q2	1.588	0.106	8.30	3.31	56.4	82.3	0.011
2022q4	1.611	0.128	8.28	3.33	56.7	83.1	0.007
2023q2	1.621	0.139	8.33	3.33	56.9	83.7	-0.000
2023q4	1.633	0.150	8.41	3.32	57.1	83.8	-0.010

Notes: Energy use is the unweighted mean kWh per cycle across distinct products offered in each retained quarter. Only the second and fourth quarters of each year are reported. Δ is the quarterly mean minus the common 2020q2 mean; negative values indicate lower energy use. Input levels are quarterly portfolio means: load size is in kg, time in hours, temperature in °C, water in liters, and technology is the mean product fixed-effect index. The source is `wm_demand_quarterly_sku.dta`; observations with `firm == SAMSUNG` are excluded. Repeated rows are reduced to the first valid `product_ids`–program observation within each quarter and are not averaged. The product samples need not be balanced across quarters, so entry and exit contribute to portfolio change. Eco uses the regulatory full-, half-, and quarter-load weights at the product-quarter level. Technology is `alpha_product_hat`, which enters log energy with coefficient +1, so lower values reduce fitted energy. The mechanical-action term is excluded.

under the pooled slopes in Table A1. The demand and marginal-cost parameters used downstream agree with Tables 4 and 5.

Demand integration uses five nodes for each of income, stain share, and the two random program tastes, giving the 625 agents per market used in estimation. Profit labels integrate over 32 joint demand and cost shock draws and re-solve Bertrand prices. Own-entrant input derivatives account for the price response through implicit differentiation of the price first-order conditions. Incumbents affect profits but are not treated as redesign opportunities.

The operating-profit approximation is a permutation-invariant set network with shared product encoders and pooled firm and market features (Zaheer et al., 2017). Smooth activations allow its own-input derivatives to enter the training loss alongside profit levels. The saved network has width 96 and 63 input features. Its 32,000 labeled profiles are split into 22,400 training, 4,800 validation, and 4,800 reserved test profiles. Observed profiles are split within quarter; simulated profiles are split by entire path to avoid placing quarters from one path in both training and validation. Training uses standardized profit and derivative losses, float64 arithmetic, AdamW, and an initial learning rate of 3×10^{-4}. Figure F1 reports the saved validation-loss sequence. The checkpoint's training-complete and acceptance flags remain false; counterfactual-support accuracy has not been established.

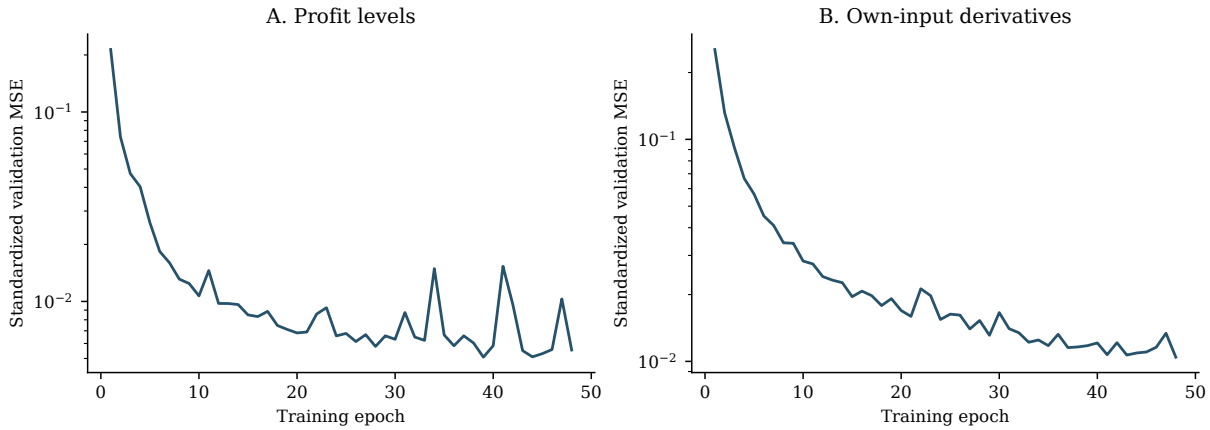


Figure F1: Operating-Profit Approximation: Saved Validation Loss

Notes: The figure reports standardized mean squared errors on the saved validation split across 48 recorded epochs. The vertical scales are logarithmic and differ across panels. Loss includes firm-profit levels and own-entrant derivatives of the same network. The saved checkpoint is provisional; this figure is not a final reserved-test or counterfactual-support validation.

A second set network predicts optimized inputs and profits net of design costs from pre-choice portfolios. Adjustable entrant inputs are prediction targets, not observed features. Entry estimation uses the saved epoch-90 checkpoint trained on design-cost evaluation 44. The reported design estimates and realized counterfactual design refinement use evaluation 47. Eco and Cotton duration coefficients are identical across these two evaluations, but technology curvature is 1.816×10^8 in

evaluation 44 versus 5.448×10^8 in evaluation 47, and the Eco-temperature linear term is 45.27 versus 45.92. Realized-design refinement does not update the entry payoffs from the earlier checkpoint; validation of both checkpoints remains incomplete.

F.2 Reference Designs, Samples, and Moments

Reference designs follow the model: duration remains anchored to the matched A^{+++} benchmark, including fitted baseline Cotton-duration references where supplied by the baseline optimization. Simultaneous technology and temperature donors are averaged. Estimation uses fixed reference data; a simulated path carries its own optimized donor history forward.

The criterion contains five mean marginal profits and three mean optimized durations for A^{+++} products. It also includes four destination-class means for each input group and 416 individual input changes. Independent first-class introductions enter the design targets. Repeated-class copies, fixed coordinates, Samsung Cotton 60°C duration, and first-transformation Eco temperature are excluded from their corresponding targets. Temperature-change moments exclude A^{+++} transitions. Technology-change moments include those transitions but require observed technology below its reference; this selection is based on observed changes, not fitted improvements. Full- and half-load Eco coordinates are pooled, and technology enters once per product. The input-change counts are 77 for Cotton 40°C duration, 67 for Cotton 60°C, 154 for Eco duration, 70 for Eco temperature, and 48 for technology.

The baseline-gradient sample retains the archived trimming of the seven largest Cotton 60°C gradients, Cotton 40°C products 741 and 774, and seven flagged Eco-duration coordinates. These are features of the completed estimator, rather than exclusions inferred from the current fit. Within each input group, the baseline weight is 1/3 and the design weight 2/3. Duration baseline weight is split equally between marginal-profit and optimized-duration restrictions. Design weight is split equally between class means and individual changes, with equal weight within each block. Each group receives one-fifth of the total criterion. Fixed scales are the empirical input standard deviations for design moments and the stored marginal-profit scales for baseline gradient moments.

Table A3 reports selected levels and energy diagnostics; Table A4 summarizes the estimation blocks. Energy rows were computed after the selected fit and did not enter its 444-moment criterion. Figure F2 uses saved eligible input predictions and eligible A^{+++} baseline predictions, retaining fitted errors and boundary observations.

Table A3: Selected Design-Cost Moments

Moment	Observed	Fitted
A. Baseline A^{+++} moments		
Cotton 40°C duration: marginal profit	29.09	27.14
Cotton 60°C duration: marginal profit	61.89	59.20
Eco duration: marginal profit	-0.93	-1.55
Eco temperature: marginal profit	45.60	45.92
Technology index: marginal profit	-263,089.43	-263,089.43
Cotton 40°C duration (min)	176.46	176.94
Cotton 60°C duration (min)	175.81	176.85
Eco duration (min)	221.29	220.28
Eco energy (kWh/cycle)	0.741	0.743
Cotton 40°C energy (kWh/cycle)	0.991	0.947
Cotton 60°C energy (kWh/cycle)	1.421	1.523

B. Mean inputs and energy consumption by destination class (Observed / Fitted)

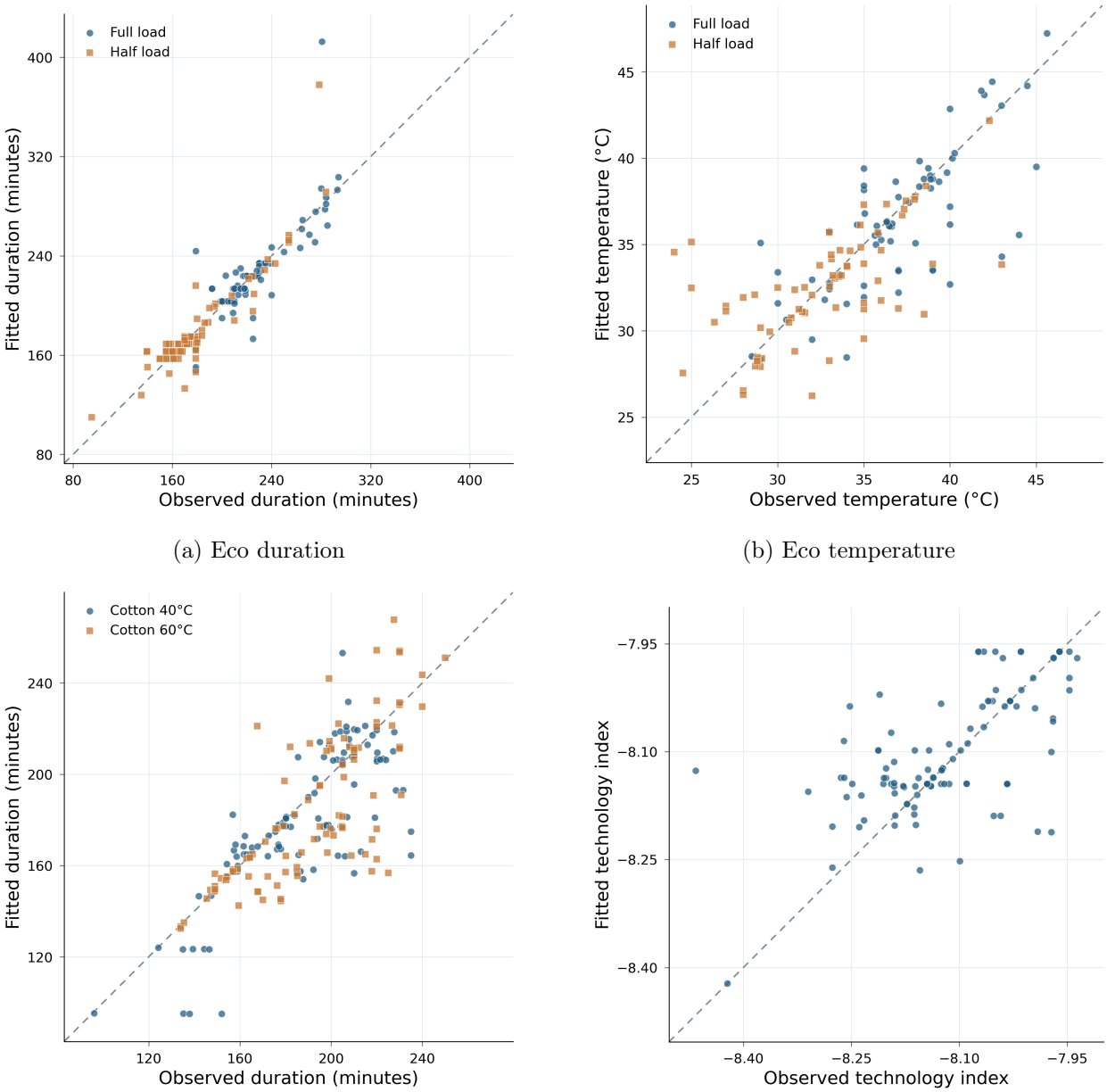
	A	*B*	*C*	*D*
Cotton 40°C duration (min)	191.20 / 187.23	194.75 / 179.11	199.48 / 188.52	183.94 / 168.91
Cotton 60°C duration (min)	194.38 / 187.56	195.73 / 185.36	202.95 / 188.97	201.65 / 186.85
Eco duration (min)	191.20 / 192.00	190.05 / 189.15	190.21 / 191.43	188.74 / 187.76
Eco temperature (°C)	31.46 / 32.02	35.43 / 33.36	36.40 / 35.04	38.50 / 33.72
Technology index	-8.1250 / -8.1292	-8.1425 / -8.1048	-8.1313 / -8.1024	-8.1112 / -8.0814
Eco energy (kWh/cycle)	0.482 / 0.472	0.546 / 0.526	0.630 / 0.578	0.723 / 0.636
Cotton 40°C energy (kWh/cycle)	1.120 / 1.055	1.059 / 0.966	1.133 / 1.002	0.998 / 0.905
Cotton 60°C energy (kWh/cycle)	1.672 / 1.700	1.565 / 1.574	1.600 / 1.588	1.550 / 1.490

Notes: Values use the latest completed estimates, before re-estimation with energy targets. Panel A reports A^{+++} baselines; Eco energy averages full and half loads for products with observed energy. Panel B pools capacities with equal product weights and uses regulatory Eco load weights. Repeated-class copies and Samsung Cotton 60°C are excluded from Panel B. Baseline marginal profits are euros per input unit. Energy rows are diagnostics.

Table A4: Design Fit: Levels and Adjustments

Input	n	Observed	Fitted	RMSE
A. Destination-class means				
Cotton 40°C duration	4	192.34	180.94	12.31
Cotton 60°C duration	4	198.68	187.19	11.92
Eco duration	4	190.05	190.09	0.99
Eco temperature	4	35.45	33.54	2.71
Technology	4	-8.127	-8.104	0.028
B. Introduction-level changes				
Cotton 40°C duration	77	14.08	3.44	24.41
Cotton 60°C duration	67	12.60	1.64	27.25
Eco duration	154	-41.56	-41.42	5.80
Eco temperature	70	-2.78	-3.61	4.28
Technology	48	-0.078	−0.000059	0.099

Notes: Panel A gives equal-weight means and RMSEs over four destination-class cells for each input. Panel B uses individual changes from fixed references; its sample counts total 416. Duration is in minutes, temperature in degrees Celsius, and technology in index units. The technology-change sample retains observed improvements only. These are in-sample target comparisons; class-mean fit and individual-change fit use different units of observation.



(a) Eco duration

(b) Eco temperature

(c) Cotton duration

(d) Technology

Figure F2: Observed and Fitted Program Inputs

Notes: Each point compares observed and fitted values for an eligible product introduction or an A^{+++} baseline product. Colors distinguish full and half loads in the Eco panels and Cotton 40°C and Cotton 60°C in the Cotton panel. Temperature observations for new-label products exclude initial A^{+++} transformations. Repeated-class copies and Samsung Cotton 60°C introductions are excluded. Fixed quarter-load inputs are not plotted. Dashed lines denote equality.

F.3 Constrained Design Solver

For each candidate cost vector, the inner loop optimizes free first-introduction inputs conditional on rival designs, updates firms in turn, and checks unilateral improvements. It uses at most 20 passes and 300 SLSQP iterations, with solver tolerance 10^{-8} and fit tolerance 10^{-4}. Positive curvatures are searched in logs. The outer search completed 50 evaluations and selected evaluation 47, with objective 0.3268: 0.00025 from baseline restrictions, 0.05876 from class means, and 0.26776 from introduction-level changes. The inner solution is locally verified; outer convergence, global optimality, and uniqueness remain unestablished.

The implementation retains calibrated constraint offsets and margins of five minutes for duration, one EEI unit at the upper class boundary, and one unit of weighted Eco water. Fixed water, quarter-load attributes, and copied designs are audited separately rather than freely adjusted. The selected fit records four bound hits, 182 copied-design policy audit entries, and 61 fixed-water or quarter-load audit entries. Strict nominal feasibility is not assured; Appendix G reports the corresponding counterfactual audits.

F.4 Portfolio Conditionals, Gibbs Sampling, and NPL

For each complete own portfolio a_{ft}, approximate the expected payoff defined in Section 6.2 by

$$\overline{W}_{ft}(a_{ft};\mathcal{P}_{-f,t}) \simeq \frac{1}{B}\sum_{b=1}^{B} W_{ft}(a_{ft}, a_{-f,t}^{(b)}) - C_{ft}^{\text{entry}}(a_{ft}), \tag{F.1}$$

where $a_{-f,t}^{(b)}$ is a complete rival-portfolio draw. The mixed strategy is

$$\mathcal{P}_{ft}(a_{ft}) = \frac{\exp\{\overline{W}_{ft}(a_{ft};\mathcal{P}_{-f,t})/\sigma\}}{\sum_{b_{ft}\in\mathcal{F}_f(J_{f,t-1})}\exp\{\overline{W}_{ft}(b_{ft};\mathcal{P}_{-f,t})/\sigma\}}. \tag{F.2}$$

Conditioning on the firm's other lines cancels the portfolio normalizer and yields equation (23). The product of observed line conditionals defines a composite likelihood, not the joint portfolio likelihood.

Given rival beliefs, each firm's Gibbs chain visits its eligible lines in random order, draws an action from the line conditional, and records the complete portfolio after each sweep. Payoffs are averaged over rival configurations before the logit transformation. The estimation discards 100 burn-in sweeps, retains 64 portfolios with thinning one, and uses 64 rival draws. Sampling complete rival portfolios preserves dependence within firms; independence across firms is the maintained mixed-strategy assumption. Other own-line actions are fixed at their observed values in the likelihood and at their current sampled values during Gibbs updates.

For fixed beliefs, the inverse-scale and cost-over-scale parameterization makes the logit index affine. The constrained likelihood step uses three optimization starts and enforces nonnegative launch costs on the estimation support. Beliefs and parameters are then updated until relative parameter change is below 0.01 and the largest monitored conditional-probability gap is below 0.02 for two consecutive iterations, with a maximum of 20. Monitoring covers observed and probe own configurations. An independent-seed update checks monitored probabilities, but does not establish full-chain mixing.

The completed estimation converges in four iterations. Its final independent probability gap is 0.00337. Two launch-cost restrictions bind, while the shock-scale lower bound does not. The likelihood has 469 observations from a broader risk set of 510 records. Table A5 separates entry participation from class choice. Its probabilities condition on observed own actions and should not be interpreted as unconditional simulation frequencies.

Table A5: Entry Participation and Class Choice

Action	Observed (%)	Fitted (%)
No entry	63.97	66.21
Class A	20.04	15.26
Class B	5.12	6.48
Class C	5.12	6.24
Class D	5.76	5.81
Any entry	36.03	33.79
Eligible line–quarters	469	
Conditional log loss	0.8776	
Entry Brier score	0.1972	
Multiclass Brier score	0.4600	

Notes: Fitted shares average conditional choice probabilities at observed actions on other own lines and simulated rival portfolios. All percentages use eligible line–quarters as the denominator, including no entry. Class percentages do not condition on introducing a product. The sample pools transition and post-policy observations.

F.5 Conditional Inference

The reported design-cost standard errors use a local numerical Jacobian of the 444 standardized moments with ten free parameters. The stored Jacobian has full column rank. The nonclustered HC1 covariance treats quarter–product–input contributions as independent while retaining their joint contributions to overlapping moments. The firm-clustered alternative aggregates those contributions within each of seven firms. Both condition on demand, engineering, reference construction,

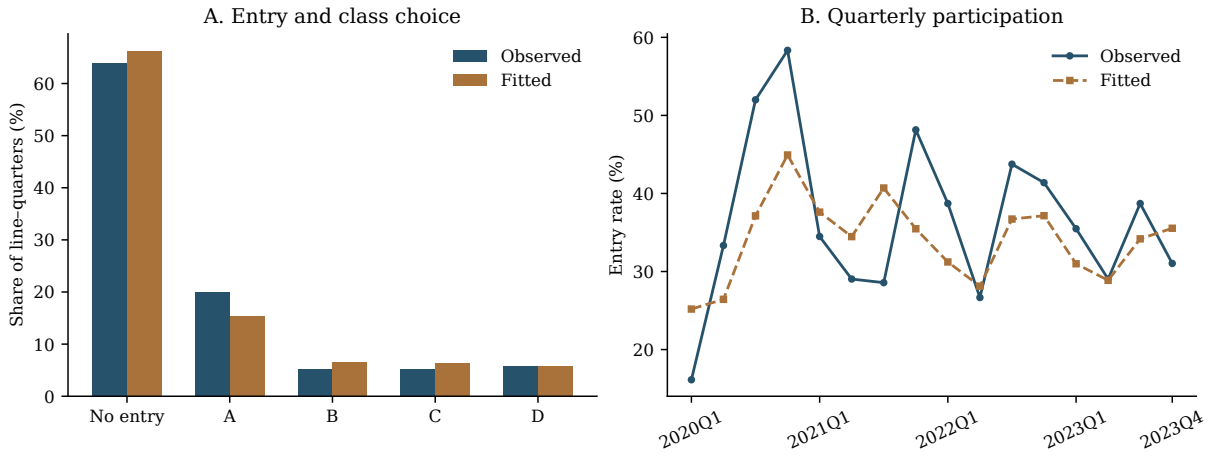


Figure F3: Entry Model Fit

Notes: The 469 eligible line–quarters pool transition and post-policy observations. Fitted shares average line-conditional probabilities at observed own other-line actions, integrating over simulated rival portfolios. The same eligible sample supplies each quarterly rate. These probabilities are not unconditional frequencies from simulated entry histories.

the profit approximation, and the selected local solutions. Neither incorporates the incomplete outer search or uncertainty from changing active constraints.

For entry costs, the sandwich uses the conditional composite-likelihood Hessian and score contributions, projected onto the tangent space of the two binding restrictions. The 13-parameter problem therefore has an 11-dimensional local tangent space. The nonclustered version uses individual line–quarter scores; the alternative aggregates scores by firm. A delta-method transformation yields euro-valued costs and the shock scale. Both condition on fitted beliefs, the selected active set, the intermediate payoff predictor, and upstream estimates. They omit uncertainty from belief feedback and active-set selection. With only seven firm clusters, clustered estimates are a sensitivity comparison rather than a definitive small-sample correction.

G Counterfactual Computation and Validation

G.1 Policy implementation and state transitions

The common simulation starts from the observed 2020Q1 incumbent set and applies the counterfactual rules to eligible introductions throughout 2020Q1–2023Q4, including the transition year. Demand during 2020 is evaluated using the model's old-/new-label regime mixture; later quarters use the post-policy regime. Entry costs use the fixed estimates for launch, quarter, incumbent count, time since the preceding introduction, and class switching. Opportunities and counts are computed from the previous state even after current survivors have been formed. All nonstationary

Table A6: Conditional Standard Errors: Clustering Sensitivity

Parameter	Estimate	HC1 SE	Firm-cluster SE
A. Design costs (euros; technology rescaled)			
Cotton 40°C curvature	35.47	7.50	12.04
Cotton 60°C curvature	69.31	74.49	81.50
Eco-duration curvature	1.31	1.75	2.12
Eco-temperature curvature	97.61	304.51	319.05
Technology curvature	544.76	3,581.98	2,855.53
Cotton 40°C linear	27.14	7.04	8.47
Cotton 60°C linear	59.20	10.64	19.58
Eco-duration linear	-1.55	3.25	4.59
Eco-temperature linear	45.92	6.44	13.96
Technology linear	-263.09	40.42	60.41
B. Entry costs (EUR million)			
Launch intercept	2.118	1.259	1.343
Transition: A	2.576	1.574	1.094
Transition: B	3.434	2.124	2.182
Transition: C/D	1.609	0.959	1.351
Post-policy: A	1.892	0.768	1.215
Post-policy: B	3.494	2.048	2.630
Post-policy: C/D	4.735	2.602	3.169
Quarter 2 adjustment	0.067	0.474	0.279
Quarter 3 adjustment	-0.637	0.502	0.601
Quarter 4 adjustment	-0.520	0.483	0.571
$\log(1+N)$	-0.759	0.391	0.434
$\log(1+G)$	1.344	0.855	0.711
Shock scale	1.617	0.853	0.957

Notes: Design-cost linear terms denote $(\kappa_r)_k$ and curvatures denote $2(C_r)_{kk}$. The point estimates are unchanged across covariance calculations. Technology uses the scaling in Table 6. Both covariance estimators condition on upstream estimates and numerical approximations. Entry inference additionally conditions on fitted beliefs and two binding restrictions. Firm clustering uses seven clusters. These are local conditional standard errors, not confidence measures incorporating outer-search, active-set, or surrogate uncertainty.

comparisons retain this timing. Legacy availability follows the observed schedule, and the post-2020 compatibility filter remains active. Policy-dependent exit or a counterfactual legacy-survival schedule is not estimated.

The Eco-only index and nominal duration caps follow Appendices A.1.2 and A.2.1. Regulatory test-load weights are distinct from household program-choice probabilities; the modeled entry classes are A through D. Removing the cap switches off this policy restriction but retains the supported design bounds.

Optimization retains calibrated offsets, numerical margins, and inherited or copied designs. Fixed water and quarter-load attributes are audited rather than adjusted, so solved portfolios need not satisfy every nominal regulatory constraint. These conventions do not establish regulatory compliance or a grandfathering rule.

Adjustable first-introduction coordinates comprise Eco technology, full- and half-load Eco temperature and duration, and Cotton 40°C/60°C duration, subject to support and brand restrictions. Quarter-load Eco coordinates, water use, Cotton 20, and EasyCare attributes are fixed; Samsung Cotton 60°C duration is also fixed. Reference and copying rules follow Appendix F.2. For unlabeled entry, the D template supplies physical inputs without an EEI target or explicit class premium. The resulting experiment is broader than restoring pre-reform labels.

Cotton 40°C/60°C electricity uses its separately calibrated engineering relation. In particular, a change in the Eco technology coordinate does not mechanically improve Cotton technology. Household program probabilities determine how engineering outcomes enter realized use. The exported aggregate outcomes are sufficient for the electricity accounting below, but do not identify a causal split between within-product engineering, program switching, purchase substitution, and portfolio availability.

G.2 Numerical protocol used in the completed runs

The removal batch contains 20 paths for each of current policy, no duration cap, and no EEI/labels, plus one frozen joint-removal solve: 61 completed shards. The additional 2,000 grid paths are documented in Appendix G.4. All nonstationary paths contain 16 quarters and three household-group rows per quarter. Matching uses path IDs 0–19 and the common base seed 20261025; quarter seeds add 10,000 times the path ID and the quarter index. A common seed reduces avoidable variation but does not force realized introductions to match when policy changes payoffs or action sets.

The entry solver uses 16 retained Gibbs draws after 20 burn-in sweeps, with a maximum of 30 belief iterations. It monitors conditional entry probabilities at three fixed own-portfolio con-

figurations per firm: all no-entry, all last available entry action, and one seeded random profile. The maximum probability change must be at most 0.02 for two consecutive iterations. All 960 evolving removal path–quarter solves satisfy that check in three or four iterations. This checks a restricted collection of conditional probabilities; it does not verify the entire joint portfolio distribution, global uniqueness, or complete Gibbs mixing. The subsequently revised protocol with 64 draws, 100 burn-in sweeps, and an independently seeded update has not been applied to these results. No independent-check statistics are therefore reported.

Entry payoffs are supplied by an optimal-outcome predictor. Realized designs use the constrained local solver and tolerances in Appendix F.3. Household outcomes are then evaluated using solved prices and purchase shares. The price routine uses a fixed-point tolerance of 10^{-11} (the runtime override), with accelerated iteration and a simple-iteration fallback, allowing 10,000 and 20,000 evaluations respectively. These are numerical tolerances, not bounds on economic approximation error.

Table A7: Recorded numerical diagnostics for the removal experiments

	Current policy	No duration cap	No EEI / labels
Completed path–quarters	320	320	320
Passed monitored entry check	320	320	320
Mean entry iterations	3.19	3.19	3.27
Maximum monitored probability gap	0.00573	0.00307	0.00544
Strict-feasibility flag true	2	18	4
Fixed-input audit entries	2,361	845	1,753
Maximum design passes	3	4	3
Median payoff discrepancy (EUR m)	3.73	3.36	1.97
95th percentile discrepancy (EUR m)	10.68	8.95	10.75
Maximum payoff discrepancy (EUR m)	26.82	14.15	14.59

Notes: Each nonstationary scenario contributes 320 path–quarter records. All recorded constraint-audit entries are flagged both fixed and audit-only, and no tolerance allowances are recorded. Audit entries can recur for the same product across quarters, so their counts are not counts of unique violations or products. The payoff discrepancy is the maximum across firms within a realized portfolio of the difference between the outcome predictor's net profit and refined net profit under the underlying profit model; the table summarizes these maxima. It is not a comparison with observed or approximation-free profit.

Most records fail strict feasibility because fixed-input constraints are audited rather than re-

paired. Payoff discrepancies are substantial, and refinement does not correct earlier entry-payoff errors. Results remain conditional on the provisional approximations and outstanding convergence and validation checks; reported Monte Carlo errors exclude approximation and parameter uncertainty.

G.3 Aggregation, decomposition, and reproducibility

Table A8 reports the accounting levels underlying the removal comparisons. The equality $R = G - D$ is satisfied in every exported path–quarter–group row to within 5.6×10^{-9} kWh, and $T = R + B$ to within 7.5×10^{-9} kWh. Subgroup totals add to the all-household values to numerical precision. Taxes and rebates are zero in these experiments, so consumer welfare with rebates equals consumer surplus. The reference basket and population are identical across the three nonstationary policies; only the frozen benchmark has a different horizon reference-all total and population.

Table A8: Horizon levels and accounting components for the removal experiments

	Current policy	No duration cap	No EEI / labels	Frozen joint removal
Consumer surplus (EUR m)	588.351	1,191.568	689.220	1,780.844
Gross purchase use G (GWh)	200.034	165.889	142.563	256.406
Displaced reference D (GWh)	184.144	197.023	138.271	256.089
Relative use R (GWh)	15.890	-31.134	4.293	0.317
Reference-all use B (GWh)	443.824	443.824	443.824	440.119
Population total T (GWh)	459.714	412.690	448.117	440.436
Decision observations (millions)	9.660	9.660	9.660	9.579

Notes: Means of sums across 16 quarterly decision cohorts. Electricity is each cohort's quarterly use, aggregated across cohorts. The frozen benchmark repeats one initial-quarter outcome 16 times; it is not an average of 20 independently solved frozen paths. A negative R means savings relative to the purchase outside reference, not negative physical consumption.

For outcome Y, the paired path contrast is $d_p = \sum_t (Y^c_{pt} - Y^0_{pt})$. I report $\bar{d} = 20^{-1} \sum_p d_p$ and $MCSE = \mathrm{sd}(d_p)/\sqrt{20}$. For the frozen benchmark, the same deterministic value replaces Y^c_{pt} for every path; its reported MCSE therefore comes entirely from variation in the current-policy paths. Per-observation welfare effects in the main distribution table divide the horizon aggregate change by the common group population sum. They are neither effects conditional on buying a machine nor gains per distinct household.

The electricity bridge is the exact identity $\Delta R = \Delta G - \Delta D$. Since the tariff and lifetime

factor are common, the electricity-expenditure contribution to consumer surplus is $-0.30A\Delta R$. Table A9 separates that contribution from the remaining corrected purchase-logsum component. This accounting does not re-solve any hybrid economy. A causal mechanism decomposition would require household–product–program probabilities, a common support for changing portfolios, and explicit definitions of the hybrid attribute and choice environments.

Table A9: Accounting decomposition of removal-policy surplus changes

	Total ΔCS	Electricity term	Remainder
No duration cap	603.22	515.68	87.54
No EEI / labels	100.87	127.18	-26.31

Notes: EUR million across 16 cohorts. The electricity term equals $-0.30A\Delta R$; the remainder is total consumer-surplus change less this term. The electricity term is 85.5% of the cap-removal gain and exceeds the total gain under EEI/label removal because the remaining component is negative.

Product-design moments are recovered from the saved solved-design histories and matched to active product IDs in the final-quarter portfolio, deduplicating region and program rows. Within-path equal-product means are averaged over the 20 paths. Class shares are defined only for the labeled scenarios. The saved exports do not provide a ready-to-report sales-weighted design distribution, price series, program-specific energy decomposition, binding-constraint share, producer-surplus series, or household losing fraction. No such quantities are inferred from the aggregate welfare totals, and none of the reported tables require rerunning the simulation.

G.4 Completed Policy Grids: Implementation and Validation

The downloaded grid contains 50 weighted-EEI policies and 50 producer-tax policies, each with 20 complete paths. Every path has 16 quarters and three household groups: 96,000 outcome rows and 32,000 path–quarter diagnostic records in total. All 2,000 completed-path files, diagnostic files, and terminal checkpoints are present. Recalculated policy means agree with the collector exports to numerical precision, and packaged dependencies match their recorded hashes. Tables A11 and A12 enumerate every tested point, including baseline ID 0.

The weighted-EEI index is

$$EEI_j(w) = \frac{100\,[w_E e_{jE} + w_{40} e_{j40} + w_{60} e_{j60}]}{S(k_j)}, \qquad w_E + w_{40} + w_{60} = 1, \quad w_r \geq 0, \tag{G.1}$$

where Eco energy uses regulatory load weights and $S(k_j)$ is the capacity denominator in Appendix A.1.2. The grid is a deterministic selection from a half-percentage-point lattice with total Cotton weight no greater than 10%. Class thresholds are unchanged. Tax policies use equal program

weights of one third in equation (28) and enter production cost as

$$mc_j^\tau = mc_j + \mathcal{T}_j. \tag{G.2}$$

The EEI entry-payoff checkpoint is epoch 302; the carbon checkpoint is epoch 27 and uses its matching carbon operating-profit and Cotton-engineering files. Both retain provisional training status. Design refinement uses cost evaluation 47 and the saved entry-cost estimates. The different Cotton calibrations preclude treating the two family baselines as one common physical economy.

Accounting verification. The grid runtime computes electricity as $G = \sum_i W_i \sum_j s_{ij} E_{ij}$ and subtracts $0.30AG$ in its surplus calculation. It does not export displaced-reference use D or total reference-inclusive electricity T. Its corrected purchase-logsum component is recovered as $CS^G + 0.30AG$; thus

$$\Delta CW^G = \Delta(CS^G + 0.30AG) - 0.30A\Delta G + \Delta\text{rebates}. \tag{G.3}$$

The coefficients and units are those in Section 8.1. Aggregate rebates equal tax revenue to within EUR 7.5×10^{-9}; welfare equals surplus plus rebates to within EUR 1.9×10^{-8}. Subgroup sums agree within 2.3×10^{-8} in the exported units. Horizon populations are identical across policies: 9.660 million weighted decision observations, of which 2.254 million have positive duration preferences. Equal market-level rebates do not remove differences in pre-transfer incidence.

At baseline ID 0, mean gross electricity is 165.89 GWh for EEI and 128.87 GWh for tax. Corresponding CS^G levels are minus EUR 969.04 million and minus EUR 643.27 million. Negative surplus levels reflect the specified utility normalization and subtraction of gross lifetime electricity expenditure; the reported comparisons are within-family changes. Because D can change with purchasing and product availability, a common aggregate outside-option correction cannot convert these grids to the removal experiments' reference-adjusted accounting. Harmonized comparisons require re-evaluating household purchase allocations and reference electricity.

Numerical checks and scope. Both grids use the same path/quarter seeds and 16-draw, 20-burn protocol described above. All recorded belief checks pass: weighted EEI takes three to five iterations, with maximum final monitored gap 0.0158; tax takes three to four, with maximum gap 0.0115. These are the recorded conditional-probability checks, not an independent-seed test of the full portfolio distribution.

Strict-feasibility flags pass in 947 EEI records and no tax records. EEI audits contain 35,629 weighted-water, 5,883 upper-EEI, and 881 lower-EEI entries, all marked fixed and audit-only. Tax audits contain 25,803 weighted-water entries marked audit-only; the tax export does not supply their fixed-coordinate flag. No numerical policy allowances are recorded. As in the removal experiments,

Table A10: Numerical diagnostics for the completed policy grids

	Weighted EEI	Producer tax
Path–quarter solves	16,000	16,000
Belief checks passed	16,000	16,000
Strict-feasibility flags passed	947	0
Audit-only constraint entries	42,393	25,803
Median maximum profit error (EUR m)	2.84	1.46
95th-percentile maximum error (EUR m)	9.48	5.03

Notes: Each family contains 16,000 path–quarter solves. Profit errors are absolute discrepancies between predicted and refined net profits, taking the maximum across firms within each realized portfolio. Audit entries may repeat across quarters. They are not counts of distinct products, binding constraints, or failed numerical solves.

completed solves do not ensure nominal feasibility, and Monte Carlo errors exclude the substantial payoff discrepancies in Table A10 and parameter uncertainty.

Product moments in Table 10 use the final active portfolio and its saved solved-input history, deduplicated across region/program rows before averaging. They are equal-product means. Neither these means nor aggregate G identify a causal design/use/purchase decomposition, a sales-weighted attribute effect, or the fraction of households losing. The frontier compares means over the tested policies under the recorded accounting; a boundary maximizer does not establish a global optimum or the ranking after an outside-reference correction.

Table A11: Complete weighted-EEI grid

ID	w_E	w_{40}	w_{60}	ΔG	MCSE	ΔCW^G	MCSE
0	1.000	0.000	0.000	0.00	0.00	0.00	0.00
1	0.990	0.010	0.000	-0.21	0.12	1.30	2.13
2	0.990	0.000	0.010	0.58	0.98	0.31	4.70
3	0.980	0.020	0.000	-0.57	0.17	4.23	2.34
4	0.980	0.010	0.010	0.23	0.96	2.98	4.44
5	0.980	0.000	0.020	-0.72	1.03	11.76	6.50
6	0.900	0.050	0.050	-6.63	1.34	64.19	9.38
7	0.900	0.100	0.000	-1.48	0.90	21.15	4.43
8	0.900	0.000	0.100	-10.00	1.69	91.59	12.93
9	0.930	0.010	0.060	-5.72	1.46	54.44	10.13
10	0.930	0.060	0.010	-1.36	1.10	16.50	7.36
11	0.940	0.025	0.035	-2.07	1.34	19.38	8.81
12	0.900	0.020	0.080	-8.71	1.52	80.29	9.67
13	0.900	0.070	0.030	-4.04	1.13	37.31	7.50
14	0.950	0.035	0.015	-0.19	1.09	2.56	7.61
15	0.900	0.035	0.065	-7.05	1.49	65.55	10.91
16	0.900	0.085	0.015	-2.71	1.03	28.52	6.64
17	0.955	0.005	0.040	-1.85	1.37	14.60	8.70
18	0.920	0.000	0.080	-7.83	1.47	71.69	10.00
19	0.920	0.050	0.030	-2.88	1.39	28.96	9.06
20	0.955	0.045	0.000	-0.09	0.97	7.39	4.60
21	0.925	0.075	0.000	-0.83	0.89	14.66	4.51
22	0.920	0.030	0.050	-5.41	1.33	50.74	8.95
23	0.900	0.010	0.090	-9.37	1.70	84.91	13.95
24	0.960	0.015	0.025	0.17	1.18	0.89	8.02
25	0.920	0.040	0.040	-4.29	1.36	43.16	8.63
26	0.900	0.060	0.040	-4.85	1.35	48.37	8.63
27	0.950	0.000	0.050	-2.79	1.42	22.09	10.06
28	0.935	0.000	0.065	-5.49	1.50	49.85	10.99
29	0.915	0.010	0.075	-7.36	1.41	66.95	9.03
30	0.940	0.015	0.045	-3.17	1.43	26.58	10.02
31	0.965	0.020	0.015	-0.57	1.07	10.65	7.16
32	0.925	0.020	0.055	-5.94	1.48	56.13	10.14
33	0.905	0.025	0.070	-7.10	1.49	65.55	10.81
34	0.965	0.030	0.005	0.09	0.99	5.11	4.85
35	0.945	0.030	0.025	-0.54	1.21	7.67	8.20
36	0.905	0.040	0.055	-6.44	1.37	61.41	9.12
37	0.935	0.045	0.020	-0.97	1.14	9.08	7.91
38	0.915	0.060	0.025	-1.86	1.23	21.97	7.89
39	0.915	0.070	0.015	-1.57	1.13	17.17	7.91
40	0.905	0.090	0.005	-1.95	1.07	22.56	6.41
41	0.970	0.000	0.030	-1.01	1.19	12.19	7.08
42	0.910	0.000	0.090	-8.57	1.72	79.38	15.25
43	0.950	0.015	0.035	-1.74	1.37	15.46	8.92
44	0.930	0.040	0.030	-2.70	1.38	26.22	8.98
45	0.945	0.045	0.010	-0.37	1.08	8.33	7.92
46	0.910	0.050	0.040	-4.52	1.33	45.17	8.39
47	0.945	0.055	0.000	-0.35	0.97	9.87	4.86
48	0.935	0.065	0.000	-0.62	0.97	12.65	4.96
49	0.990	0.005	0.005	0.67	0.96	-0.79	4.47

Each row uses 20 paired paths. ΔG is gross purchase electricity in GWh; ΔCW^G is consumer welfare with rebates in EUR million under gross-use accounting. Changes use the family-specific zero-Cotton-weight or zero-tax baseline (ID 0). MCSE is the standard error of the paired simulation mean, conditional on parameters and numerical approximations. IDs identify downloaded shards; all tested policies are shown.

Table A12: Complete producer-tax grid

ID	τ_E	τ_C	ΔG	MCSE	ΔCW^G	MCSE
0	0	0	0.00	0.00	0.00	0.00
1	10	10	-4.03	0.12	40.72	1.06
2	25	25	-9.94	0.29	95.61	2.88
3	50	50	-19.02	0.40	185.29	3.37
4	75	75	-28.78	1.00	265.98	5.82
5	100	100	-37.34	0.97	339.48	6.31
6	100	0	-8.97	0.39	87.18	4.36
7	0	100	-29.02	0.95	274.25	6.57
8	25	0	-2.37	0.28	20.56	3.35
9	0	25	-7.61	0.21	76.73	1.49
10	50	0	-4.51	0.32	40.98	3.37
11	0	50	-14.96	0.36	147.16	2.74
12	40	95	-31.42	0.97	287.50	6.18
13	95	40	-20.19	0.57	191.39	4.18
14	15	75	-23.73	0.89	226.13	5.35
15	65	25	-13.28	0.33	128.66	3.51
16	65	100	-34.33	1.01	314.37	6.46
17	100	65	-28.05	0.99	261.78	6.28
18	25	50	-16.96	0.34	165.45	2.88
19	75	0	-6.74	0.33	64.01	3.78
20	40	70	-24.70	0.97	228.16	5.23
21	75	50	-21.42	0.56	203.03	4.02
22	45	20	-10.17	0.31	99.50	3.29
23	85	20	-13.54	0.37	131.63	3.69
24	20	95	-29.71	0.93	274.35	5.58
25	55	80	-28.36	0.97	263.02	5.92
26	85	90	-34.08	0.92	309.39	5.91
27	0	70	-20.35	0.45	200.09	4.11
28	40	35	-14.10	0.35	136.12	3.16
29	60	65	-25.05	0.97	232.14	5.54
30	80	35	-17.69	0.54	167.94	3.71
31	100	25	-16.24	0.65	154.10	4.60
32	0	85	-24.74	0.83	240.38	5.83
33	100	80	-32.52	0.96	298.59	5.67
34	10	35	-11.59	0.24	115.02	1.87
35	10	60	-18.35	0.41	180.35	2.98
36	25	65	-21.32	0.49	205.73	3.21
37	30	80	-26.81	0.97	247.85	5.76
38	35	10	-6.37	0.28	59.12	3.08
39	55	35	-15.27	0.34	148.63	3.15
40	60	10	-8.55	0.34	82.94	3.69
41	85	60	-25.55	0.97	238.43	5.89
42	10	90	-27.36	0.92	257.44	5.85
43	15	20	-7.59	0.22	75.98	1.47
44	15	45	-14.90	0.34	146.33	2.34
45	30	40	-14.64	0.31	142.57	2.68
46	30	100	-31.87	0.94	295.27	7.16
47	35	55	-19.03	0.37	185.74	3.12
48	45	60	-21.49	0.54	204.71	3.52
49	45	85	-29.04	0.98	267.66	5.89

Each row uses 20 paired paths. ΔG is gross purchase electricity in GWh; ΔCW^G is consumer welfare with rebates in EUR million under gross-use accounting. Changes use the family-specific zero-Cotton-weight or zero-tax baseline (ID 0). MCSE is the standard error of the paired simulation mean, conditional on parameters and numerical approximations. IDs identify downloaded shards; all tested policies are shown.